

# Relation Algebra

## §1. Notation

Traditionally Adjunction  $F \dashv G$  is defined as  $F(x) \leq y$  iff  $x \leq G(y)$ , and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

By define  $X : * \mapsto x$ ,  $Y : * \mapsto y$  and use diagram order, application can be replaced by composition:  $XF \leq Y$  iff  $X \leq YG$

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• write  $F$  as  $/$ ,  $G$  as  $\backslash$

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• render  $a \leq b$  as  $\boxed{\begin{smallmatrix} a \\ b \end{smallmatrix}}$

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• **rewrite**  $X \leq XF$  as  $1 \leq F$

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(1a)

$$\boxed{\frac{X/}{Y}} = \boxed{\frac{X}{Y\backslash}}$$

adj

$$\begin{aligned} \boxed{\frac{X/}{X/}} &\xRightarrow{\text{adj}} \boxed{\frac{X}{X\wedge}} \xRightarrow{\text{rewrite}} \boxed{\frac{*}{\wedge}} \xRightarrow[\text{unit}]{\text{put } \backslash \text{ on left}} \boxed{\frac{\backslash}{\wedge\backslash}} \\ &\xRightarrow[\text{put } / \text{ on right}]{\text{put } / \text{ on right}} \boxed{\frac{/}{\wedge/}} \\ \boxed{\frac{Y\backslash}{Y\backslash}} &\xRightarrow{\text{adj}} \boxed{\frac{Y\vee}{Y}} \xRightarrow{\text{rewrite}} \boxed{\frac{\vee}{*}} \xRightarrow[\text{counit}]{\text{put } / \text{ on left}} \boxed{\frac{\vee/}{/}} \\ &\xRightarrow[\text{put } \backslash \text{ on right}]{\text{put } \backslash \text{ on right}} \boxed{\frac{\vee\backslash}{\backslash}} \end{aligned}$$

now we have  $\wedge/ = /$  and  $\vee\backslash = \backslash$ , and you just discovered string diagrams.

## §1.1. Adjunctions

(1.1a)

| $F \dashv G$                                           | $F$ 's type                                                           | monad $FG$                        | $1 \vDash FG$                                                        | $GFE \vDash 1$                                                       | $F$ preserves $u$                                        | $G$ preserves $n$                                             | $FGF=F$                                                         | $GFG=G$                                                         |
|--------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|
| $\triangleleft \dashv \triangleleft$                   | $A \rightarrow A \otimes A$                                           | $\triangleleft \triangleleft = 1$ | $1 \vDash \triangleleft \triangleleft$                               | $\triangleleft \triangleleft \vDash 1$                               | $(RuS) \triangleleft = R \triangleleft uS \triangleleft$ | $(RnS) \triangleright = R \triangleright nS \triangleright$   | $\triangleleft \triangleleft \triangleleft = \triangleleft$     | $\triangleright \triangleleft \triangleright = \triangleright$  |
| $\multimap \dashv \multimap$                           | $A \rightarrow I$                                                     | $\multimap \multimap = T$         | $1 \vDash \multimap \multimap$                                       | $\multimap \multimap \vDash 1$                                       | $(RuS) \multimap = R \multimap uS \multimap$             | $(RnS) \multimap = R \multimap nS \multimap$                  | $\multimap \multimap \multimap = \multimap$                     | $\multimap \multimap \multimap = \multimap$                     |
| $\circ \dashv \circ$                                   | $(A \rightarrow B) \rightarrow (B \rightarrow A)$                     | $\circ \circ = 1$                 | $R = R \circ \circ$                                                  | $R = R \circ \circ$                                                  | $(RuS) \circ = R \circ uS \circ$                         | $(RnS) \circ = R \circ nS \circ$                              | $R \circ \circ \circ = R \circ$                                 | $R \circ \circ \circ = R \circ$                                 |
| $\multimap \dashv \triangleleft \dashv \multimap$      | $(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$ | $1$                               | $(\multimap \triangleleft \otimes 1) (1 \otimes \multimap) = 1$      | $(1 \otimes \multimap \triangleleft) (\triangleright \otimes 1) = 1$ | $=$                                                      | $=$                                                           | $=$                                                             | $=$                                                             |
| $\triangleright \dashv \multimap \dashv \triangleleft$ | $(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$ | $1$                               | $(1 \otimes \multimap \triangleleft) (\triangleright \otimes 1) = 1$ | $(\multimap \triangleleft \otimes 1) (1 \otimes \triangleright) = 1$ | $=$                                                      | $=$                                                           | $=$                                                             | $=$                                                             |
| $\Delta \dashv n$                                      | $(A \rightarrow B) \rightarrow (A \rightarrow B)^2$                   | $R \mapsto RnR = R$               | $R \vDash RnR$                                                       | $RnS \vDash R$                                                       | $\Delta(RuS) = \Delta Ru \Delta S$                       | $(RnT) n (SnU) = (RnS) n (TnU)$                               | $RnR = R$                                                       | $RnR = R$                                                       |
| $u \dashv \Delta$                                      | $(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$                   | $(R, S) \mapsto (RuS, RuS)$       | $R \vDash RuS$                                                       | $RuR \vDash R$                                                       | $(RuT) u (SuU) = (RuS) u (TuU)$                          | $\Delta(RnS) = \Delta Rn \Delta S$                            | $RuR = R$                                                       | $RuR = R$                                                       |
| $\perp \dashv !$                                       | $\{*\} \rightarrow (A \rightarrow B)$                                 | —                                 | —                                                                    | $\perp \vDash R$                                                     | —                                                        | —                                                             | —                                                               | —                                                               |
| $\cdot S \dashv /S$                                    | $(A \rightarrow B) \rightarrow (A \rightarrow C)$                     | $S/S$                             | $R \vDash (RS)/S$                                                    | $(T/S)S \vDash T$                                                    | $(RuT)S = RSuTS$                                         | $(RnT)/S = R/SnT/S$                                           | $((RS)/S)S = RS$                                                | $((T/S)S)/S = T/S$                                              |
| $S \cdot \dashv S \setminus$                           | $(B \rightarrow C) \rightarrow (A \rightarrow C)$                     | $S \setminus S$                   | $R \vDash S \setminus (SR)$                                          | $S(S \setminus T) \vDash T$                                          | $S(RuT) = SRuST$                                         | $S \setminus (RnT) = S \setminus RnS \setminus T$             | $S(S \setminus (SR)) = SR$                                      | $S \setminus (S(S \setminus T)) = S \setminus T$                |
| $Rn \dashv R \Rightarrow$                              | $(A \rightarrow B) \rightarrow (A \rightarrow B)$                     | $X \mapsto R \Rightarrow (XnR)$   | $X \vDash R \Rightarrow (XnR)$                                       | $Rn(R \Rightarrow Y) \vDash Y$                                       | $Rn(XuY) = (RnX)u(RnY)$                                  | $R \Rightarrow (XnY) = (R \Rightarrow X) n (R \Rightarrow Y)$ | $Rn(R \Rightarrow (XnR)) = XnR$                                 | $R \Rightarrow (Rn(R \Rightarrow Y)) = R \Rightarrow Y$         |
| $\mathcal{D} \dashv \cdot T$                           | $(A \rightarrow B) \rightarrow \text{Cor } A$                         | $R \mapsto (\mathcal{D}R)T$       | $R \vDash (\mathcal{D}R)T$                                           | $\mathcal{D}(AT) \vDash A$                                           | $\mathcal{D}(RuS) = \mathcal{D}Ru \mathcal{D}S$          | $(AnB)T = ATnBT$                                              | $\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$                   | $(\mathcal{D}(AT))T = AT$                                       |
| $\mathcal{R} \dashv T \cdot$                           | $(A \rightarrow B) \rightarrow \text{Cor } B$                         | $R \mapsto T(\mathcal{R}R)$       | $R \vDash T(\mathcal{R}R)$                                           | $\mathcal{R}(TA) \vDash A$                                           | $\mathcal{R}(RuS) = \mathcal{R}Ru \mathcal{R}S$          | $T(AnB) = TAnTB$                                              | $\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$                   | $T(\mathcal{R}(TA)) = TA$                                       |
| $\cdot f \dashv \cdot f^\circ$                         | $(A \rightarrow B) \rightarrow (A \rightarrow C)$                     | $ff^\circ$                        | $1 \vDash ff^\circ$                                                  | $f^\circ f \vDash 1$                                                 | $(RuS)f = RfuSf$                                         | $(RnS)f^\circ = Rf^\circ nSf^\circ$                           | $ff^\circ f = f$                                                | $f^\circ ff^\circ = f^\circ$                                    |
| $f^\circ \dashv \cdot f \cdot$                         | $(A \rightarrow C) \rightarrow (B \rightarrow C)$                     | $ff^\circ$                        | $1 \vDash ff^\circ$                                                  | $f^\circ f \vDash 1$                                                 | $f^\circ(XuY) = f^\circ Xu f^\circ Y$                    | $f(XnY) = fXn fY$                                             | $ff^\circ f = f$                                                | $f^\circ ff^\circ = f^\circ$                                    |
| $i \dashv E$                                           | $\text{Map} \dashv \text{Rel}$                                        | $E$                               | $\frac{1}{\exists} : A \rightarrow EA$                               | $\exists : EB \rightarrow B$                                         | —                                                        | —                                                             | $\frac{\exists}{\exists} = 1$                                   | $\frac{R}{\exists} \exists = R$                                 |
| $\cdot \exists \dashv \frac{f}{\exists}$               | $\text{Map}(A, EB) \rightarrow (A \rightarrow B)$                     | $1$                               | $\frac{f \exists}{\exists} = f$                                      | $\frac{R \exists}{\exists} = R$                                      | —                                                        | —                                                             | $\frac{\frac{f \exists}{\exists}}{\exists} = f \exists$         | $\frac{\frac{R \exists}{\exists}}{\exists} = \frac{R}{\exists}$ |
| $\frac{f}{\exists} \dashv \cdot \exists$               | $(A \rightarrow B) \rightarrow \text{Map}(A, EB)$                     | $1$                               | $\frac{R}{\exists} \exists = R$                                      | $\frac{f \exists}{\exists} = f$                                      | —                                                        | —                                                             | $\frac{R}{\exists} \frac{\exists}{\exists} = \frac{R}{\exists}$ | $\frac{f \exists}{\exists} \frac{\exists}{\exists} = f \exists$ |

## §1.2. Composing adjunctions

(1.2a)

$$\begin{array}{c}
\boxed{\begin{array}{c} X \\ f(R/T) \end{array}} \xLeftrightarrow[f^\circ \cdot \dashv f \cdot] \boxed{\begin{array}{c} f^\circ X \\ R/T \end{array}} \xLeftrightarrow[\cdot T \dashv /T] \boxed{\begin{array}{c} (f^\circ X)T \\ R \end{array}} \\
\text{f a map}
\end{array}$$

$$\begin{array}{c}
\xLeftrightarrow[\text{associativity}] \boxed{\begin{array}{c} f^\circ(XT) \\ R \end{array}} \xLeftrightarrow[f^\circ \cdot \dashv f \cdot] \boxed{\begin{array}{c} XT \\ fR \end{array}} \xLeftrightarrow[\cdot T \dashv /T] \boxed{\begin{array}{c} X \\ (fR)/T \end{array}}
\end{array}$$

$$\begin{array}{c}
\xRightarrow[\text{indirect equality}]{} \boxed{\begin{array}{c} f(R/T) \\ (fR)/T \end{array}}
\end{array}$$

|                       | $\cdot T$                                   | $T \cdot$                                         | $\cdot g$                                         | $g \cdot$                                        | $Tn$                                                      | $\circ$                                                                                                              | $\circ \triangleleft$                                                                                                | $\Delta$                                                                |
|-----------------------|---------------------------------------------|---------------------------------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| $\cdot S$             | $R/(ST) = (R/T)/S$                          | $T \setminus (R/S) = (T \setminus R)/S$           | $R/(Sg) = (Rg^\circ)/S$                           | $g(R/S) = (gR)/S$                                | —                                                         | $(Y/S)^\circ = S^\circ \setminus (Y^\circ)$                                                                          | $(RS)^\wedge = R^\wedge (S \otimes 1)$                                                                               | $(T_1 n T_2)/S = T_1/S n T_2/S$                                         |
| $S \cdot$             | $S \setminus (R/T) = (S \setminus R)/T$     | $(TS) \setminus R = S \setminus (T \setminus R)$  | $S \setminus (Rg^\circ) = (S \setminus R)g^\circ$ | $(g^\circ S) \setminus R = S \setminus (gR)$     | —                                                         | $(S \setminus Y)^\circ = Y^\circ/S^\circ$                                                                            | $((S \otimes 1)R)^\wedge = SR^\wedge$                                                                                | $S \setminus (T_1 n T_2) = S \setminus T_1 n S \setminus T_2$           |
| $\cdot f$             | $R/(fT) = (R/T)f^\circ$                     | $T \setminus (Rf^\circ) = (T \setminus R)f^\circ$ | $(fg)^\circ = g^\circ f^\circ$                    | —                                                | —                                                         | $(fY)^\circ = Y^\circ f^\circ$                                                                                       | —                                                                                                                    | $(T_1 n T_2)f^\circ = T_1 f^\circ n T_2 f^\circ$                        |
| $f \cdot$             | $f(R/T) = (fR)/T$                           | $(Tf^\circ) \setminus R = f(T \setminus R)$       | —                                                 | $(fg)^\circ = g^\circ f^\circ$                   | —                                                         | $(fY)^\circ = Y^\circ f^\circ$                                                                                       | —                                                                                                                    | $f(T_1 n T_2) = fT_1 n fT_2$                                            |
| $Rn$                  | —                                           | —                                                 | —                                                 | —                                                | $(RnT) \Rightarrow Y = R \Rightarrow (T \Rightarrow Y)$   | $(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$                                                            | —                                                                                                                    | $R \Rightarrow (T_1 n T_2) = (R \Rightarrow T_1) n (R \Rightarrow T_2)$ |
| $\circ$               | $(Y/T)^\circ = T^\circ \setminus (Y^\circ)$ | $(T \setminus Y)^\circ = Y^\circ/T^\circ$         | $(gY)^\circ = Y^\circ g^\circ$                    | $(gY)^\circ = Y^\circ g^\circ$                   | $(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$ | $Y^\circ \Rightarrow Y$                                                                                              | $(R^\circ)^\circ = (R^\circ \otimes 1)(1 \otimes \triangleright \circ)$<br>both bends: the $\circ$ section's display | $(T_1 n T_2)^\circ = T_1^\circ n T_2^\circ$                             |
| $\circ \triangleleft$ | $(RT)^\wedge = R^\wedge (T \otimes 1)$      | $((T \otimes 1)R)^\wedge = TR^\wedge$             | —                                                 | —                                                | —                                                         | $(R^\circ)^\circ = (R^\circ \otimes 1)(1 \otimes \triangleright \circ)$<br>both bends: the $\circ$ section's display | —                                                                                                                    | —                                                                       |
| $u$                   | $(X_1 u X_2)T = X_1 T u X_2 T$              | $T(X_1 u X_2) = TX_1 u TX_2$                      | $(X_1 u X_2)g = X_1 g u X_2 g$                    | $g^\circ(X_1 u X_2) = g^\circ X_1 u g^\circ X_2$ | $Tn(X_1 u X_2) = (TnX_1)u(TnX_2)$                         | $(X_1 u X_2)^\circ = X_1^\circ u X_2^\circ$                                                                          | —                                                                                                                    | —                                                                       |
| $\perp$               | $\perp T = \perp$                           | $T \perp = \perp$                                 | $\perp g = \perp$                                 | $g^\circ \perp = \perp$                          | $Tn \perp = \perp$                                        | $\perp^\circ = \perp$                                                                                                | —                                                                                                                    | —                                                                       |

### §1.3. Adjoint triples

Beyond  $u \dashv \Delta \vdash n$ , which §1.1 already carries as two rows, the table holds one adjoint triple with content:  $\exists_f \dashv f^* \dashv \forall_f$  along a map  $f$ , once on each side of the composite.  $f : A \rightarrow B$  throughout this subsection.

$$\begin{aligned} \cdot f \dashv \cdot f^\circ \dashv /f^\circ \\ f^\circ \cdot \dashv f \cdot \dashv f \setminus \end{aligned} \quad (1.3a)$$

The first two links of each chain are the map shunting rules, and the third is §1.1's division row read at a chosen divisor:  $\cdot S \dashv /S$  at  $S := f^\circ$ , and  $S \cdot \dashv S \setminus$  at  $S := f$ .

The third link is not the first over again. For a map  $f$ ,  $/f$  collapses to  $\cdot f^\circ$ ; being a map is exactly what that collapse spends  $R/(fS) = (R/S)f^\circ$ . —  $f^\circ$  is not a map, so  $/f^\circ$  does not collapse in turn, and the three operators stay distinct. On  $\text{Rel}(\mathbb{I}, A) = \text{EA}$  they are the image triple:

| operator        | acts by                                   | name                      |
|-----------------|-------------------------------------------|---------------------------|
| $\cdot f$       | $S \mapsto f[S]$                          | direct image, $\exists_f$ |
| $\cdot f^\circ$ | $T \mapsto f^{-1}[T]$                     | inverse image, $f^*$      |
| $/f^\circ$      | $S \mapsto \{b : f^{-1}(b) \subseteq S\}$ | $\forall_f$               |

§1.1's  $\mathcal{D} \dashv \cdot T$  is this same chain along the projection  $A \otimes B \rightarrow A$ , read through  $\text{Rel}(A, B) = E(A \otimes B)$ :  $\mathcal{D}R = \{(a, a) : \exists b. aRb\}$ ,  $AT$  is the relation that ignores  $b$  altogether, and the third link is  $R \mapsto 1 n R/T = \{(a, a) : \forall b. aRb\}$ .

$$\mathcal{D} \dashv \cdot T \dashv 1 n \cdot /T \quad (1.3c)$$

Three is where it stops, and one map breaks both ends. Take  $A = \{a_1, a_2\}$  and  $B = \{b\}$ :  $f[\{a_1\}] n f[\{a_2\}] = \{b\}$  while  $f[\{a_1\} n \{a_2\}] = \emptyset$ , so  $\cdot f$  does not preserve meets and has no left adjoint, and the same two subsets give  $\forall_f(\{a_1\} u \{a_2\}) = \{b\}$  against  $\forall_f\{a_1\} u \forall_f\{a_2\} = \emptyset$ , so  $/f^\circ$  does not preserve joins and has no right adjoint. A monic  $f$  restores the binary case and no more —  $\tau f$  still reaches only  $\text{im } f$ , and  $\forall_f \perp = B \setminus \text{im } f$  is still not  $\perp$ . The chain extends only when  $f^\circ$  is a map as well, and then  $/f^\circ = \cdot f^\circ = \cdot f$  and it repeats forever. For an  $S$  with neither  $S$  nor  $S^\circ$  a map there is no triple at all:  $\cdot S \dashv /S$  is two links and stops.

The rows that chain forever say nothing by it:  $\circ$  is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and  $\circ \dashv \circ \dashv \circ$  has no end, and §1.1's other row of that kind,  $\circ \triangleleft \dashv \triangleright \circ$ , goes the same way.

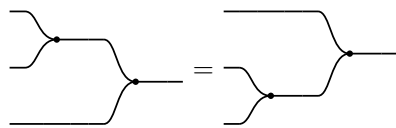
## §2. Relations

$\mathbf{Rel}$  is a poset-enriched category with  $(\otimes, \circ)$  where  $\otimes$  is commutative cartesian product, and converse  $\circ^{-1}$ , and

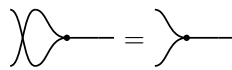
$\mathbf{C} : (\triangleright : A \otimes A \rightarrow A, \circ : I \rightarrow A)$  a commutative monoid with  $\mathbf{C}^\circ : \mathbf{C}$ . We write  $\mathbf{C}^\circ$  as  $(\triangleleft : A \rightarrow A \otimes A, \circ : A \rightarrow I)$

(2a)

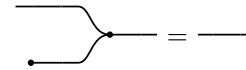
(2b)



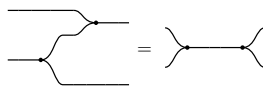
$\triangleright$  associative



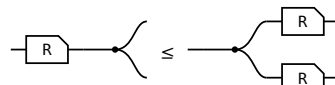
$\triangleright$  commutative



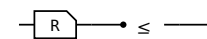
$\circ$  is its unit



Frobenius, one half — the other is its  $\circ$



$R \triangleleft \leq \triangleleft (R \otimes R)$

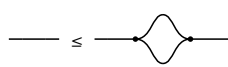


$R \circ \leq \circ$

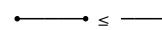
### §2.1. $\forall$ object $\mathbf{A}$ , $(\mathbf{A}, \triangleleft, \circ) \dashv (\mathbf{A}, \triangleright, \circ)$



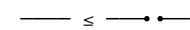
$\triangleleft \leq 1$



$1 \leq \triangleleft$



$\circ \leq 1$



$1 \leq \circ$

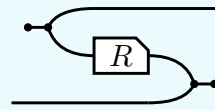
(2.1a)

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In  $\mathbf{Rel}$  it reads  $\{(a, a)\} \subseteq A \times A$  — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

### §3. $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

(3a)

$\circ$  is primitive, part of the data the first section lists. It turns both of  $R$ 's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are  $R^\circ$ , not its definition:



$$R^\circ = (\circ \triangleleft 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \circ)$$

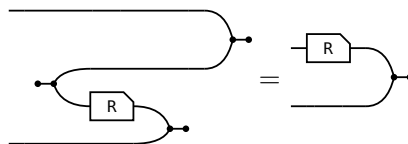
where  $\circ \triangleleft : I \rightarrow a \otimes a$  opens a pair of wires out of nothing and  $\triangleright \circ : b \otimes b \rightarrow I$  closes one, so the input of  $R^\circ$  is where the output of  $R$  was. Taking that formula as the definition would now be circular:  $\triangleleft$  and  $\triangleright$  are  $\triangleright^\circ$  and  $\triangleleft^\circ$ . The laws of  $\circ$  are that it is a contravariant 2-functor  $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ :

$$(i) \ 1^\circ = 1 \quad (ii) \ (RS)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

#### §3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:

(3.1a)

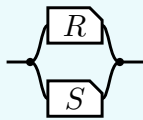


$R^\circ$  is DEFINED as the bending of  $(R \otimes 1) \triangleright \circ$ , so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the  $\circ \triangleleft$  bends the  $a$  strand down and the  $\triangleright \circ$  brings it back up, while the  $b$  strand rides through untouched. Nothing else is spent below.

§4.  $\cap$  is a commutative idempotent monoid on every hom-set

(4a)

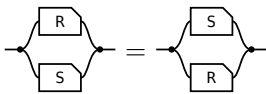
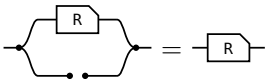
The **meet** of  $R, S : a \rightarrow b$ , the paper's **convolution**, is  $R \cap S := \langle (R \otimes S) \rangle$  — copy the input, run  $R$  and  $S$  on the two copies, merge the results — so what comes out is what both of them do.



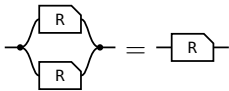
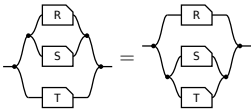
transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow  $\top = \text{---}\bullet\text{---}\bullet\text{---}$  (the paper's Lemma 4.11).

(4b)



**unit:** one half of  $\top$  per end — the merge's unit law absorbs the  $\bullet$ , the copy's counit law the  $\bullet$  **commutative:**  $\sigma$  crosses  $R \otimes S$  by naturality and is absorbed by cocommutativity and commutativity



**associative:** coassociativity and associativity;  $\otimes$  re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2 being strict here

So  $\leq$  is the order this monoid induces.  $R \cap S \leq R$  comes from the unit, and idempotency turns anything under both  $S$  and  $T$  into something under  $S \cap T$ , since  $R = R \cap R \leq S \cap T$ .

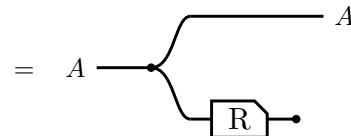
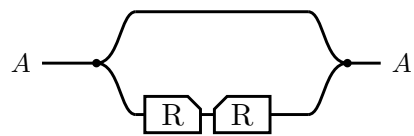
And one law relating  $\cap$  to composition, which is **not** an equation:

(4c)

| semi-distributivity, and what supplies it                                                                                                                         | picture |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $R \cap (S \cap T) \leq R \cap S \cap R \cap T$ — the lax copy law. Equality exactly when $R$ is single valued: the Maps section's $F(R \cap S) = F R \cap F S$ . |         |

## §5. Domain and range

The **domain**  $\text{Dom}(R) \triangleq 1 \cap R R^\circ$  and the **range**  $\text{Ran } R \triangleq \text{Dom}(R^\circ)$ .



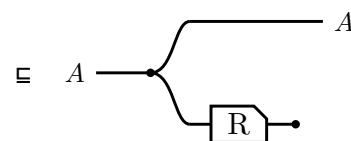
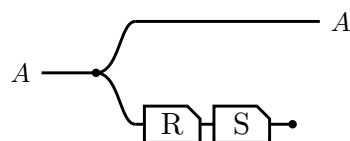
$\triangleright$  lands the return leg back on the value  $\triangleleft$  handed out,  
so  $R^\circ \triangleright$  cuts to  $\circ$  — Frobenius

Running  $R$  and throwing the result away leaves only the fact that  $R$  could fire, and  $\text{Ran } R$  is the same picture with the box mirrored. In  $\text{Rel}$  both steps are  $\{(a, a) : \exists b. a R b\}$ .

|                                                                                   |
|-----------------------------------------------------------------------------------|
| $\text{Dom}(R) \triangleq 1 \cap R R^\circ$                                       |
| $\text{Dom}(R) \subseteq A \iff R \subseteq A R$ , for $A$ coreflexive            |
| $\text{Dom}(RS) \subseteq \text{Dom}(R)$                                          |
| $\text{Dom}(R \cap S) = 1 \cap S R^\circ$                                         |
| $R \text{ entire} \iff \text{Dom}(R) = 1 \iff 1 \subseteq R R^\circ$              |
| $R \text{ simple} \iff R^\circ R \subseteq 1$                                     |
| $R \text{ a map} \iff R \text{ entire and simple}$                                |
| $R, S \text{ entire} \implies RS \text{ entire}$ — likewise simple, likewise maps |
| $RS \text{ entire} \implies R \text{ entire}$                                     |

### §5.1. Sliding the discard

$\text{Dom}(RS) \subseteq \text{Dom}(R)$ , and a single glyph for  $\text{Dom}$  would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.



$S^\circ \circ \circ$ , the lax axiom for  $\circ$  in the first section  
— the discard slides back past  $S$

Equality is  $S$  entire, which is the same picture read as  $\text{Dom}(R) = 1 \iff R \text{ entire}$ .

## §6. Maps

|                                                                                               |                                                          |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>surjective</b><br>$\mathbb{1} \subseteq R^\circ R$ , that is, $R^\circ$ entire.            | <b>single valued</b><br>$R^\circ R \subseteq \mathbb{1}$ |
| <b>entire</b><br>$\mathbb{1} \subseteq RR^\circ$ . With <b>single valued</b> , a <b>map</b> . | <b>injective</b><br>$RR^\circ \subseteq \mathbb{1}$      |

(6a)

|                                                         |                        |
|---------------------------------------------------------|------------------------|
| <b>law</b><br><br>$F(RnS) = FRnFS$<br>$F$ single valued | <b>picture</b><br><br> |
|---------------------------------------------------------|------------------------|

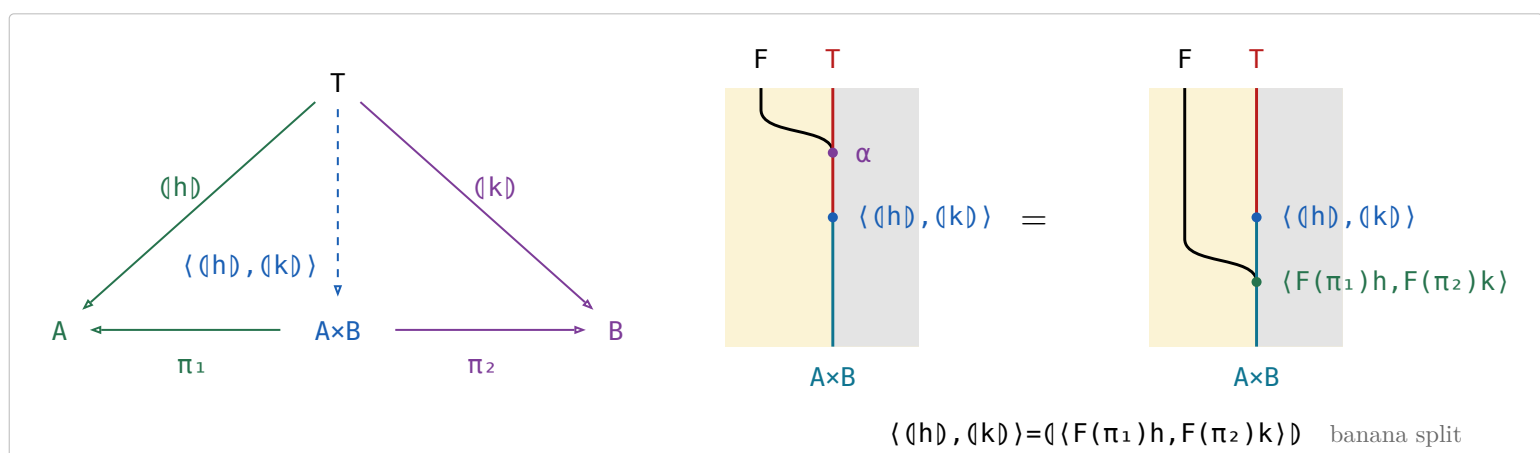
(6b)

§7. Reduce in  $\mathcal{S}et$ 

| definition                                                                                                                                                                         | type                                                        | note                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| $\text{listr } A ::= \text{nil} \mid \text{cons } (A, \text{listr } A)$                                                                                                            | $\text{Set} \rightarrow \text{Set}$                         | The cons-lists over $A$ , the datatype every row below folds (B&dM p. 55).                                   |
| $\text{sum} \triangleq ([\text{zero}, \text{plus}])$                                                                                                                               | $\text{listr Nat} \rightarrow \text{Nat}$                   | $\text{plus}(a, b) = a + b$ .                                                                                |
| $\text{length} \triangleq ([\text{zero}, \pi_2 \text{ succ}])$                                                                                                                     | $\text{listr } A \rightarrow \text{Nat}$                    | $\pi_2$ drops the head and keeps the count of the tail, $\text{succ}$ adds one for the head.                 |
| $\text{average} \triangleq (\text{sum}, \text{length}) \text{ div}$                                                                                                                | $\text{listr Nat} \rightarrow \text{Real}$                  | $\text{div}(m, n) = m/n$ , with $\text{div}(0, 0) = 0$ so <b>average</b> is total. Traverses the list twice. |
| banana-split law<br>$\langle \langle h \rangle, \langle k \rangle \rangle = \langle \langle F(\pi_1)h, F(\pi_2)k \rangle \rangle$                                                  | $T \rightarrow A \times B$                                  | Any fork of folds is a single fold, hence one traversal — $F$ the base functor.                              |
| what it reduces to<br>$\alpha \langle \langle h \rangle, \langle k \rangle \rangle = F(\langle \langle h \rangle, \langle k \rangle \rangle) \langle F(\pi_1)h, F(\pi_2)k \rangle$ | $FT \rightarrow A \times B$                                 | All that (11.6.1a) leaves to check: the fork satisfies the defining equation.                                |
| the instance<br>$\langle \text{sum}, \text{length} \rangle = ([\text{zeros}, \text{pluss}])$                                                                                       | $\text{listr Nat} \rightarrow \text{Nat} \times \text{Nat}$ | $\text{pluss}(a, (b, n)) = (a + b, n + 1)$ , so <b>average</b> runs in one pass.                             |
| $\text{preds } n = [n, n-1, \dots, 1]$                                                                                                                                             | $\text{Nat} \rightarrow [\text{Nat}]$                       | B&dM Ex 3.6 (p. 57): apply Ex 3.4 to write <b>preds</b> as $\langle k \rangle \pi_1$ .                       |

(7a)

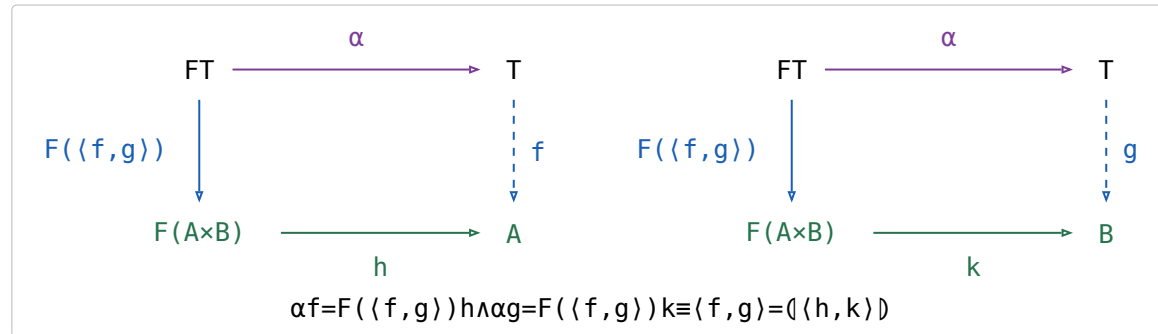
(7b)





## §7.1. Fokkinga's mutual recursion theorem

- $F\text{-Alg}(\mathcal{A})$  — the algebras and their homomorphisms — has binary products whenever  $\mathcal{A}$  does, and the forgetful  $U : F\text{-Alg}(\mathcal{A}) \rightarrow \mathcal{A}$  creates them: the product of  $h : FA \rightarrow A$  and  $k : FB \rightarrow B$  is carried by  $A \times B$ , with structure  $\langle F(\pi_1)h, F(\pi_2)k \rangle : F(A \times B) \rightarrow A \times B$ .
- $\pi_1$  and  $\pi_2$  are homomorphisms out of it, and  $\langle p, q \rangle : X \rightarrow A \times B$  is a homomorphism **iff**  $p$  and  $q$  both are — substitute the structure and compare the two forks component by component.
- The banana-split law of (7a) IS that product's universal property read at the initial algebra:  $\langle h \rangle$  and  $\langle k \rangle$  are the unique homomorphisms to  $(A, h)$  and  $(B, k)$ , so their fork is the unique homomorphism into the product, which is  $\langle \langle F(\pi_1)h, F(\pi_2)k \rangle \rangle$ .
- Fokkinga's theorem is **strictly more general** and is not that product: in (7.1a) the two arrows leave  $F(A \times B)$ , not  $FA$  and  $FB$ , so each sees BOTH components. The product is the case where they factor as  $F(\pi_1)h$  and  $F(\pi_2)k$ ; B&dM's Ex 3.4 is the other special case (p. 58). What it says: **any algebra on a product carrier is folded by a fork**.



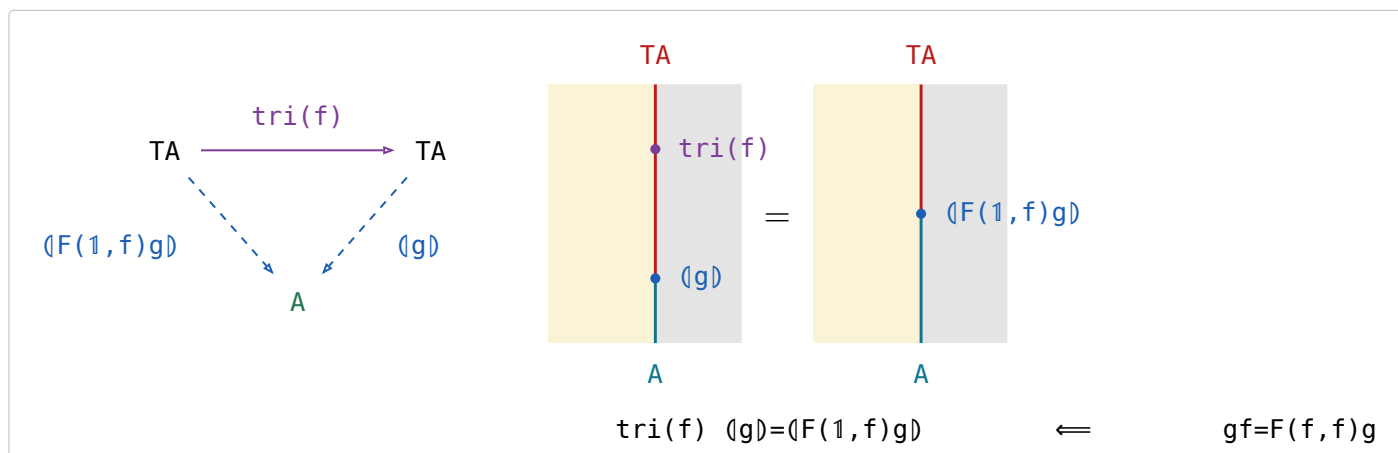
(7.1a)

## §7.2. Ruby triangles

| stage                       | definition                                                                                                                            |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| informally (p. 58)          | $\text{tri}(f) [a_0, a_1, \dots, a_i, \dots, a_n] = [a_0, f a_1, \dots, f^i a_i, \dots, f^n a_n]$                                     |
| for cons-lists (p. 59)      | $\text{tri}(f) = \langle [\text{nil}, (1 \times \text{listr}(f)) \text{ cons}] \rangle$                                               |
| with the base functor named | $F(A, B) = 1 + A \times B$ , $\alpha = [\text{nil}, \text{cons}]$ , so $\text{tri}(f) = \langle F(1, \text{listr}(f)) \alpha \rangle$ |
| abstractly (p. 59)          | $F$ a bifunctor with initial type $(\alpha, T)$ : $\text{tri}(f) = \langle F(1, T(f)) \alpha \rangle$                                 |

(7.2a)

For the definition to make sense  $f : A \rightarrow A$  is required, and then  $\text{tri}(f) : TA \rightarrow TA$ .



(7.2b)

(7.2b) is Horner's rule, generalised from cons-lists to any initial type; B&dM call it that because for cons-lists it is the schoolbook method for evaluating a polynomial (p. 58). Fusion reduces it to  $F(1, T(f)) \alpha \langle g \rangle = F(1, \langle g \rangle) F(1, f) g$  (p. 59).

## §7.3. Depth of a tree

|                  |                                                                                                                                                                                                                                                |        |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| the datatype     | $\text{tree } A ::= \text{tip } A \mid \text{bin } (\text{tree } A, \text{tree } A)$ , base functor $F(A,B)=A+B \times B$ ,<br>initial type $([\text{tip}, \text{bin}], \text{tree})$                                                          | (7.3a) |
| the fold         | $\llbracket [g, h] \rrbracket$ is the unique $f$ with $f(\text{tip } a) = g \ a$ and $f(\text{bin } (x, y)) = h \ (f \ x, f \ y)$                                                                                                              |        |
| $\text{tree}(f)$ | $\text{tree}(f) = \llbracket F(f, 1) \rrbracket [\text{tip}, \text{bin}]$ ; pointwise $\text{tree}(f)(\text{tip } a) = \text{tip } (f \ a)$ and<br>$\text{tree}(f)(\text{bin } (x, y)) = \text{bin } (\text{tree}(f) \ x, \text{tree}(f) \ y)$ |        |
| $\text{max}$     | $\text{max} = \llbracket [1, \text{bmax}] \rrbracket$ , where $\text{bmax } (a, b)$ is the larger of $a$ and $b$                                                                                                                               |        |
| $\text{depths}$  | $\text{depths} = \text{tree}(\text{zero}) \ \text{tri}(\text{succ})$ — replaces every tip by its depth in the tree, <b>zero</b> the constant<br>function returning 0 and <b>succ</b> the successor                                             |        |
| $\text{depth}$   | $\text{depth} = \text{depths} \ \text{max}$                                                                                                                                                                                                    |        |

## §8. / is all of

(8a)

$$x (R/S) y \iff \forall p. y S p \rightarrow x R p$$

$$x (S \setminus R) y \iff \forall p. p S x \rightarrow p R y$$

/compares images:  $S(y) \subseteq R(x)$ . \ compares preimages:  $S^\circ(x) \subseteq R^\circ(y)$ .

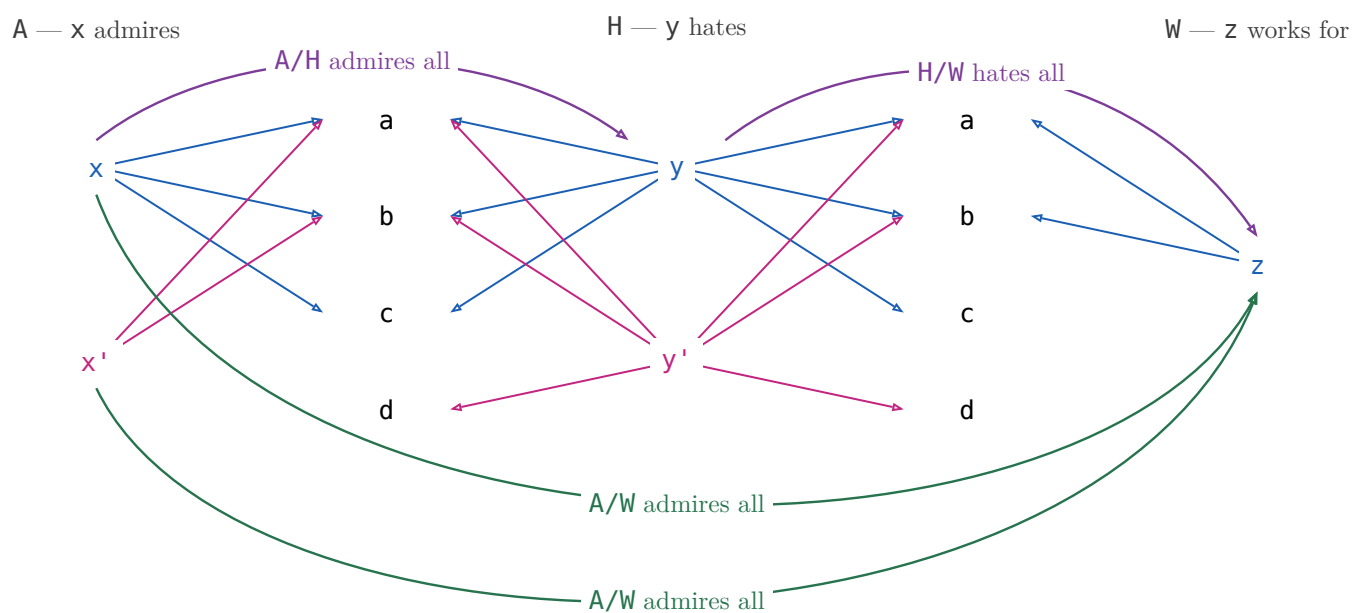
example: A admires, H hates, W works for

$x (A/H) y$  —  $x$  admires everyone  $y$  hates.

$x (H \setminus A) y$  — everyone who hates  $x$  admires  $y$ .

§8.1.  $(R/S)(S/W) \sqsubseteq R/W$ 

(8.1a)



$$x (A/H) y \text{ — } x \text{ admires everyone } y \text{ hates}$$

$$y (H/W) z \text{ — } y \text{ hates everyone } z \text{ works for}$$

$$x (A/W) z \text{ — } x \text{ admires everyone } z \text{ works for}$$

(8.1b)

$(A/H)(H/W)$  is a path:  $x \rightarrow y \rightarrow z$ , and that is all of it.  $A/W$  also holds of  $x'$ , who admires everyone  $z$  works for — but  $x'$  does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of  $(R/S)(S/W) \sqsubseteq R/W$ .

|                                                                                                                                                                                                                         |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| $X \in R/S \iff X \subseteq R$<br>$X$ is any $X$ -to- $Y$ pairing; one that only pairs $X$ with a $y$ such that $x$ admires everyone $y$ hates lies inside $R/S$ , and $R/S$ is the largest such.                       |  |
| $X \in S \setminus R \iff S X \subseteq R$<br>The mirror — divide on the left when $X$ comes first.                                                                                                                     |  |
| $(R/S) S \subseteq R$<br>There is a $y$ such that $x$ admires everyone $y$ hates, and $p$ is one of the people $y$ hates — then $x$ admires $p$ too. Strict at $S=\emptyset$ : $R/S$ is everyone, $(R/S) S=\emptyset$ . |  |
| $S (S \setminus R) \subseteq R$<br>The mirror.                                                                                                                                                                          |  |
| <b>associate:</b> $R/(S_1 S_2) = (R/S_2)/S_1$<br>A friend's enemy is two hops: divide by the far end first.                                                                                                             |  |
| $(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$<br>The mirror.                                                                                                                                                |  |
| <b>maps:</b> $f (R/S) = (f R)/S$<br>Rename $x$ before or after dividing — the licence to write $f R/S$ .                                                                                                                |  |
| $R/(f S) = (R/S) f^\circ$<br>Rename $y$ : a map leaves a denominator as $f^\circ$ outside the box.                                                                                                                      |  |
| $(R/S) (S/W) \subseteq R/W$<br>Someone who admires all of a hate-set that already covers everyone $Z$ works for admires those people too.                                                                               |  |
| $1 \in R/R$<br>$R/R$ runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only $a$ and $b$ admire each other's idols too, and still stay two people.                          |  |
| $(R/R) (R/R) = R/R$<br>$R/R$ is the preorder <b>admires at least as much as</b> , and a preorder is idempotent. Freyd writes $\sqsubseteq$ ; with $1 \in R/R$ above it is an equality.                                  |  |
| $(R/R) R = R$<br>Reaching $p$ through someone whose idols $x$ fully admires is reaching $p$ directly, since $x$ admires their own idols.                                                                                |  |
| $R/1 = R$<br>Dividing by $1$ : $p$ 's set is just $\{p\}$ , so admiring all of it is admiring $p$ .                                                                                                                     |  |
| $R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$<br>Admiring a combined hate-set is admiring each set in full.                                                                                                                     |  |
| $S \setminus (R/W) = (S \setminus R)/W$<br>Which is why $S \setminus R/W$ needs no bracket.                                                                                                                             |  |

Fifteen laws, fifteen pictures, and not one shows a generator:  $\eta$ ,  $\upsilon$ ,  $\circ$  and composition are what the Frobenius generators build, and  $/$  is none of those — it is posited, with nothing to unfold.

# §9. $\frac{R}{S}$

$\frac{R}{S} \triangleq (R/S) \cap (S/R)^\circ$ . In Rel  $x$  and  $y$  has the same image:  $\forall p. (x R p \iff y S p)$

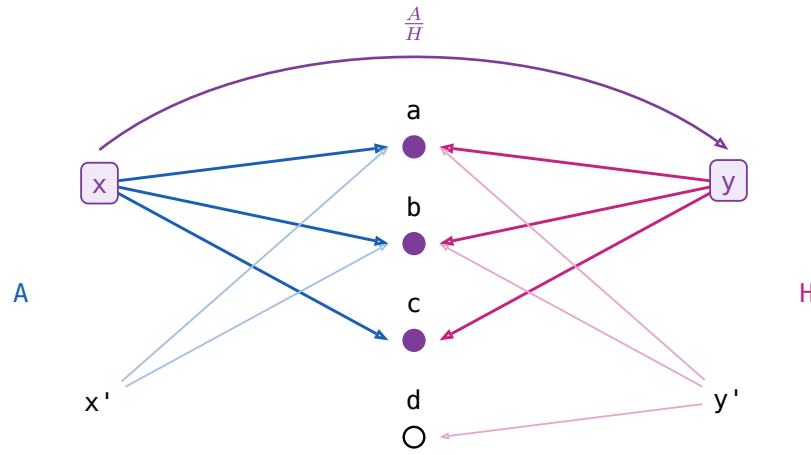
(9a)

$x (A/H) y$  —  $x$  admires everyone  $y$  hates  
 $x ((H/A)^\circ) y$  —  $x$  admires only people  $y$  hates  
 $x ((A/H) \cap (H/A)^\circ) y$  —  $x$  admires only and all whom  $y$  hates

(9b)

$x$  admires exactly whom  $y$  hates:  $/$  is **all of**,  $\frac{R}{S}$  is **only all of**.

(9c)



| law                                                                              | the reading                                                                                                                                        |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| $X \subseteq \frac{R}{S} \iff XS \subseteq R \text{ and } X^\circ R \subseteq S$ | $X$ may pair $x$ with $y$ only when $x$ admires exactly whom $y$ hates. Both halves must typecheck, so the operation is <b>partial</b> .           |
| $(\frac{R}{S})^\circ = \frac{S}{R}$                                              | Matching is symmetric.                                                                                                                             |
| $\frac{R}{S} \frac{S}{W} \subseteq \frac{R}{W}$                                  | And transitive.                                                                                                                                    |
| $\frac{R}{S} S \subseteq R$                                                      | $(\exists y. x(\frac{R}{S})y \wedge ySp) \rightarrow xRp$<br>$x$ <b>only</b> admires whom $y$ hates<br>$\frac{R}{S} S = \text{Dom}(\frac{R}{S}) R$ |
| $\frac{R}{R} R = R$                                                              | $(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$<br>$x$ and $y$ admire the same people<br>$y=x$ always qualifies ( $1 \in R \circ R$ below)      |
| $1 \subseteq \frac{R}{R}$                                                        | $x(\frac{R}{R})y$ if $x$ and $y$ admires the same peoples.                                                                                         |
| $(\frac{R}{R})^2 = \frac{R}{R}$                                                  | So the relation <b>admires the same people</b> is an equivalence relation.                                                                         |
| $X \subseteq \frac{R}{R} \iff XR \subseteq R$ , for symmetric $X$                | The largest symmetric arrow that leaves $R$ alone.                                                                                                 |
| $\frac{R}{1}$ is the <b>simple part</b> of $R$                                   | The people who admire exactly one person and nobody else. It equals $R$ only when $R$ is simple, unlike $R/1=R$ .                                  |
| $\text{Dom } \frac{R}{S} = 1 \cap (R/S) (S/R)$                                   | Its domain is the <b>domain of simplicity</b> of $R$ .                                                                                             |

(9d)

## §10. Freyd's Power allegories

(10a)

A **power allegory** is a division allegory with one unary operation on arrows,  $\exists$  **epsilon**, subject to

$$\begin{aligned} \exists_R \square &= R\square, & \exists_R &= \exists_R \square \\ \mathbb{1} &\subseteq (R/\exists_R)(\exists_R/R) & \exists_R &\text{ is } \mathbf{thick} \\ \exists_R &= \mathbb{1} & \exists_R &\text{ is } \mathbf{straight} \end{aligned}$$

$R\square$  is  $R$ 's target, an identity arrow. For  $R : A \rightarrow B$  write  $\exists : EA \rightarrow B$ , dropping the subscript.

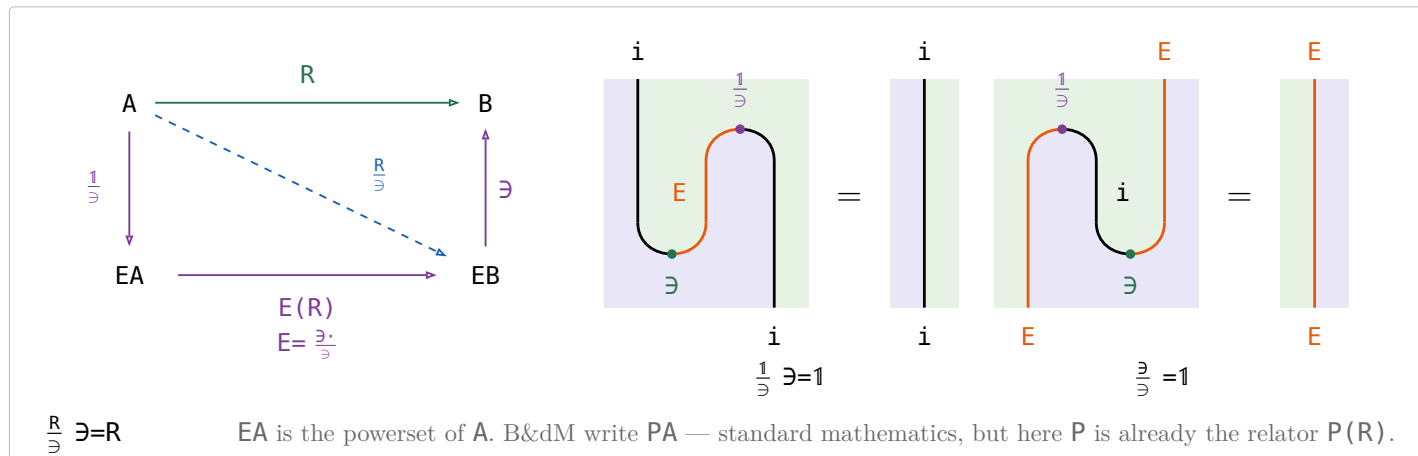
$$\epsilon \triangleq \exists^\circ : A \rightarrow EA$$

(10b)

| law                                                                                            | the reading                                                                                                                         |
|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1 $\exists_R \square = R\square, \exists_R = \exists_R \square$                                | One $\exists$ per object, not per arrow.                                                                                            |
| 2 $\exists$ is <b>thick</b>                                                                    | <b>Comprehension:</b> every $x$ has a set of exactly the people $x$ admires.                                                        |
| 3 $\frac{R}{\exists} : A \rightarrow EB$ , for $R : A \rightarrow B$                           | convert a relation to a function. $\mathbf{a} \frac{R}{\exists} = \{b \mid a R b\}$                                                 |
| 4 $\frac{R}{\exists}$ is a <b>map</b>                                                          |                                                                                                                                     |
| 5 $\frac{R}{\exists} \exists = R$                                                              |                                                                                                                                     |
| 6 $F \sqsubseteq \frac{F\exists}{\exists}$ , $F$ simple                                        | A partial choice of sets is inside the total one.                                                                                   |
| 7 $\frac{1}{\exists}$ , the <b>singleton map</b> , monic                                       | The one-person set.                                                                                                                 |
| 8 $\frac{\exists}{\exists} = \mathbb{1}$                                                       | Make the set of a set, then read it back one level down.                                                                            |
| 9 <b>fusion:</b> $\frac{fR}{\exists} = f \frac{R}{\exists}$ , $f$ a map                        | Naturality of the unit, $f \frac{1}{\exists} = \frac{1}{\exists} (E(f))$ .                                                          |
| 10 $\frac{f}{\exists} = f \frac{1}{\exists}$ , $f$ a map                                       | Rename first or take singletons first — the fusion row above at $R=1$ .                                                             |
| 11 $\frac{R}{S} = \frac{R}{\exists} \left( \frac{S}{\exists} \right)^\circ$                    | $x$ and $y$ match when $R$ sends $x$ and $S$ sends $y$ to the same set.                                                             |
| 12 $E(R) \triangleq \frac{\exists R}{\exists}$ , $E \triangleq \frac{\exists \cdot}{\exists}$  | $E(R) : EA \rightarrow EB$ , image of a set of $A$                                                                                  |
| 13 $\frac{R}{\exists} = \frac{1}{\exists} E(R)$                                                | $\frac{1}{\exists} : x \mapsto \{x\}$                                                                                               |
| 14 $\frac{S}{\exists} E(R) = \frac{S}{\exists} \frac{\exists R}{\exists} = \frac{SR}{\exists}$ | absorption — the monad's composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{SR}{\exists}$ , §10.2               |
| 15 $\text{subset} \triangleq \exists/\exists : EA \rightarrow EA$                              | $xs \text{ subset } ys \iff \forall a. ys \exists a \rightarrow xs \exists a$ , that is $ys \subseteq xs$ , not $xs \subseteq ys$ . |

### §10.1. $\mathbf{i} \vdash \mathbf{E}$ Power Allegory defined as adjunction

(10.1a)



$i$  is the inclusion  $\text{Map}(\mathcal{A}) \rightarrow \mathcal{A}$ , doing nothing, so  $E$  is both the functor and the monad  $iE$ .

## §10.2. $\mathcal{A} \cong \text{Kleisli}(\mathbf{E})$

$$\mathbf{E}(R) \triangleq \frac{\exists R}{\exists}, \frac{1}{\exists} : A \rightarrow EA, \text{ union} = \mathbf{E}(\exists) = \frac{\exists \exists}{\exists} : E(EA) \rightarrow EA \quad (10.2a)$$

$$f \diamond g \triangleq f \mathbf{E}(g) \text{ union} : A \rightarrow EC, \text{ for } f : A \rightarrow EB \text{ and } g : B \rightarrow EC$$

the monad is on  $\text{Map}(\mathcal{A})$ , not on the allegory:  $\mathbf{E}$  is a relator on all relations, but  $\frac{1}{\exists}$ , **union** and  $f \mathbf{E}(g) \text{ union}$  are maps, and the Kleisli construction happens where they live.

$$\begin{aligned} \boxed{\frac{S}{\exists} \diamond \frac{R}{\exists}} & \xrightarrow{\text{definition of } \diamond, \text{ union} = \mathbf{E}(\exists)} \boxed{\frac{S}{\exists} \mathbf{E}(\frac{R}{\exists}) \mathbf{E}(\exists)} \xrightarrow{\mathbf{E} \text{ a functor, } E(X)E(Y) = E(XY)} \boxed{\frac{S}{\exists} \mathbf{E}(\frac{R}{\exists} \exists)} \\ & \xrightarrow{(10b), \frac{R}{\exists} \exists = R} \boxed{\frac{S}{\exists} \mathbf{E}(R)} \xrightarrow{(10b), \text{ absorption}} \boxed{\frac{SR}{\exists}} \end{aligned} \quad (10.2b)$$

the first three steps only unfold  $\diamond$  —  $\mathbf{E}$  is a functor, so  $\mathbf{E}(\frac{R}{\exists})\text{union} = \mathbf{E}(\frac{R}{\exists} \exists) = \mathbf{E}(R)$  and the **union** is gone. Absorption, the row of (10b), is the whole law, and it is the functoriality of  $\frac{1}{\exists}$ .

$1$  goes to  $\frac{1}{\exists}$ , the Kleisli identity, by definition — so  $\frac{1}{\exists}$  is an isomorphism of categories  $\mathcal{A} \cong \text{Kleisli}(\mathbf{E})$ , and §10.1's  $i \dashv \mathbf{E}$  is the Kleisli adjunction.  $i$  is the identity on objects, so every object of the allegory is free.

## §11. Relator

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from  $R$  to  $S$  IS  $R \sqsubseteq S$ . A **relator**  $F : \mathcal{C} \rightarrow \mathcal{D}$  is a 2-functor between allegories:

$$F(1) = 1 \quad F(RS) = F(R)F(S) \quad R \sqsubseteq S \implies F(R) \sqsubseteq F(S)$$

Preserving  $\circ$  is **not** asked for —  $\circ$  is an identity-on-objects involution  $\mathcal{C}^{\circ} \rightarrow \mathcal{C}$ , no part of the 2-category.

| the statement                                                                       | B&dM                      | (11b) |
|-------------------------------------------------------------------------------------|---------------------------|-------|
| For $f$ a map, $F(f)$ is a map and $F(f^{\circ}) = F(f)^{\circ}$ .                  | Lemma 5.1                 |       |
| Over a <b>tabular</b> allegory a functor is a relator $\iff$ it preserves $\circ$ . | Theorem 5.1               |       |
| $F(R^{\circ}) = F(R)^{\circ}$ for every $R$ , so $F(R)^{\circ}$ needs no bracket.   | after Theorem 5.1, p. 113 |       |
| Two relators agreeing on maps are equal.                                            | Corollary 5.1             |       |
| $F(X \cap Y) = F(X) \cap F(Y)$ for $X, Y$ coreflexive.                              | Ex 5.2                    |       |
| $F(R \cap S) \sqsubseteq F(R) \cap F(S)$ , and strictly.                            | Ex 5.2, the restriction   |       |
| $F(\text{Dom}(R)) = \text{Dom}(F(R))$ for $F$ preserving $\circ$ .                  | —                         |       |

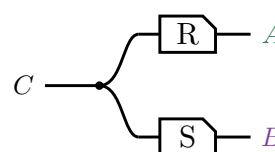
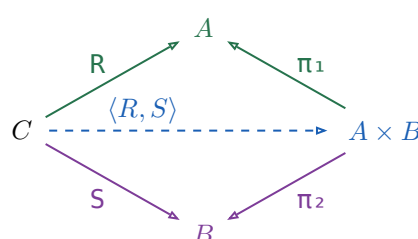
The **power relator**  $P$  —  $xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$  — is where the fourth is strict: for  $R = \{(a_1, b_1), (a_2, b_2)\}$  and  $S = \{(a_1, b_2), (a_2, b_1)\}$  the pair  $(\{a_1, a_2\}, \{b_1, b_2\})$  is in  $P(R) \cap P(S)$ , while  $R \cap S = \emptyset$ .

### §11.1. Fork $\langle R, S \rangle$

The **fork** of  $R : C \rightarrow A$  and  $S : C \rightarrow B$  is  $\langle R, S \rangle \triangleq R \pi_1^{\circ} \cap S \pi_2^{\circ}$ , where  $(\pi_1, \pi_2)$  is the tabulation of  $\tau$ .

$$\langle R, S \rangle \pi_1 = \text{Dom}(S) R$$

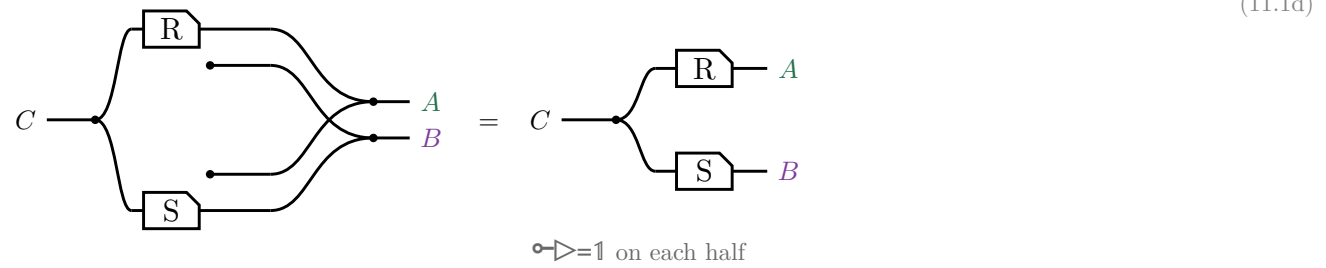
$$\langle R, S \rangle \pi_2 = \text{Dom}(R) S$$





A domain is coreflexive, so  $\langle R, S \rangle \pi_1 \sqsubseteq R$ , with equality exactly when  $S$  is entire; for maps both triangles commute and  $\langle f, g \rangle$  is unique. In  $\mathbf{Rel}$ ,  $c \langle R, S \rangle (a, b)$  iff  $c R a$  and  $c S b$  — copy  $c$ , then  $R$  on one strand and  $S$  on the other, which is  $\triangleleft (R \otimes S)$  on the right.

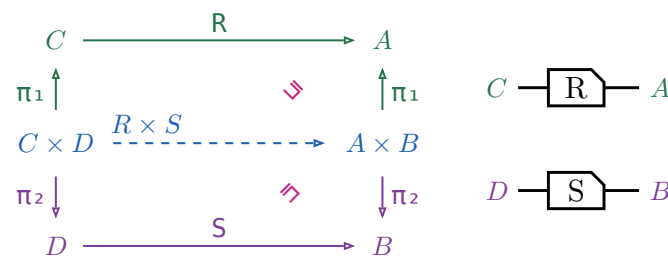
No  $\circ$  survives the translation.  $\pi_1 = 1 \otimes \circ$  discards the second component, so  $\pi_1 \circ = 1 \otimes \circ$  **creates** one out of nothing, and  $n$  is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:



### §11.1.1. Relational product $R \times S$

$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$ , a relator in each argument but no longer a categorical product.

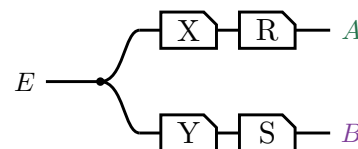
(11.1.1a)



Right-then-up is  $(R \times S) \pi_1$ , up-then-right is  $\pi_1 R$ , and  $(R \times S) \pi_1 \sqsubseteq \pi_1 R$ , equality when  $S$  is entire. In  $\mathbf{Rel}$ ,  $(c, d) (R \times S) (a, b)$  iff  $c R a$  and  $d S b$  — two strands side by side, no copy dot:  $R \times S = R \otimes S$ .

### §11.1.2. Absorption

For  $X : E \rightarrow C$  and  $Y : E \rightarrow D$ ,  $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$ . Both sides are this picture:



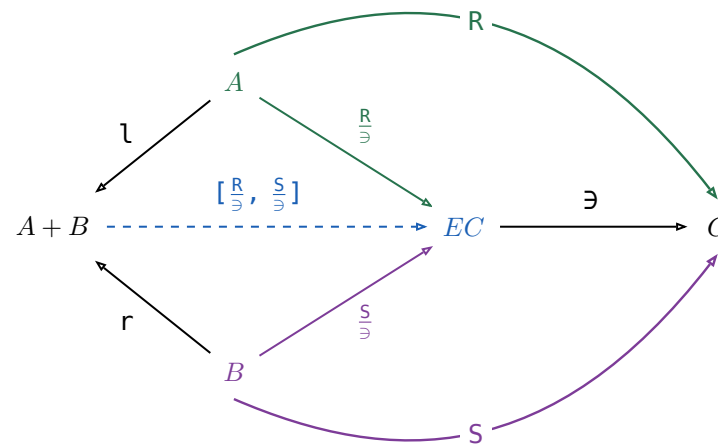
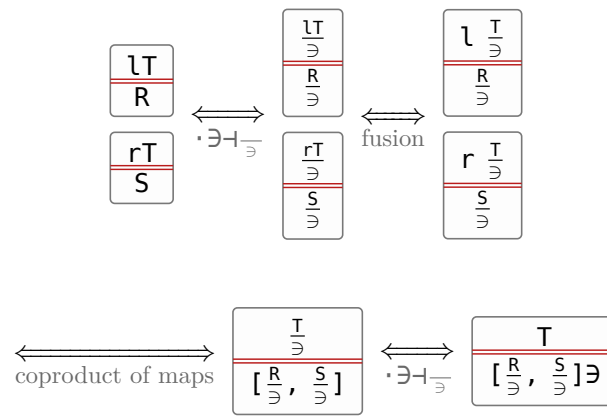
| the law                                                                                                                                                                                                                                                                                                                                                                      | picture |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1 $R \times S = (\pi_1 R, \pi_2 S)$                                                                                                                                                                                                                                                                                                                                          |         |
| 2 $(X, Y) (R \times S) = (XR, YS)$<br>both sides are the same strokes —<br>(11.1.2a). Its $\sqsubseteq$ half is Ex 5.8, that<br>half at $S := \mathbb{1}$ and at $R := \mathbb{1}$<br>B&dM's (5.4), (5.5) — stages of<br>their proof of this row; Ex 5.6 is<br>the $(R \times S) (U \times V) = (RU) \times (SV)$ it<br>yields, at $R := \mathbb{1}$ and $V := \mathbb{1}$ . |         |
| 3 $(R, S) \pi_1 = \text{Dom}(S) R$<br>(11.1b)                                                                                                                                                                                                                                                                                                                                |         |
| 4 $(R, S) \pi_2 = \text{Dom}(R) S$<br>(11.1b)                                                                                                                                                                                                                                                                                                                                |         |
| 5 $(X, Y) (R, S)^\circ = (XR^\circ) \cap (YS^\circ)$                                                                                                                                                                                                                                                                                                                         |         |
| 6 $(R, S)^\circ (P, Q) \sqsubseteq (R^\circ P) \times (S^\circ Q)$                                                                                                                                                                                                                                                                                                           |         |
| 7 $f(R, S) = (fR, fS)$<br>f a map; it fails for an arbitrary<br>arrow                                                                                                                                                                                                                                                                                                        |         |
| 8 $F(R \times S) \text{unzip}(F) = \text{unzip}(F)$<br>$(F(R) \times F(S))$<br>$\text{unzip}(F) \triangleq (F(\pi_1), F(\pi_2))$ , a<br>map                                                                                                                                                                                                                                  |         |
| 9 $g = \text{curry}(f) \iff (g \times \mathbb{1}) \text{eval} = f$<br>reading $\times$ as the relational<br>product, does $\text{Rel}$ have<br>exponentials?                                                                                                                                                                                                                 |         |

## §11.2. Coproduct $[R, S] : A + B \rightarrow C$

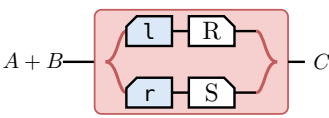
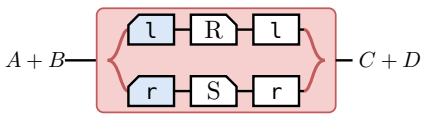
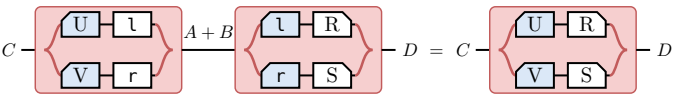
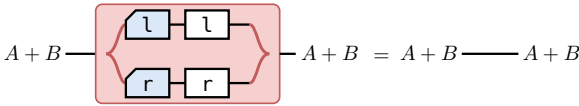
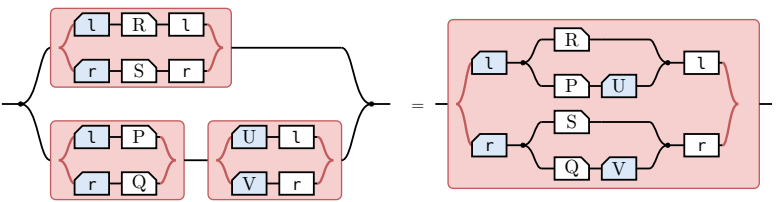
| the statement                                                                                                                                                                                                       | picture | (11.2a) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------|
| $[R, S] \triangleq \mathfrak{l} \circ R \mathfrak{u} r \circ S$<br>The tape is the union — a particle entering at $A+B$ takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint. |         |         |
| $[R, S] = [\frac{R}{\mathfrak{S}}, \frac{S}{\mathfrak{S}}] \mathfrak{S}$                                                                                                                                            |         |         |
| $R+S \triangleq [R\mathfrak{l}, S\mathfrak{r}]$                                                                                                                                                                     |         |         |
| $\mathfrak{l}[R, S]=R$ , $\mathfrak{r}[R, S]=S$ , and $[R, S]$ is the only such arrow                                                                                                                               |         |         |
| $\mathfrak{l}\mathfrak{l}^\circ = \mathfrak{l} = \mathfrak{r}\mathfrak{r}^\circ$                                                                                                                                    |         |         |
| $\mathfrak{l}\mathfrak{r}^\circ = \mathfrak{l} = \mathfrak{r}\mathfrak{l}^\circ$                                                                                                                                    |         |         |
| $\mathfrak{l}^\circ \mathfrak{l} \mathfrak{u} \mathfrak{r}^\circ \mathfrak{r} = \mathfrak{l}$                                                                                                                       |         |         |
| $[U, V]^\circ [R, S] = U^\circ R \mathfrak{u} V^\circ S$                                                                                                                                                            |         |         |

### §11.2.1. $[R, S] \triangleq [\frac{R}{\mathfrak{S}}, \frac{S}{\mathfrak{S}}] \mathfrak{S}$

The universal-property row is not free:  $\mathfrak{l}, \mathfrak{r}$  were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because  $\mathfrak{S}$  sends an arrow  $A \rightarrow C$  to a map  $A \rightarrow EC$  reversibly, so the map coproduct can be applied underneath it. For any  $T : A+B \rightarrow C$ ,



Nothing here holds only up to  $\mathfrak{E}$ : every triangle commutes on the nose, which is the difference from the fork above. The border spells  $[R, S] = [\frac{R}{\mathfrak{S}}, \frac{S}{\mathfrak{S}}] \mathfrak{S}$ , and pushing  $\mathfrak{S}$  into the union that  $[\cdot, \cdot]$  on maps already is turns that back into the definition,  $(\mathfrak{l}^\circ \frac{R}{\mathfrak{S}} \mathfrak{u} \mathfrak{r}^\circ \frac{S}{\mathfrak{S}}) \mathfrak{S} = \mathfrak{l}^\circ R \mathfrak{u} r^\circ S$ .

| B&dM    | the law                                                                                                                                                                                                                           | picture                                                                             |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| (5.9)   | $[R, S] = (\mathbf{l} \circ R) \cup (\mathbf{r} \circ S)$<br>(11.2a)'s first row                                                                                                                                                  |  |
| (5.10)  | $R + S = [R\mathbf{l}, S\mathbf{r}]$                                                                                                                                                                                              |  |
| (5.11)  | $[U, V] \circ [R, S] = (U \circ R) \cup (V \circ S)$<br>(11.2a)'s last row                                                                                                                                                        |   |
| Ex 5.12 | $X \triangleq [\mathbf{1}, \mathbf{1}] = \mathbf{l} \circ$ and<br>$Y \triangleq [\mathbf{1}, \mathbf{1}] = \mathbf{r} \circ$ ,<br>so $(X\mathbf{l}) \cup (Y\mathbf{r}) = [\mathbf{l}, \mathbf{r}] = \mathbf{1}$<br>which is (5.9) |   |
| Ex 5.13 | prove (5.11), and say why duality<br>does not carry it over from the<br>product law                                                                                                                                               |                                                                                     |
| Ex 5.14 | $(R + S) \cap ([P, Q][U, V] \circ)$<br>$= (R \cap (PU \circ)) + (S \cap (QV \circ))$<br>[P, Q][U, V] \circ is a full 2x2 of<br>composites; R+S is diagonal, so the<br>meet cuts the two off-diagonal<br>branches                  |  |

§11.3. The power relator  $P(R)$ 

(11.3a)

For  $R : A \rightarrow B$ ,

$$P(R) \triangleq ((\exists R)/\exists) \cap ((\exists R^\circ)/\exists)^\circ : EA \rightarrow EB$$

$$E(R) \triangleq \frac{\exists R}{\exists} = ((\exists R)/\exists) \cap (\exists/(\exists R))^\circ$$

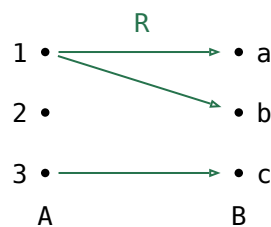
$$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$$

Every element of  $xs$  is related by  $R$  to some element of  $ys$ , and conversely.

(11.3b)

|                                     | in Rel                                                                                                 | in words                                                                          | $\text{im}(xs)$              | partner                                       |
|-------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------|-----------------------------------------------|
| $(\exists R)/\exists$               | $\forall y \in ys. \exists x \in xs. x \ R \ y$                                                        | $\forall y. \text{ some } xs \ R \ y$                                             | $ys \subseteq \text{im}(xs)$ | every $y$ has a partner                       |
| $(\exists/(\exists R))^\circ$       | $\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$                                        | every $y$ with some $xs \ R \ y$ is in $ys$                                       | $ys \supseteq \text{im}(xs)$ | $ys$ leaves out no partner                    |
| $E(R)$                              | $ys = \{y \mid \exists x \in xs. x \ R \ y\}$                                                          | $ys = \text{every } y \text{ with some } xs \ R \ y$                              | $ys = \text{im}(xs)$         | every $y$ has a partner, and none is left out |
| $((\exists R^\circ)/\exists)^\circ$ | $\forall x \in xs. \exists y \in ys. x \ R \ y$                                                        | $\forall x. x \ R \ \text{some } ys$                                              | —                            | every $x$ has a partner                       |
| $P(R)$                              | $\forall y \in ys. \exists x \in xs. x \ R \ y$ and<br>$\forall x \in xs. \exists y \in ys. x \ R \ y$ | $\forall y. \text{ some } xs \ R \ y$ and<br>$\forall x. x \ R \ \text{some } ys$ | —                            | every $x$ and every $y$ has a partner         |

(11.3c)



| $xs$          | $(\exists R)/\exists$  | $(\exists/(\exists R))^\circ$ | $E(R)$      | $((\exists R^\circ)/\exists)^\circ$ | $P(R)$        |
|---------------|------------------------|-------------------------------|-------------|-------------------------------------|---------------|
| $\emptyset$   | $\emptyset$            | all 8 subsets of $abc$        | $\emptyset$ | all 8 subsets of $abc$              | $\emptyset$   |
| $\{1\}$       | $\emptyset, a, b, ab$  | $ab, abc$                     | $ab$        | $a, b, ab, ac, bc, abc$             | $a, b, ab$    |
| $\{2\}$       | $\emptyset$            | all 8 subsets of $abc$        | $\emptyset$ | none                                | none          |
| $\{3\}$       | $\emptyset, c$         | $c, ac, bc, abc$              | $c$         | $c, ac, bc, abc$                    | $c$           |
| $\{1, 2\}$    | $\emptyset, a, b, ab$  | $ab, abc$                     | $ab$        | none                                | none          |
| $\{1, 3\}$    | all 8 subsets of $abc$ | $abc$                         | $abc$       | $ac, bc, abc$                       | $ac, bc, abc$ |
| $\{2, 3\}$    | $\emptyset, c$         | $c, ac, bc, abc$              | $c$         | none                                | none          |
| $\{1, 2, 3\}$ | all 8 subsets of $abc$ | $abc$                         | $abc$       | none                                | none          |

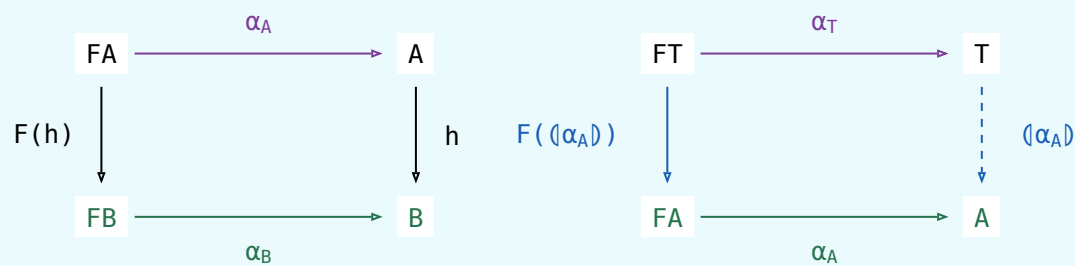
(11.3d)

| law                                                                                          | the reading                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $X \subseteq P(R) \iff X \subseteq \exists R \text{ and } X^\circ \subseteq \exists R^\circ$ | One containment, and the same one at $R^\circ$ — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$ , and $R \subseteq S \implies P(R) \subseteq P(S)$ .                                                                                                                        |
| $P(1) = \frac{\exists}{\exists} = 1$                                                         | The straightness axiom verbatim: extensionality is $P$ 's unit law.                                                                                                                                                                                                                                              |
| $P(f) = \frac{\exists f}{\exists}$ , for $f$ a map                                           | In Rel, $xs \ P(f) \ ys \iff ys = \{f \ a \mid a \in xs\}$ . The half at $f^\circ$ says every $a \in xs$ has its $f \ a$ on $ys$ ; $f$ has just the one image per $a$ , so that already says $ys$ contains everything $xs$ reaches, which is the fraction's second half. For a map the two definitions coincide. |
| $P(RS) = P(R)P(S)$                                                                           | $\exists$ is the division cancellation laws. $\subseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(RS)$ .                                                                                                                                                            |

## §11.4. Initial algebra

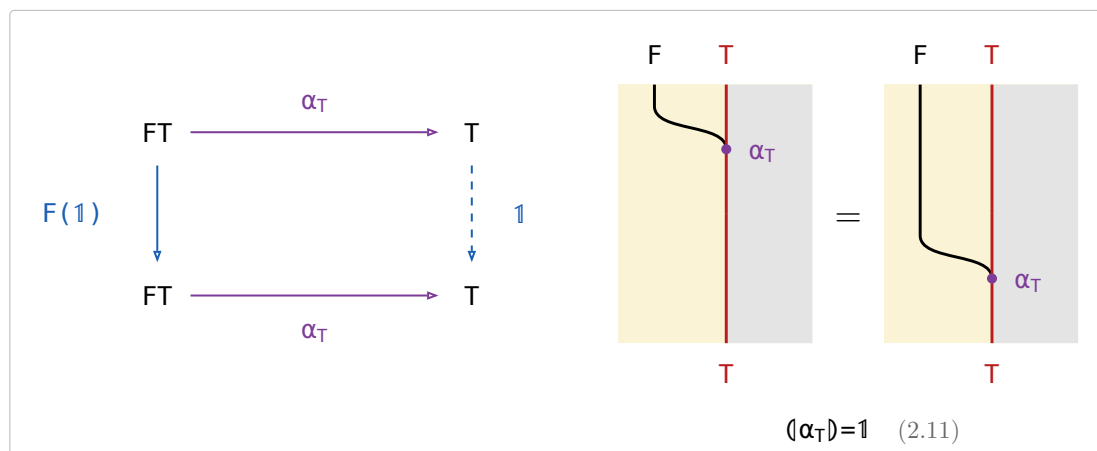
An **F-algebra** on  $A$  is a map  $\alpha_A : FA \rightarrow A$ . An **F-homomorphism** from  $\alpha_A$  to  $\alpha_B$  is a map  $h : A \rightarrow B$  with  $\alpha_B h = F(h) \alpha_A$ . The **initial algebra**  $\alpha_T : FT \rightarrow T$  is the F-algebra with exactly one F-homomorphism  $\langle \alpha_A \rangle : T \rightarrow A$  to every F-algebra  $\alpha_A$ .

(11.4a)



### §11.4.1. Reflection

(11.4.1a)

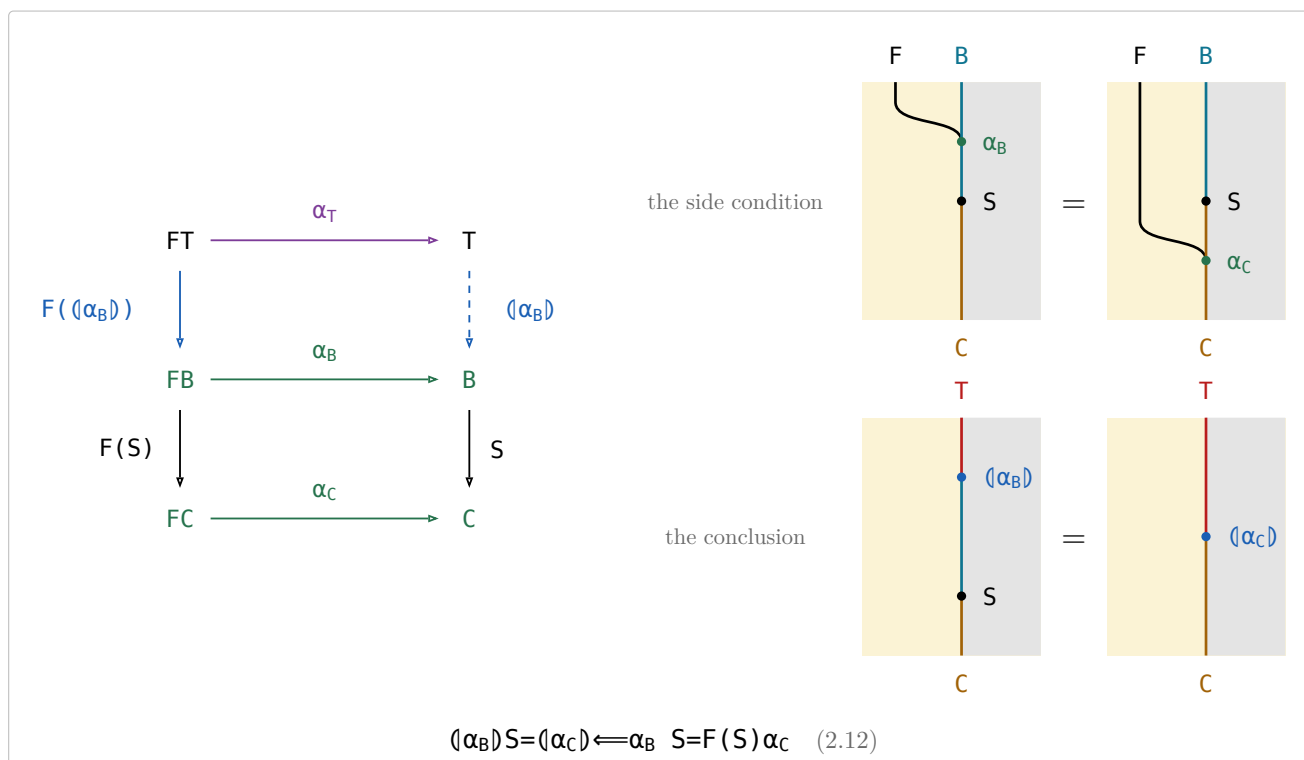


Taking a value apart with  $\alpha_T$  and putting it straight back is doing nothing.

### §11.4.2. Fusion

Fusion rewrites  $\langle \alpha_B \rangle S$  through a second algebra  $\alpha_C : FC \rightarrow C$  along an arrow  $S : B \rightarrow C$ .

(11.4.2a)



§11.5. Type relator

Let  $F$  be a binary relator with initial type  $(\alpha, T)$ , so  $T$  is a type functor.  $F(R, S)$  is its action on a pair, and  $F(X)$  abbreviates  $F(1, X)$ , the  $F$  of the reduce section. For every object  $A$  the initial algebra is  $\alpha : F(A, TA) \rightarrow TA$ , among the maps.  $T$  acts on an arrow  $R : A \rightarrow B$  by

$$T(R) = (F(R, 1) \alpha) : TA \rightarrow TB$$

(11.5a)

| name                   | law                               | what it says                                                                                                                                                                                                |
|------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the defining equation  | $T(R) = (F(R, 1) \alpha)$         | Rebuild the structure with $\alpha$ , applying $R$ to the parameter on the way.                                                                                                                             |
| functor                | $T(1) = 1$ and $T(R)T(S) = T(RS)$ | Acting by the identity changes nothing, and two actions in a row are one action.                                                                                                                            |
| type functor fusion    | $T(R)(Q) = (F(R, 1)Q)$            | A relator action followed by a fold is a single fold — the intermediate structure is never built. The side condition holds because $F$ is a bifunctor — $F(R, 1)F(1, (Q)) = F(R, (Q)) = F(1, (Q))F(R, 1)$ . |
| naturality of $\alpha$ | $\alpha T(R) = F(R, T(R)) \alpha$ | Building and then mapping is the same as mapping the parts and then building, so $\alpha$ is natural from $G(R) = F(R, T(R))$ to $T$ .                                                                      |
| type relator           | $T(R)^\circ = T(R^\circ)$         | A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.                                                                                                        |

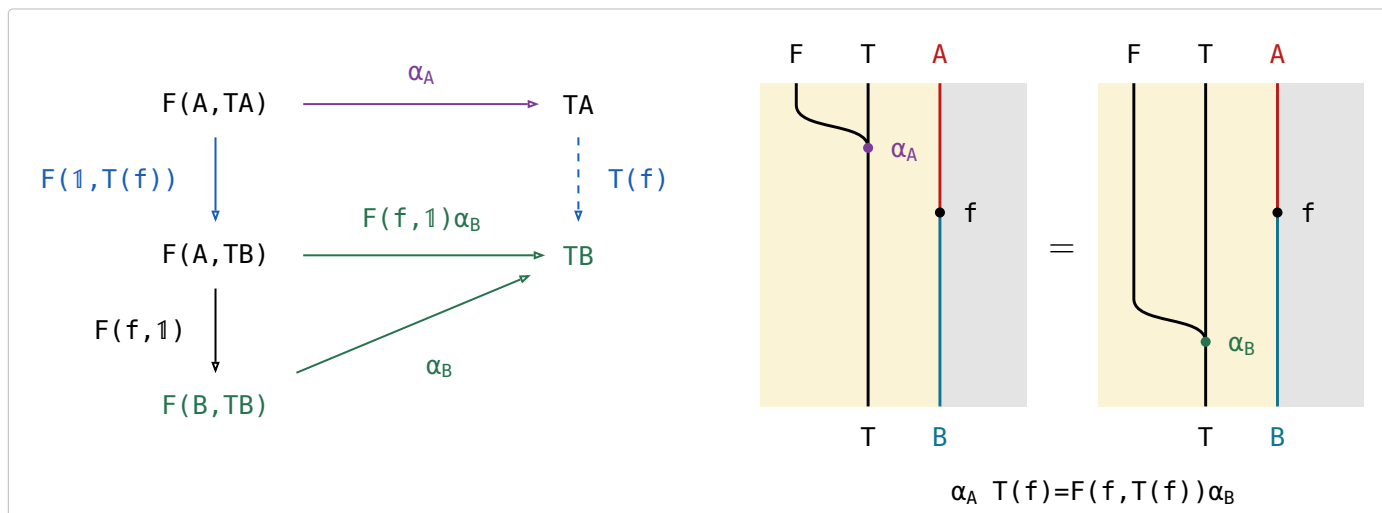
(11.5b)

## §11.5.1. Type functor

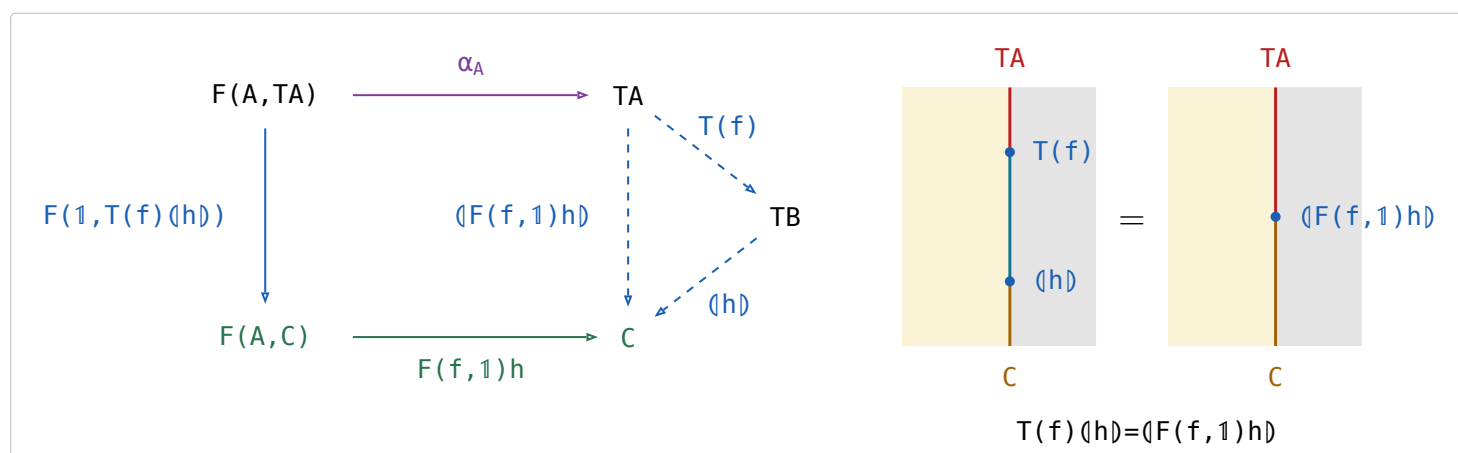
Let  $F$  be a bifunctor taking both the parameter  $A$  and the recursive position  $TA$ , with an initial algebra  $\alpha_A : F(A, TA) \rightarrow TA$  for every object  $A$ . Then  $T$  is a functor, acting on a map  $f : A \rightarrow B$  by

$$T(f) \triangleq (F(f, 1) \alpha_B) : TA \rightarrow TB$$

(11.5.1a)



(11.5.1b)



(11.5.1c)

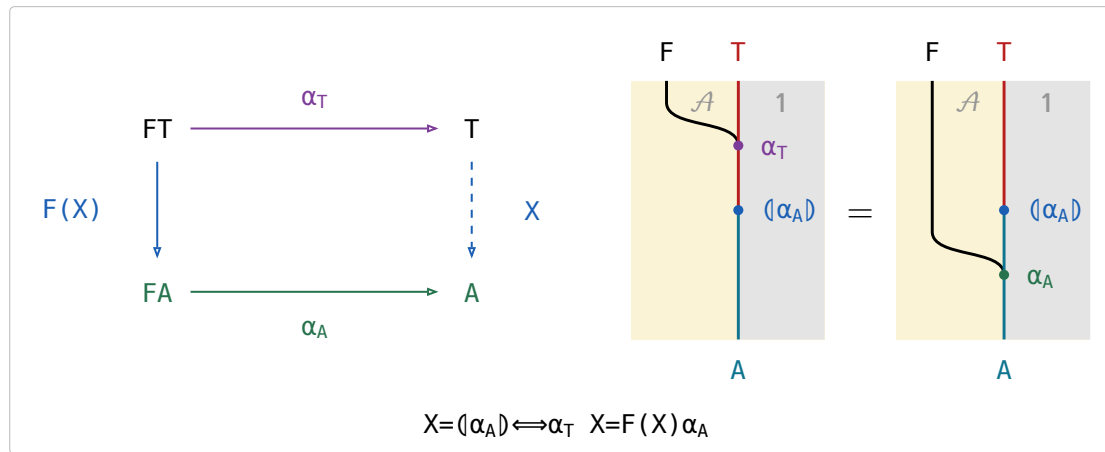


## §11.6. Reduce

let  $F$  be a relator and has **initial algebra**  $\alpha_T : FT \rightarrow T$  in the subcategory of functions.  $\alpha_T$  is also initial in the allegory:

(11.6a)

### §11.6.1. The defining equation



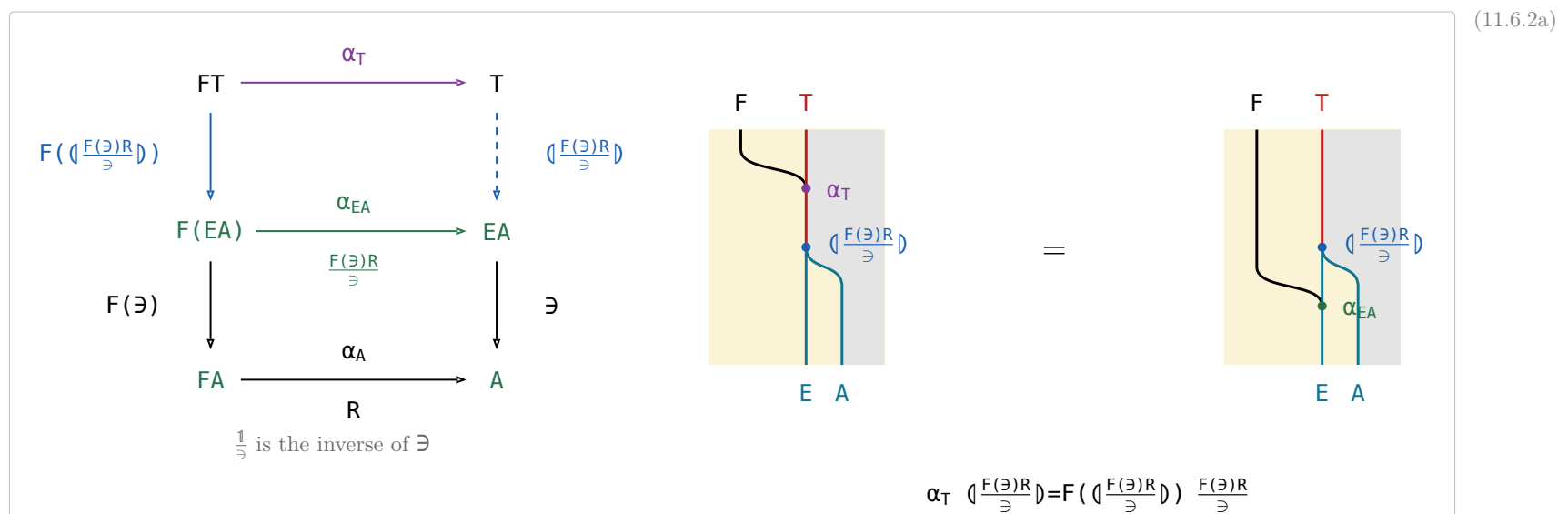
(11.6.1a)

An  $F$ -algebra is a **weakened** natural transformation. A transformation  $F \Rightarrow 1$  on  $\mathcal{A}$  would need a component  $FX \rightarrow X$  at every object and a commuting square at every arrow, but  $F$ -Algebra only need it works on  $T$  and  $A$ .

| $\llbracket [c, f] \rrbracket = \text{reduce}(c, f)$ |                                                                                                                                         |                                                                                                                                                |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| part                                                 | Nat                                                                                                                                     | [A]                                                                                                                                            |
| datatype                                             | $\text{Nat} ::= \text{zero} \mid \text{succ Nat}$                                                                                       | $[A] ::= \text{nil} \mid \text{cons } (A, [A])$                                                                                                |
| base functor $F$                                     | $F(X) = 1 + X$                                                                                                                          | $F(X) = 1 + A \times X$                                                                                                                        |
| initial algebra $\alpha$                             | $\alpha = [\text{zero}, \text{succ}]$<br>$: 1 + \text{Nat} \rightarrow \text{Nat}$                                                      | $\alpha = [\text{nil}, \text{cons}]$<br>$: 1 + A \times [A] \rightarrow [A]$                                                                   |
| the fold, pointwise                                  | $\llbracket [c, f] \rrbracket \text{ zero} = c$<br>$\llbracket [c, f] \rrbracket (\text{succ } n) = f (\llbracket [c, f] \rrbracket n)$ | $\llbracket [c, f] \rrbracket \text{ nil} = c$<br>$\llbracket [c, f] \rrbracket (\text{cons } (a, x)) = f (a, \llbracket [c, f] \rrbracket x)$ |

(11.6.1b)

### §11.6.2. $\llbracket (R) \rrbracket = (\llbracket \frac{F(\exists)R}{\exists} \rrbracket) \exists$



(11.6.2a)

$$\begin{array}{c}
 \frac{\alpha_T X}{F(X)R} \iff \frac{\alpha_T X}{\frac{F(X)R}{\exists}} \iff \frac{\alpha_T X}{\frac{F(\frac{X}{\exists})R}{\exists}} \\
 \\
 \iff \frac{\alpha_T \frac{X}{\exists}}{F(\frac{X}{\exists}) \frac{F(\exists)R}{\exists}} \iff \frac{\frac{X}{\exists}}{\llbracket \frac{F(\exists)R}{\exists} \rrbracket} \iff \frac{X}{\llbracket \frac{F(\exists)R}{\exists} \rrbracket \exists}
 \end{array}$$

relator, fusion twice      reduce of maps

(11.6.2b)

§11.6.3.  $\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$ 

(11.6.3a)

$\varphi$  a **monotonic** mapping of the hom-set  $A \rightarrow B$  into itself:  $X \sqsubseteq Y \implies \varphi(X) \sqsubseteq \varphi(Y)$ .

$(\mu X : \varphi(X))$  the least  $X : A \rightarrow B$  with  $\varphi(X) \sqsubseteq X$ .

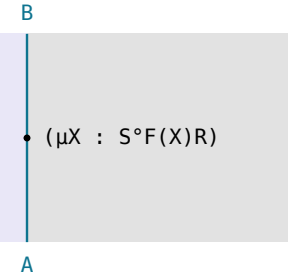
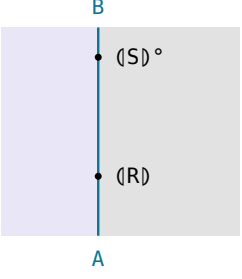
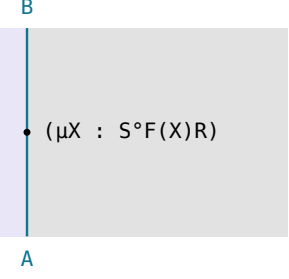
| the law                                                                                                 | what it says                                                                                                                                                                   | (11.6.3b) |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| $\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$<br>Theorem 6.1, lean:AOP.A6_2.mu | to bound $(\mu X : \varphi(X))$ above, exhibit one $Y$ the body does not grow past — the half §11.6.4 and every chapter after it uses                                          |           |
| $\varphi((\mu X : \varphi(X))) = (\mu X : \varphi(X))$<br>Theorem 6.1, lean:AOP.A6_2.mu_fixed           | <b>Knaster–Tarski:</b> the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$ , so the least prefix point and the least fixed point are one relation |           |

§11.6.4.  $(S)^\circ (R) = (\mu X : S^\circ F(X) R)$ 

(11.6.4a)

| $S^\circ F((S)^\circ (R)) R = (S)^\circ (R)$<br>Theorem 6.2, lean:AOP.A6_3.hylo_fixed; $R : FA \rightarrow A$ , $S : FB \rightarrow B$ , $\alpha : FT \rightarrow T$ initial                                                                                                                     |               | (11.6.4a) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------|
| circuit                                                                                                                                                                                                                                                                                          | Hinze–Marsden |           |
| $B \text{ --- } S^\circ \text{ --- } F((S)^\circ (R)) \text{ --- } R \text{ --- } A$<br>$S^\circ F((S)^\circ (R)) R$<br>the body at $(S)^\circ (R)$                                                                                                                                              |               |           |
| $B \text{ --- } S^\circ \text{ --- } F((S)^\circ) \text{ --- } F((R)) \text{ --- } R \text{ --- } A$<br>$= S^\circ F((S)^\circ) F((R)) R$<br>$F(RS) = F(R) F(S) \text{ — (11a)}$                                                                                                                 |               |           |
| $B \text{ --- } S^\circ \text{ --- } F((S)^\circ) \text{ --- } \alpha \text{ --- } (R) \text{ --- } A$<br>$= S^\circ F((S)^\circ) \alpha (R)$<br>(11.6.1a) at $R: F((R)) R = \alpha(R)$                                                                                                          |               |           |
| $B \text{ --- } (S)^\circ \text{ --- } \alpha^\circ \text{ --- } \alpha \text{ --- } (R) \text{ --- } A$<br>$= (S)^\circ \alpha^\circ \alpha (R)$<br>(11.6.1a) at $S$ conversed: $(S)^\circ \alpha^\circ = S^\circ F((S)^\circ)^\circ$ , and $F((S)^\circ)^\circ = F((S)^\circ) \text{ — (11b)}$ |               |           |
| $B \text{ --- } (S)^\circ \text{ --- } (R) \text{ --- } A$<br>$= (S)^\circ (R)$<br>$\alpha^\circ \alpha = 1$ , Lambek — lean:AOP.A6_2.InitialAlgebra.recip_alpha_alpha                                                                                                                           |               |           |

| $S^\circ F(X) R \sqsubseteq X \implies (S)^\circ (R) \sqsubseteq X$<br>Theorem 6.2, lean:AOP.A6_3.hylo_le_of_prefixed                                                                                                                                                                                                          |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                        | Hinze–Marsden |
| $B \vdash \boxed{(S)^\circ} \boxed{(R)} \vdash \boxed{X} \vdash A$ $(S)^\circ (R) \sqsubseteq X$ the conclusion                                                                                                                                                                                                                |               |
| $T \vdash \boxed{(R)} \vdash \boxed{(S)^\circ \setminus X} \vdash A$ $\Leftrightarrow (R) \sqsubseteq (S)^\circ \setminus X$ (1.1a)'s $S \cdot \neg S \setminus$ at $(S)^\circ$                                                                                                                                                |               |
| $T \vdash \boxed{\alpha^\circ} \boxed{F((S)^\circ \setminus X)} \boxed{R} \vdash \boxed{(S)^\circ \setminus X} \vdash A$ $\Leftarrow \alpha^\circ F((S)^\circ \setminus X) R \sqsubseteq (S)^\circ \setminus X$ (6.2) $(R) = (\mu X : \alpha^\circ F(X) R)$ — (11.6.1a) and (11.6.3b);<br>lean:AOP.A6_2.relCata_le_of_prefixed |               |
| $B \vdash \boxed{(S)^\circ} \boxed{\alpha^\circ} \boxed{F((S)^\circ \setminus X)} \boxed{R} \vdash \boxed{X} \vdash A$ $\Leftrightarrow (S)^\circ \alpha^\circ F((S)^\circ \setminus X) R \sqsubseteq X$ (1.1a)'s $S \cdot \neg S \setminus$ at $(S)^\circ$                                                                    |               |
| $B \vdash \boxed{S^\circ} \boxed{F((S)^\circ)} \boxed{F((S)^\circ \setminus X)} \boxed{R} \vdash \boxed{X} \vdash A$ $\Leftrightarrow S^\circ F((S)^\circ) F((S)^\circ \setminus X) R \sqsubseteq X$ $(S)^\circ \alpha^\circ = S^\circ F((S)^\circ)$ — (11.6.4a)                                                               |               |
| $B \vdash \boxed{S^\circ} \boxed{F(X)} \boxed{R} \vdash \boxed{X} \vdash A$ $\Leftarrow S^\circ F(X) R \sqsubseteq X$ $F(RS) = F(R)F(S)$ — (11a) — and $(S)^\circ ((S)^\circ \setminus X) \sqsubseteq X$ — (1.1a)                                                                                                              |               |

| $(S)^\circ(R) = (\mu X : S^\circ F(X)R)$<br>Theorem 6.2, lean:AOP.A6_3.hylo_eq_mu                                                                                                                                                                                                                                         |                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| circuit                                                                                                                                                                                                                                                                                                                   | Hinze–Marsden                                                                        |
| $B \dashv \boxed{(\mu X : S^\circ F(X)R)} \dashv A$ $(\mu X : S^\circ F(X)R)$ (11.6.3a) at $\varphi(X) := S^\circ F(X)R$                                                                                                                                                                                                  |   |
| $B \dashv \boxed{(S)^\circ} \boxed{(R)} \dashv A$ $\sqsubseteq (S)^\circ(R)$ (11.6.3b)'s $\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$ at $Y := (S)^\circ(R)$ , whose $S^\circ F((S)^\circ(R))R = (S)^\circ(R)$ is (11.6.4a); lean:AOP.A6_2.mu_le_of_fixed                                       |   |
| $B \dashv \boxed{(\mu X : S^\circ F(X)R)} \dashv A$ $\sqsubseteq (\mu X : S^\circ F(X)R)$ (11.6.4b) at $X := (\mu X : S^\circ F(X)R)$ , whose $S^\circ F((\mu X : S^\circ F(X)R))R \sqsubseteq (\mu X : S^\circ F(X)R)$ is (11.6.3b)'s $\varphi((\mu X : \varphi(X))) = (\mu X : \varphi(X))$ ; lean:AOP.A6_2.mu_prefixed |  |

## §12. Combinatorial functions

| definition                                                                                               | type                                  | note                                                                                                                                      | (12a) |
|----------------------------------------------------------------------------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------|
| $[A] ::= \text{nil} \mid \text{cons } (A, [A])$                                                          | $\mathcal{A} \rightarrow \mathcal{A}$ | B&dM's <b>listr</b> under the short name it keeps from p. 125 on.                                                                         |       |
| $\text{list}(R) \triangleq ([\text{nil}, (R \otimes 1) \text{ cons}])$                                   | $[A] \rightarrow [B]$                 | The relator's action on $R : A \rightarrow B$ : one $R$ per element, the shape untouched.                                                 |       |
| $\text{subseq} \triangleq ([\text{nil}, \text{cons} \pi_2])$                                             | $[A] \rightarrow [A]$                 | $xs \text{ subseq } ys$ : $ys$ is $xs$ with elements dropped — <b>cons</b> keeps the head, $\pi_2$ drops it.                              |       |
| $\text{prefix} \triangleq ([\text{nil}, \text{nilucons}])$<br>$= \text{cat}^\circ \pi_1 = \text{init}^*$ | $[A] \rightarrow [A]$                 | $ys$ is an initial segment of $xs$ ; the first <b>nil</b> is where it stops early. $\text{init} \triangleq \text{snoc}^\circ \pi_1$ .     |       |
| $\text{suffix} \triangleq \text{cat}^\circ \pi_2 = \text{tail}^*$                                        | $[A] \rightarrow [A]$                 | The dual, $\text{tail} \triangleq \text{cons}^\circ \pi_2$ ; as a reduce it needs <b>snoc</b> -lists.                                     |       |
| $\text{segment} \triangleq \text{suffix prefix}$                                                         | $[A] \rightarrow [A]$                 | A contiguous stretch of $xs$ : a suffix, then a prefix of that.                                                                           |       |
| $\text{partition} \triangleq \text{concat}^\circ$                                                        | $[A] \rightarrow [[A]]$               | <b>cat</b> restricted to a non-empty first argument, so $ys$ is a list of non-empty segments of $xs$ .                                    |       |
| $\text{concat} \triangleq ([\text{nil}, \text{cat}])$                                                    | $[[A]] \rightarrow [A]$               | Joins the segments back up, which is why its converse splits a list.                                                                      |       |
| <b>inits</b>                                                                                             | $[A] \rightarrow [[A]]$               | Implements $\frac{\text{prefix}}{\ni}$ , listing the prefixes by increasing length.                                                       |       |
| <b>tails</b>                                                                                             | $[A] \rightarrow [[A]]$               | Implements $\frac{\text{suffix}}{\ni}$ by decreasing length — the opposite order.                                                         |       |
| $\text{filter}(p) \triangleq \frac{\text{subseq list}(p)}{\ni} \text{est}(R^\circ)$                      | $[A] \rightarrow [A]$                 | The longest subsequence of $xs$ whose every element passes $p$ . Ex 7.41; $\text{est}(R^\circ)$ is (13.1a)                                |       |
| $R \triangleq \text{length} \leq \text{length}^\circ$                                                    | $[A] \rightarrow [A]$                 | The preorder <b>filter</b> and <b>takewhile</b> maximise over: the longer list wins. $\geq \triangleq \leq^\circ$                         |       |
| $\text{takewhile}(p) \triangleq \frac{\text{prefix list}(p)}{\ni} \text{est}(R^\circ)$                   | $[A] \rightarrow [A]$                 | The same with <b>prefix</b> for <b>subseq</b> : the longest prefix whose every element passes $p$ . Ex 7.39                               |       |
| $\text{mss} \triangleq \frac{\text{segment sum}}{\ni} \text{est}(\geq)$                                  | $[\text{Int}] \rightarrow \text{Int}$ | Maximum segment sum. <b>segment=suffix prefix</b> splits it into $\frac{\text{prefix sum}}{\ni} \text{est}(\geq)$ on each suffix. Ex 7.40 |       |

### §12.1. $\frac{\text{subseq}}{\ni} = ([\text{nil } \frac{1}{\ni}, (\frac{1 \times \ni}{\ni} \text{E}(\text{cons}), \pi_2) \text{ cup}])$

$$\text{cup} \triangleq \frac{\pi_1 \ni \cup \pi_2 \ni}{\ni} : EA \times EA \rightarrow EA, \text{ so } \frac{R \cup S}{\ni} = (\frac{R}{\ni}, \frac{S}{\ni}) \text{ cup.}$$

(12.1a)

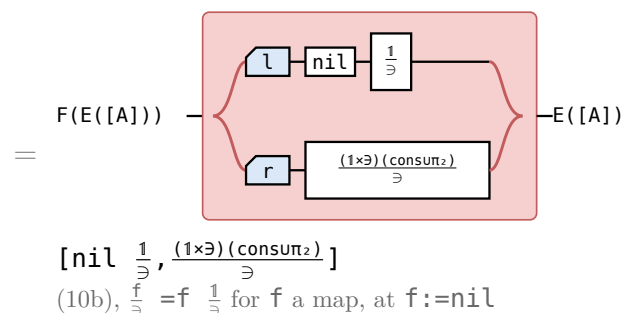
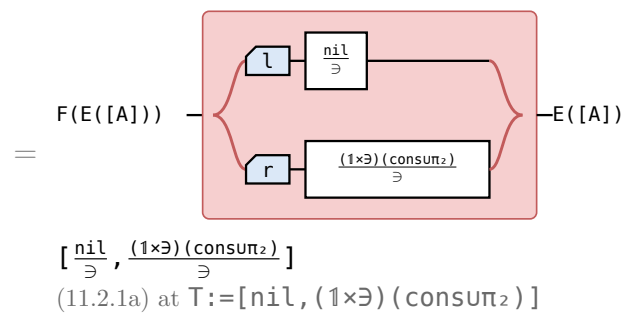
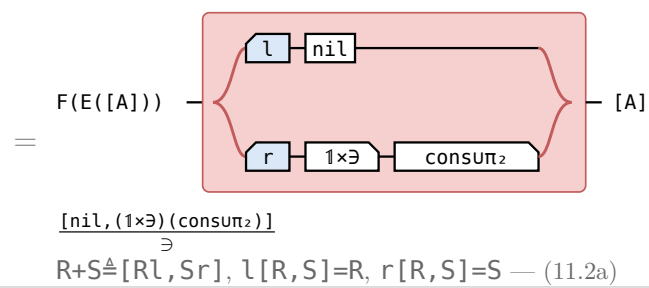
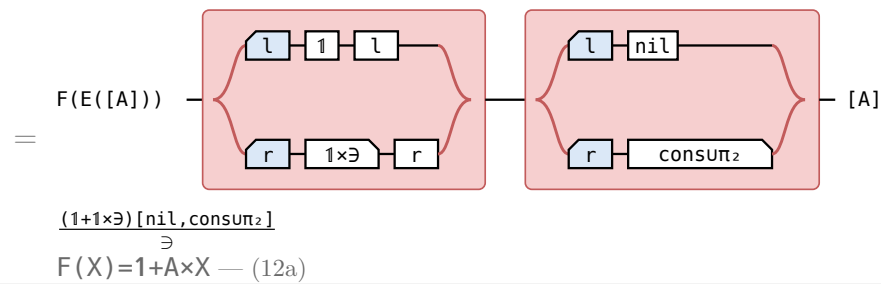
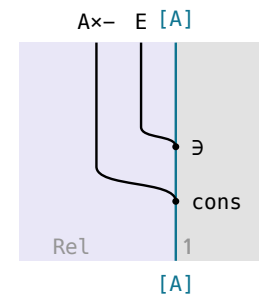
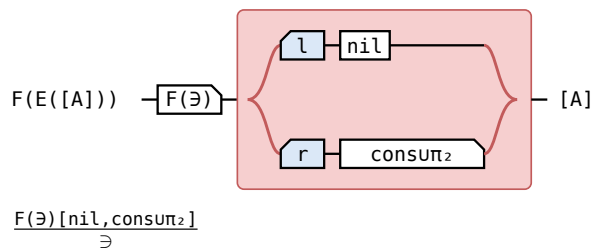
$$\frac{F(\exists)[\text{nil}, \text{cons}\pi_2]}{\exists} = [\text{nil} \quad \frac{1}{\exists}, \frac{(1 \times \exists)(\text{cons}\pi_2)}{\exists}]$$

(11.6.2b) at  $\text{subseq} = ([\text{nil}, \text{cons}\pi_2])$  — (12a)

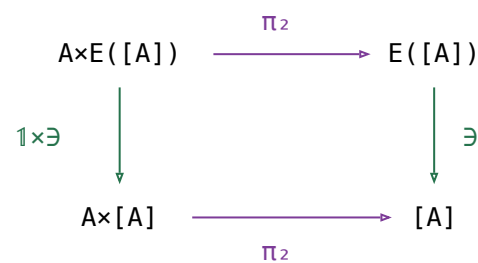
(12.1b)

**circuit** — the fork is  $F([A]) = 1 + A \times [A]: \text{nil}$  above, the pair below

**Hinze–Marsden** — the  $\text{cons}$  operand



(12.1c)



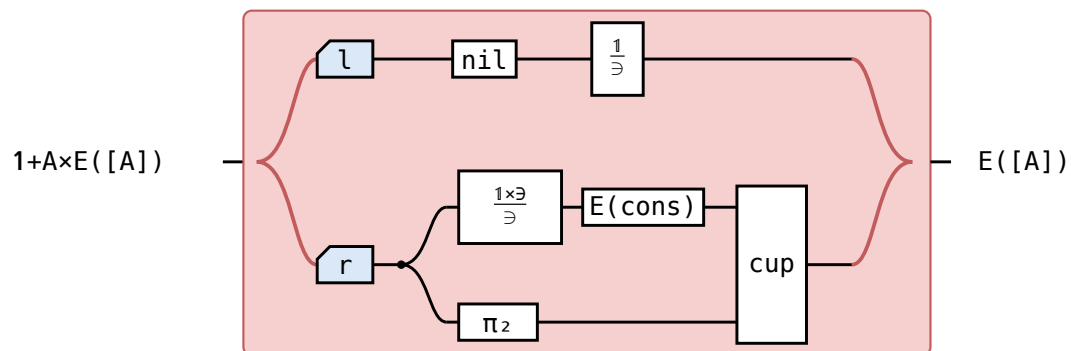
$$\frac{(1 \times \exists)(\text{cons} \pi_2)}{\exists} = \left( \frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup}$$

B&dM §5.6, p. 124

(12.1d)

| circuit — the tape IS the u, and neither branch injects                                                                                                                                                                                                                                                                                                                                   | Hinze–Marsden — the cons operand                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><math>A \times E([A]) \rightarrow [A]</math></p> <p><math>\frac{(1 \times \exists)(\text{cons} \pi_2)}{\exists}</math><br/>(12.1b)'s second branch</p>                                                                                                                                                                                                                                 | <p><math>A \times - E [A]</math></p> <p><math>\exists</math><br/><math>\text{cons}</math><br/><math>\pi_2</math><br/><math>[A]</math></p>                                                       |
| <p><math>A \times E([A]) \rightarrow [A]</math></p> <p><math>\frac{(1 \times \exists) \text{cons} (1 \times \exists) \pi_2}{\exists}</math><br/><math>T(X_1 \cup X_2) = T X_1 \cup T X_2</math> — (1.2b)</p>                                                                                                                                                                              |                                                                                                                                                                                                 |
| <p><math>A \times E([A]) \rightarrow [A]</math></p> <p><math>\frac{(1 \times \exists) \text{cons} \pi_2 \exists}{\exists}</math><br/><math>(1 \times \exists) \pi_2 = \pi_2 \exists</math> — (11.1.1b) at <math>\pi_2</math>, an equality because <math>1</math> is entire</p>                                                                                                            |                                                                                                                                                                                                 |
| <p><math>A \times E([A]) \rightarrow E([A])</math></p> <p><math>\left( \frac{(1 \times \exists) \text{cons}}{\exists}, \frac{\pi_2 \exists}{\exists} \right) \text{cup}</math><br/><math>\frac{R \cup S}{\exists} = \left( \frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup}</math> — (12.1a)</p>                                                                                   | <p><math>A \times - E [A]</math></p> <p><math>1 / \exists</math><br/><math>\exists</math><br/><math>\text{cons}</math><br/><math>\pi_2</math><br/><math>\exists</math><br/><math>[A]</math></p> |
| <p><math>A \times E([A]) \rightarrow E([A])</math></p> <p><math>\left( \frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup}</math><br/>(10b), absorption <math>\frac{S}{\exists} E(R) = \frac{SR}{\exists}</math> at <math>S := 1 \times \exists, R := \text{cons}</math>; fusion and <math>\frac{\exists}{\exists} = 1</math> on the <math>\pi_2</math> operand</p> |                                                                                                                                                                                                 |

(12.1e)


 B&dM §5.6, p. 124, which writes  $P\text{cons}$ ;  $\text{cons}$  is a map, and there  $P(\text{cons}) = E(\text{cons})$  — (11.3d).

## §13. Optimisation Problems

### §13.1. $\text{est}(R) \triangleq \exists n(\epsilon \backslash R^\circ)$

(13.1a)

For  $R : A \rightarrow A$ ,  $\text{est}(R) \triangleq \exists n(\epsilon \backslash R^\circ) : EA \rightarrow A$ .  $\text{est}(R)$  is B&dM's  $\min R$ ,  $\text{est}(R^\circ)$  his  $\max R$

$$xs \ (\text{est}(R)) \ x \iff x \in xs \wedge (\forall y \in xs. x R y)$$

$$xs \ (\text{est}(R)) \ x \iff (x \text{ in } xs) \text{ and all } x R \backslash : xs \text{ in } q$$

$$\text{est}(R) = \exists n \text{ all } R^\circ \text{ all } R \triangleq \epsilon \backslash R, q's \text{ all; the chains below keep it written } \epsilon \backslash$$

$$E(R) \triangleq \frac{\exists R}{\exists} : EA \rightarrow EB, xs \ E(R) \ ys \iff ys = \{y \mid \exists x \in xs. x R y\} \text{ the image of } xs, (10b)$$

$$P(R) : EA \rightarrow EB, xs \ P(R) \ ys \iff (\forall x \in xs. \exists y \in ys. x R y) \wedge (\forall y \in ys. \exists x \in xs. x R y) \text{ every } x \text{ and every } y \text{ has a partner, (11.3b)}$$

| the law                                                                                                        | what it says                                                                                                  | (13.1b) |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------|
| $X \sqsubseteq \text{est}(R) \iff X \sqsubseteq \exists$ and $X^\circ \sqsubseteq R$                           | in the set, and below every element of it                                                                     |         |
| $\frac{1}{\exists} (\epsilon \backslash R) = R$                                                                | bounding a singleton is bounding its element                                                                  |         |
| $\frac{S}{\exists} (\epsilon \backslash R) = S^\circ \backslash R$                                             | bound $S$ 's image without building the set                                                                   |         |
| $\text{union} \triangleq \frac{\exists \exists}{\exists} : E(EA) \rightarrow EA$                               | flattens a set of sets                                                                                        |         |
| $\text{union} (\epsilon \backslash R) = \epsilon \backslash (\epsilon \backslash R)$                           | bound a union by bounding each member set                                                                     |         |
| $\frac{1}{\exists} \text{est}(R) = 1 \cap R$                                                                   | a singleton's minimum is its element, where $R$ is reflexive<br>$\frac{S}{\exists} \text{est}(R)$ at $S := 1$ |         |
| $\frac{S}{\exists} \text{est}(R) = S \cap (S^\circ \backslash R^\circ)$                                        | an $S$ -value that points to every $S$ -value                                                                 |         |
| $\frac{S}{\exists} \text{est}(R) = \frac{S}{\exists} \text{est}(R \cap S^\circ S)$                             | only $R$ between values $S$ gives one argument counts — context                                               |         |
| $E(S) \text{est}(R) = (\exists S) \cap ((\exists S)^\circ \backslash R^\circ)$                                 | the same for the image of a set<br>$\frac{S}{\exists} \text{est}(R)$ at $S := \exists S$                      |         |
| $P(f) \text{est}(R) = \text{est}(f R f^\circ) \ f$                                                             | shunt a function through a minimum                                                                            |         |
| $P(S) \text{est}(R) = (\exists S) \cap (\epsilon \backslash (S R^\circ))$<br>$R$ reflexive                     | fusion with the power relator<br>$\exists$ is the only proof here that tabulates                              |         |
| $P(S) \text{est}(R) \sqsubseteq (\exists S) \cap (\epsilon \backslash (S R^\circ))$                            | the half of the row above that costs nothing                                                                  |         |
| $P(\text{est}(R)) \text{est}(R) \sqsubseteq \text{union} \text{est}(R)$<br>$R$ a preorder                      | a minimum in each set, then a minimum of those                                                                |         |
| $P(\text{est}(R)) \text{est}(R) = P(\text{Dom}(\text{est}(R))) \text{ union } \text{est}(R)$<br>$R$ a preorder | the same as an equality, once empty sets are dropped                                                          |         |

#### §13.1.1. $X \sqsubseteq \text{est}(R) \iff X \sqsubseteq \exists$ and $X^\circ \sqsubseteq R$

(13.1.1a)

$$\boxed{\begin{array}{c} X \\ \text{est}(R) \end{array}} \xleftrightarrow{\Delta \dashv \cap} \boxed{\begin{array}{c} X \\ \exists \\ X \\ \epsilon \backslash R^\circ \end{array}} \xleftrightarrow{T \cdot \dashv T \backslash} \boxed{\begin{array}{c} X \\ \exists \\ \epsilon X \\ R^\circ \end{array}} \xleftrightarrow{\circ} \boxed{\begin{array}{c} X \\ \exists \\ X^\circ \exists \\ R \end{array}}$$

#### §13.1.2. $\frac{1}{\exists} (\epsilon \backslash R) = R$

(13.1.2a)

$$\boxed{\begin{array}{c} X \\ \frac{1}{\exists} (\epsilon \backslash R) \end{array}} \xleftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\begin{array}{c} 1^\circ X \\ \frac{1}{\exists} \epsilon \backslash R \end{array}} \xleftrightarrow{T \cdot \dashv T \backslash} \boxed{\begin{array}{c} \epsilon \frac{1}{\exists} X \\ R \end{array}} \xleftrightarrow{\circ} \boxed{\begin{array}{c} (\frac{1}{\exists} \exists)^\circ X \\ R \end{array}} \xleftrightarrow{i \dashv E} \boxed{\begin{array}{c} X \\ R \end{array}}$$

#### §13.1.3. $\frac{S}{\exists} (\epsilon \backslash R) = S^\circ \backslash R$

$\frac{S}{\exists}$  gathers the  $S$ -image of a point into one set and  $\epsilon \backslash R$  asks that every member of that set be  $R$ -related to the target, so the set cancels and  $S^\circ \backslash R$  asks it of the  $S$ -image directly.

(13.1.3a)

$$\boxed{\begin{array}{c} X \\ \frac{S}{\exists} (\epsilon \backslash R) \end{array}} \xleftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\begin{array}{c} \frac{S}{\exists}^\circ X \\ \frac{S}{\exists} \epsilon \backslash R \end{array}} \xleftrightarrow{T \cdot \dashv T \backslash} \boxed{\begin{array}{c} \epsilon \frac{S}{\exists}^\circ X \\ R \end{array}} \xleftrightarrow{\circ} \boxed{\begin{array}{c} (\frac{S}{\exists} \exists)^\circ X \\ R \end{array}} \xleftrightarrow{\cdot \exists \dashv \frac{S}{\exists}} \boxed{\begin{array}{c} S^\circ X \\ R \end{array}} \xleftrightarrow{T \cdot \dashv T \backslash} \boxed{\begin{array}{c} X \\ S^\circ \backslash R \end{array}}$$

#### §13.1.4. $\text{union} (\epsilon \backslash R) = \epsilon \backslash (\epsilon \backslash R)$



$$\begin{array}{c} \boxed{\begin{array}{c} X \\ \text{union } (\in \backslash R) \end{array}} \end{array} \xleftrightarrow{f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} \text{union}^\circ X \\ \in \backslash R \end{array}} \end{array} \xleftrightarrow{T \cdot \neg T \backslash} \begin{array}{c} \boxed{\begin{array}{c} \in \text{union}^\circ X \\ R \end{array}} \end{array} \xleftrightarrow{\circ} \begin{array}{c} \boxed{\begin{array}{c} (\text{union} \exists)^\circ X \\ R \end{array}} \end{array} \xleftrightarrow{\cdot \exists \neg \neg} \begin{array}{c} \boxed{\begin{array}{c} (\exists \exists)^\circ X \\ R \end{array}} \end{array} \xleftrightarrow{\circ, T \cdot \neg T \backslash} \begin{array}{c} \boxed{\begin{array}{c} X \\ \in \backslash (\in \backslash R) \end{array}} \end{array} \quad (13.1.4a)$$

### §13.1.5. $\frac{S}{\exists} \text{est}(R) = S \cap (S^\circ \backslash R^\circ)$

$$\begin{array}{c} \boxed{\begin{array}{c} X \\ \frac{S}{\exists} \text{est}(R) \end{array}} \end{array} \xleftrightarrow{f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} \frac{S^\circ X}{\exists} \\ \text{est}(R) \end{array}} \end{array} \xleftrightarrow{\Delta \neg \cap} \begin{array}{c} \boxed{\begin{array}{c} \frac{S^\circ X}{\exists} \\ \in \backslash R^\circ \end{array}} \end{array} \xleftrightarrow{f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} X \\ \frac{S}{\exists} (\in \backslash R^\circ) \end{array}} \end{array} \xleftrightarrow{\cdot \exists \neg \neg, (13.1.3a)} \begin{array}{c} \boxed{\begin{array}{c} X \\ S \\ X \\ S^\circ \backslash R^\circ \end{array}} \end{array} \xleftrightarrow{\Delta \neg \cap} \begin{array}{c} \boxed{\begin{array}{c} X \\ S \cap (S^\circ \backslash R^\circ) \end{array}} \end{array} \quad (13.1.5a)$$

### §13.1.6. $\frac{S}{\exists} \text{est}(R) = \frac{S}{\exists} \text{est}(R \cap S^\circ S)$

$$\begin{array}{c} \boxed{\begin{array}{c} X \\ \frac{S}{\exists} \text{est}(R \cap S^\circ S) \end{array}} \end{array} \xleftrightarrow{(13.1.5a), \circ} \begin{array}{c} \boxed{\begin{array}{c} X \\ S \cap (S^\circ \backslash (R^\circ \cap S^\circ S)) \end{array}} \end{array} \xleftrightarrow{\Delta \neg \cap, T \cdot \neg T \backslash} \begin{array}{c} \boxed{\begin{array}{c} X \\ S \\ S^\circ X \\ R^\circ \cap S^\circ S \end{array}} \end{array} \xleftrightarrow{\Delta \neg \cap, S^\circ \cdot \text{monotone}} \begin{array}{c} \boxed{\begin{array}{c} X \\ S \\ S^\circ X \\ R^\circ \end{array}} \end{array} \xleftrightarrow{T \cdot \neg T \backslash, \Delta \neg \cap} \begin{array}{c} \boxed{\begin{array}{c} X \\ S \cap (S^\circ \backslash R^\circ) \end{array}} \end{array} \xleftrightarrow{(13.1.5a)} \begin{array}{c} \boxed{\begin{array}{c} X \\ \frac{S}{\exists} \text{est}(R) \end{array}} \end{array} \quad (13.1.6a)$$

### §13.1.7. $P(f) \text{est}(R) = \text{est}(f R f^\circ) f$

$$\begin{array}{c} \boxed{\begin{array}{c} P(f) \text{est}(R) \\ f \text{ a map} \end{array}} \end{array} \xleftrightarrow{P=E \text{ on maps, (13.1.5a)}} \begin{array}{c} \boxed{\begin{array}{c} (\exists f) \cap ((\exists f)^\circ \backslash R^\circ) \end{array}} \end{array} \xleftrightarrow{\circ, T \cdot \neg T \backslash, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} (\exists f) \cap (\in \backslash (f R^\circ)) \end{array}} \end{array} \xleftrightarrow{\text{modular law, } f \text{ simple}} \begin{array}{c} \boxed{\begin{array}{c} (\exists \cap ((\in \backslash (f R^\circ)) f^\circ)) f \end{array}} \end{array} \xleftrightarrow{\cdot f \neg f^\circ, \circ, \text{est}} \begin{array}{c} \boxed{\begin{array}{c} \text{est}(f R f^\circ) f \end{array}} \end{array} \quad (13.1.7a)$$

### §13.1.8. $P(S) \text{est}(R) \in (\exists S) \cap (\in \backslash (S R^\circ))$

$$\begin{array}{c} \boxed{\begin{array}{c} P(S) \text{est}(R) \\ (\exists S) \cap (\in \backslash (S R^\circ)) \end{array}} \end{array} \xleftrightarrow{\Delta \neg \cap, T \cdot \neg T \backslash} \begin{array}{c} \boxed{\begin{array}{c} P(S) \text{est}(R) \\ \exists S \\ \in P(S) \text{est}(R) \\ S R^\circ \end{array}} \end{array} \xleftrightarrow{\text{UP of est}} \begin{array}{c} \boxed{\begin{array}{c} P(S) \exists \\ \exists S \\ \in P(S) \\ S \in \end{array}} \end{array} \quad (13.1.8a)$$

### §13.1.9. $P(\text{est}(R)) \text{est}(R) \in \text{union est}(R)$

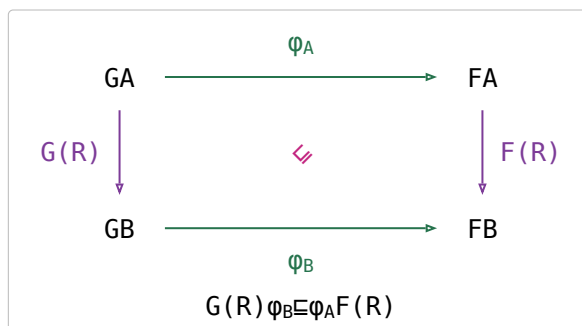
$$\begin{array}{c} \boxed{\begin{array}{c} P(\text{est}(R)) \text{est}(R) \\ \text{union est}(R) \\ R \text{ a preorder} \end{array}} \end{array} \xleftrightarrow{(13.1.5a)} \begin{array}{c} \boxed{\begin{array}{c} P(\text{est}(R)) \text{est}(R) \\ (\exists \exists) \cap ((\exists \exists)^\circ \backslash R^\circ) \end{array}} \end{array} \xleftrightarrow{\circ, \Delta \neg \cap, T \cdot \neg T \backslash} \begin{array}{c} \boxed{\begin{array}{c} P(\text{est}(R)) \text{est}(R) \\ \exists \exists \\ \in \in P(\text{est}(R)) \text{est}(R) \\ R^\circ \end{array}} \end{array} \xleftrightarrow{\text{UP of est, } R \text{ transitive}} \begin{array}{c} \boxed{\begin{array}{c} P(\text{est}(R)) \exists \\ \exists \text{est}(R) \\ \in P(\text{est}(R)) \text{est}(R) \\ \text{est}(R) \in \end{array}} \end{array} \quad (13.1.9a)$$

## §13.2. Lax natural transformations (LaT)

For relators  $G, F : \mathcal{C} \rightarrow \mathcal{D}$  and components  $\varphi_A : GA \rightarrow FA$ ,  $\varphi$  is **lax at**  $R : A \rightarrow B$  when  $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$  — both sides  $GA \rightarrow FB$ , the left through the component at  $B$ , the right through the one at  $A$ . (13.2a)

$\varphi : G \Rightarrow F$  is a **lax natural transformation** (LaT) when it is lax at **every**  $R$ . (5.13)

Lax at every **map** already gives LaT, and at a map the inequation is an equality  $G(f)\varphi = \varphi F(f)$ : laxness is about relations only. Theorem 5.2



(13.2b)

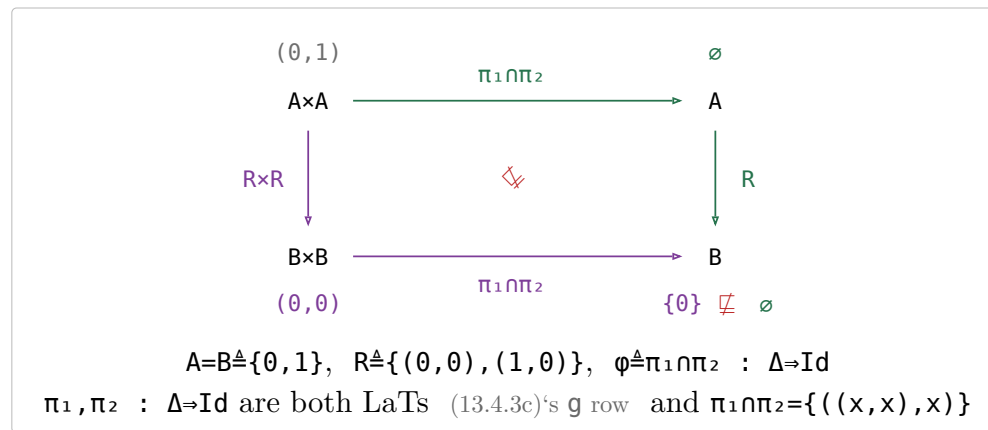
| closed under                              | commutative diagram                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | string diagram |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| composition<br>$\psi\phi$                 | <p> <math>H(R)\psi_B \sqsubseteq \psi_A G(R)</math> and <math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> give <math>H(R)</math><br/> <math>(\psi_B\phi_B) \sqsubseteq (\psi_A\phi_A) F(R)</math> </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                |
| horizontal composition<br>$\chi\circ\phi$ | <p> <math>\phi : G \Rightarrow F</math> with <math>G, F : \mathcal{C} \rightarrow \mathcal{D}</math> and <math>\chi : L \Rightarrow K</math> with <math>L, K : \mathcal{D} \rightarrow \mathcal{E}</math> give <math>\chi \circ \phi : L \circ G \Rightarrow K \circ F</math>, the family <math>A \mapsto \chi_{GA} K(\phi_A)</math><br/> <math>L(G(R)) \chi_{GB} \sqsubseteq \chi_{GA} K(G(R))</math> is <math>\chi</math> lax at <math>G(R)</math>; then<br/> <math>K(G(R)\phi_B) \sqsubseteq K(\phi_A F(R))</math> is <math>K</math> applied to <math>\phi</math>'s own inequation<br/> <math>\chi := \mathbb{1}_K</math> gives <math>K(\phi) : K \circ G \Rightarrow K \circ F</math>, <math>K</math> composed on the outside; <math>\phi := \mathbb{1}_G</math><br/> gives <math>\chi G : L \circ G \Rightarrow K \circ G</math>, <math>G</math> composed on the inside<br/> two candidates, <math>\chi_{GA} K(\phi_A)</math> and <math>L(\phi_A) \chi_{FA}</math>, ordered by <math>\sqsubseteq</math>; this row is<br/> the first </p> |                |
| union<br>$\phi \cup \psi$                 | <p> <math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> and <math>G(R)\psi_B \sqsubseteq \psi_A F(R)</math> give <math>G(R)</math><br/> <math>(\phi_B \cup \psi_B) \sqsubseteq (\phi_A \cup \psi_A) F(R)</math> </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                |
| a relator $K$<br>$K(\phi)$                | <p> <math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> gives <math>K(G(R)) K(\phi_B) \sqsubseteq K(\phi_A) K(F(R))</math> </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |
| product<br>$\phi \times \psi$             | <p> <math>\phi : G \Rightarrow F</math> and <math>\psi : G' \Rightarrow F'</math> give <math>\phi \times \psi : G \times G' \Rightarrow F \times F'</math><br/> <math>(R \times S) (U \times V) = (RU) \times (SV)</math> and monotonicity in both slots — the<br/> row above's <math>K</math> applied to the inequation, run on a bifunctor; the fork<br/> is the derived case <math>\langle \phi, \psi \rangle = \triangleleft (\phi \times \psi)</math>, and its ONE cost is the lax<br/> copy law <math>R \triangleleft \triangleleft (R \times R)</math> — (2b) </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                |
| coproduct<br>$\phi + \psi$                | <p> <math>\phi : G \Rightarrow F</math> and <math>\psi : G' \Rightarrow F'</math> give <math>\phi + \psi : G + G' \Rightarrow F + F'</math><br/> <math>(R + S) (U + V) = (RU) + (SV)</math> and monotonicity in both slots; the co-<br/> fork is the derived case <math>[\phi, \psi] = (\phi + \psi) \nabla</math>, and <math>\nabla</math> costs nothing<br/> the step it would need is <math>(\phi_A F(R)) \cap (\psi_A F(R)) \sqsubseteq (\phi \cap \psi)_A F(R)</math>, the<br/> wrong direction of (4c)<br/> a counterexample: (13.2.1a) </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                |
| meet — <b>fails</b><br>$\phi \cap \psi$   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                |

Each row is the general slide  $R_A X \sqsubseteq X' R_B$  and  $R_B Y \sqsubseteq Y' R_C$  give  $R_A (XY) \sqsubseteq (X' Y') R_C$  at  $R_A, R_B := G(R), F(R)$  and  $X, X' := \phi_B, \phi_A$ , quantified over  $R$ ; at  $X' = X$  with  $R_A, R_B$  endorelations it is the monotonicity reading instead — the  $\forall R$  sits outside the law, and is the only difference

### §13.2.1. meet is not closed in LaT

counterexample —  $\varphi, \psi : \mathbf{G} \Rightarrow \mathbf{F}$  lax natural does NOT give  $\varphi \cap \psi$  lax natural

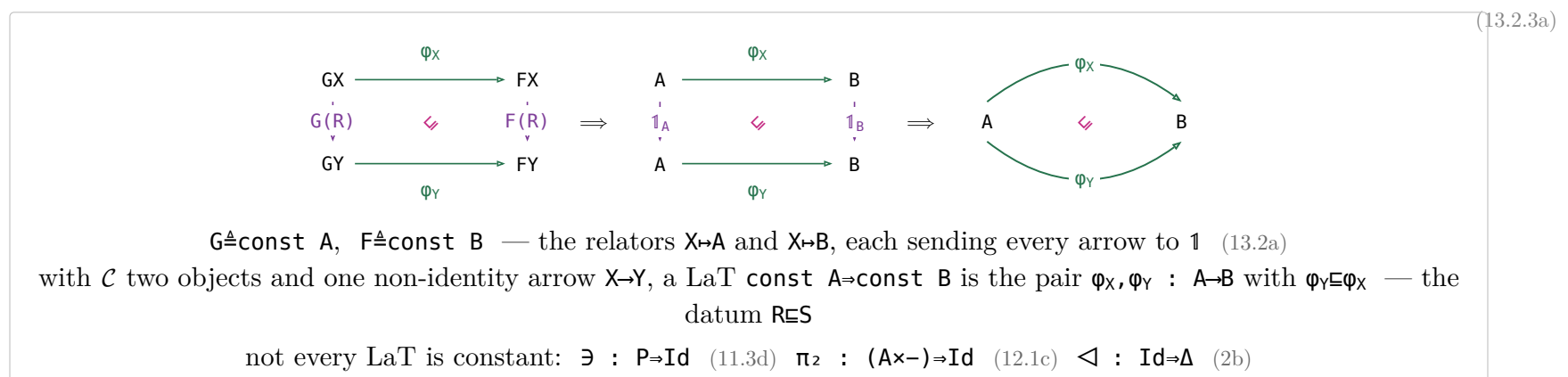
(13.2.1a)



### §13.2.2. Two stacked towers

|             | lower — inside ONE allegory      | upper — between allegories                                                                                |
|-------------|----------------------------------|-----------------------------------------------------------------------------------------------------------|
| 0-cell      | an object $A$                    | an allegory $\mathcal{C}$ — a whole lower tower                                                           |
| 1-cell      | a relation $R : A \rightarrow B$ | a relator $F : \mathcal{C} \rightarrow \mathcal{D}$                                                       |
| 2-cell      | $R \sqsubseteq S$                | a LaT $\varphi : \mathbf{G} \Rightarrow \mathbf{F}$ (13.2a)                                               |
| order       | $R \sqsubseteq S$                | $\varphi \sqsubseteq \psi$ componentwise                                                                  |
| union       | $R \cup S$                       | $\varphi \cup \psi$ , survives (13.2c)                                                                    |
| product     | $R \times S$ (11.1.1a)           | $\varphi \times \psi$ , survives; a TENSOR, not a categorical product; SYMMETRIC (11.1b) (13.2c)          |
| coproduct   | $R + S$                          | $\varphi + \psi$ , survives; a BIPRODUCT (13.2c)                                                          |
| meet        | $R \cap S$                       | $\varphi \cap \psi$ componentwise, fails (13.2.1a)                                                        |
| converse    | $R^\circ$                        | $\varphi^\circ$ , fails, oplax: $\varphi_A^\circ \mathbf{G}(R) \sqsubseteq \mathbf{F}(R) \varphi_B^\circ$ |
| zero object | $z$ with $1 \ z = 0$             | the constant relator at such a $z$ , initial and terminal                                                 |

### §13.2.3. RES is a special case of LaT

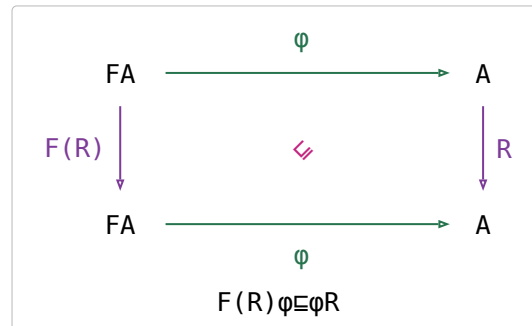


### §13.3. Monotonic algebras

An  $F$ -algebra  $\varphi : FA \rightarrow A$  is **monotonic on**  $R : A \rightarrow A$  when it is lax at  $R$  at  $G := F$ ,  $F := \text{Id}$ :  $F(R)\varphi \sqsubseteq \varphi R$ .  $R$  is an **endorelation**, so  $B=A$  and the two components  $\varphi_A, \varphi_B$  are the one arrow  $\varphi$  — an algebra, not a family.

(13.3a)

For a map  $f : FA \rightarrow A$  that is  $f \circ F(R) f \sqsubseteq R$  (1.1a)'s  $f \circ \cdot \dashv f \cdot$  at  $X := F(R) f, Y := R$ , equivalently  $F(R) \sqsubseteq f R f \circ \cdot f \dashv \cdot f \circ$  then  $f \circ \cdot \dashv f \cdot$ .  
 $(\leq x \leq) + \sqsubseteq + \leq$  — addition on  $\text{Nat}$  is monotonic on  $\leq$ , which at the point level reads  $c = a + b \wedge a \leq a' \wedge b \leq b' \implies c \leq a' + b'$ .



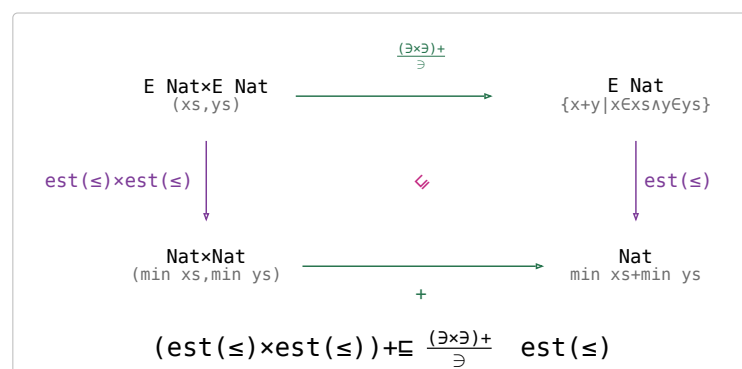
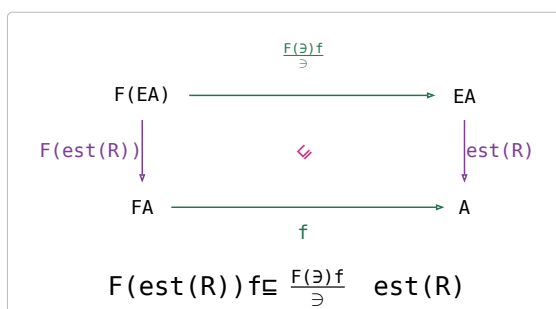
(13.3b)

#### §13.3.1. Function $f$ is monotonic on $R$ iff it distributes over $R$

$f : FA \rightarrow A$  **distributes over**  $R$  if  $F(\text{est}(R)) f \sqsubseteq \frac{F(\exists)f}{\exists} \text{est}(R)$ .

(13.3.1a)

$+$  distributes over  $\leq$ , at the point level  $\min xs + \min ys = \min\{x+y \mid x \in xs \wedge y \in ys\}$  for  $xs, ys$  non-empty and  $\min \triangleq \text{est}(\leq)$ .



(13.3.1b)

$$f \circ F(R) f \sqsubseteq R \iff F(\text{est}(R)) f \sqsubseteq \frac{F(\exists) f}{\exists} \text{est}(R)$$

Theorem 7.1,  $f$  a map,  $R$  reflexive

circuit

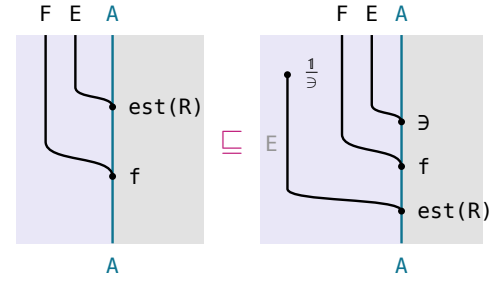
$$F(EA) \sqsubseteq \boxed{F(\text{est}(R))} \sqsubseteq \boxed{f} \sqsubseteq A \quad \sqsubseteq \quad \boxed{\frac{F(\exists) f}{\exists}} \sqsubseteq \boxed{\text{est}(R)}$$

$$F(\text{est}(R)) f \sqsubseteq \frac{F(\exists) f}{\exists} \text{est}(R)$$

$$\frac{S}{\exists} = \frac{1}{\exists} E(S), \quad S \triangleq F(\exists) f$$

 def.  $f$  distributes over  $R$  — (13.3.1a), (10.1a)

Hinze–Marsden

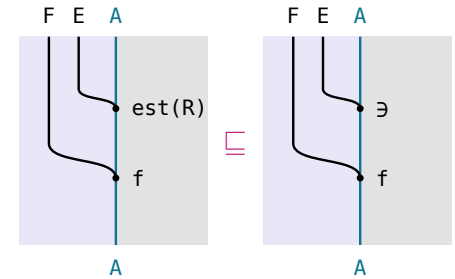


$$F(\text{est}(R)) f \sqsubseteq F(\exists) f$$

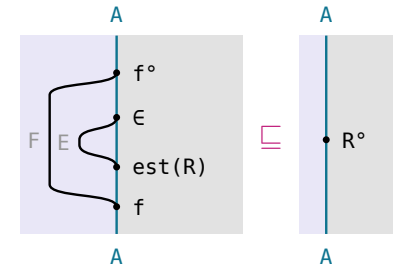
and

$$\Leftrightarrow (F(\exists) f) \circ F(\text{est}(R)) f \sqsubseteq R^\circ$$

 (13.1.5a) at  $F(\exists) f$ ;  $F(\text{est}(R)) f \sqsubseteq (F(\exists) f) \circ R^\circ \iff (F(\exists) f) \circ F(\text{est}(R)) f \sqsubseteq R^\circ$  — (8.1c)

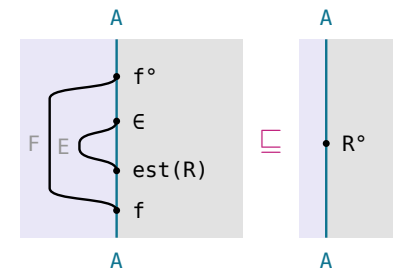
 $\text{est}(R) \triangleq \exists n(\epsilon \setminus R^\circ)$  — (13.1a)


and



$$\Leftrightarrow A \sqsubseteq \boxed{F(\exists) f} \sqsubseteq \boxed{F(\text{est}(R)) f} \sqsubseteq \boxed{R^\circ} \sqsubseteq A$$

$$(F(\exists) f) \circ F(\text{est}(R)) f \sqsubseteq R^\circ$$

 $\text{est}(R) \sqsubseteq \exists$  — (13.1a); the first conjunct is free


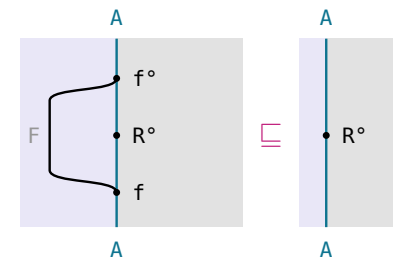
$$\Leftrightarrow A \sqsubseteq \boxed{f} \sqsubseteq \boxed{F(\epsilon \text{ est}(R))} \sqsubseteq \boxed{R^\circ} \sqsubseteq A$$

$$f \circ F(\epsilon \text{ est}(R)) f \sqsubseteq R^\circ$$

 converse, relators:  $(F(\exists) f)^\circ = f^\circ F(\epsilon)$  and  $F(\epsilon) F(\text{est}(R)) = F(\epsilon \text{ est}(R))$ 

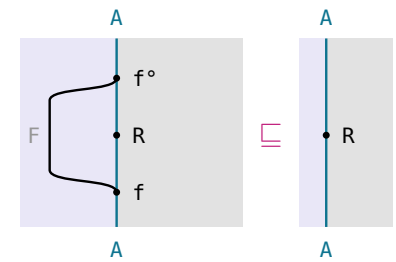
$$\Leftrightarrow A \sqsubseteq \boxed{f} \sqsubseteq \boxed{F(R^\circ)} \sqsubseteq \boxed{R^\circ} \sqsubseteq A$$

$$f \circ F(R^\circ) f \sqsubseteq R^\circ$$

 $\epsilon \text{ est}(R) = R^\circ$ ,  $R$  reflexive — (13.1a)


$$\Leftrightarrow A \sqsubseteq \boxed{f} \sqsubseteq \boxed{F(R)} \sqsubseteq \boxed{R} \sqsubseteq A$$

$$f \circ F(R) f \sqsubseteq R$$

 $f$  a map, so monotonic on  $R$  and  $R^\circ$  together




§13.3.3.  $\text{takewhile}(p) = ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \rightarrow \text{nil})])$ 

(13.3.3a)

| name                     | definition                                                            | type                                                   | example                                                                                                | in words                                                                                        |
|--------------------------|-----------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| F                        | $FX = 1 + A \times X$                                                 | $\mathcal{A} \rightarrow \mathcal{A}$                  |                                                                                                        | nothing, or a head and a tail                                                                   |
| nil,<br>cons             | $[A] ::= \text{nil} \mid \text{cons } (A, [A])$<br>(12a)              | $1 \rightarrow [A],$<br>$A \times [A] \rightarrow [A]$ | $\text{cons}(3, [1, 2]) = [3, 1, 2]$                                                                   | the empty list; a head onto a tail                                                              |
| $\alpha$                 | $[\text{nil}, \text{cons}]$                                           | $F([A]) \rightarrow [A]$                               |                                                                                                        | both constructors as one map                                                                    |
| p                        | a coreflexive                                                         | $A \rightarrow A$                                      | $p \triangleq \text{even} \rightarrow 2 \rightarrow p \rightarrow 2$ , and<br>$3 \notin \text{Dom}(p)$ | $\{(a, a) \mid a \text{ passes the test}\}$                                                     |
| R                        | $\text{length} \leq \text{length}^\circ$ , a<br>preorder              | $[A] \rightarrow [A]$                                  | $[1] \text{ R } [1, 2]$                                                                                | $xs \text{ R } ys \iff \text{length } xs \leq \text{length } ys$                                |
| $\rightarrow \text{nil}$ | the constant nil — the<br>second nil of prefix                        | $A \times [A] \rightarrow [A]$                         | $(\rightarrow \text{nil})(3, [1, 2]) = \text{nil}$                                                     | drop the pair, return nil                                                                       |
| prefix                   | $([\text{nil}, \rightarrow \text{nil} \cup \text{cons}])$<br>(12a)    | $[A] \rightarrow [A]$                                  | $[3, 1, 2] \text{ prefix } [3, 1]$                                                                     | $xs \text{ prefix } ys \iff \exists zs. xs = ys \# zs$<br>— at each cons, stop or keep the head |
| S                        | $[\text{nil}, \rightarrow \text{nil} \cup (p \times 1) \text{ cons}]$ | $F([A]) \rightarrow [A]$                               | $(4, [2]) \text{ S } [4, 2]$ , and<br>$(3, [2]) \text{ S } \text{nil}$ only                            | prefix's algebra with one extra p<br>— stop, or keep a head that passes p                       |
| $(g \rightarrow X, Y)$   | X where g is defined and Y<br>where it is not                         | $A \rightarrow B$ at X, Y :<br>$A \rightarrow B$       |                                                                                                        | the test picks the branch                                                                       |

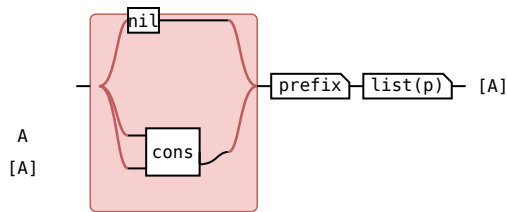
$\text{nil} \text{ R } \top$ ,  $\text{nil} \text{ R}^\circ = \text{nil}$  nil is the shortest list — below every list, and above only itself, so it loses every  $\text{est}(\text{R}^\circ)$



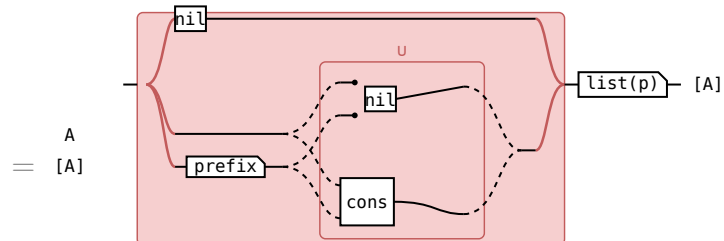
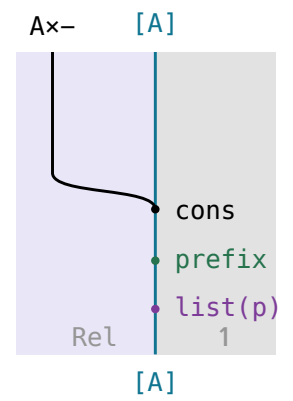
$$\alpha \text{ prefix list}(p) = F(\text{prefix list}(p))S$$

**circuit** — the fork is  $F([A]) = 1 + A \times [A]$ : nil above, the pair below

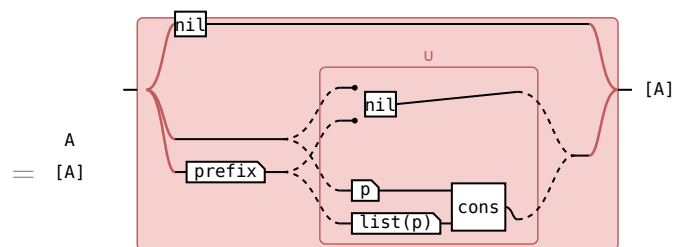
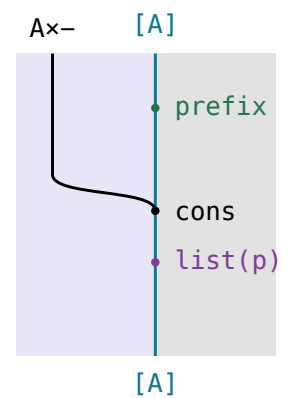
**Hinze–Marsden**



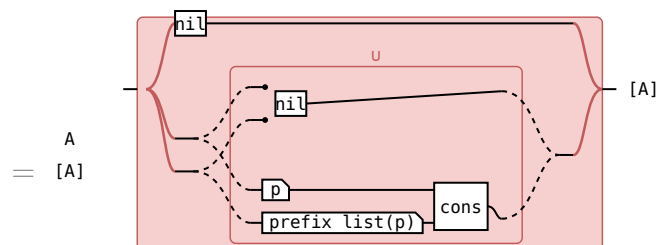
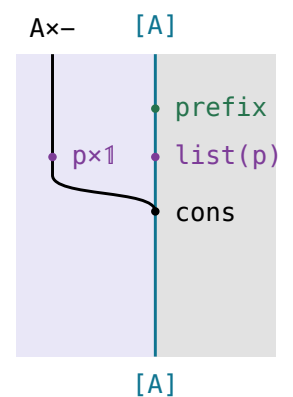
$\alpha \text{ prefix list}(p)$



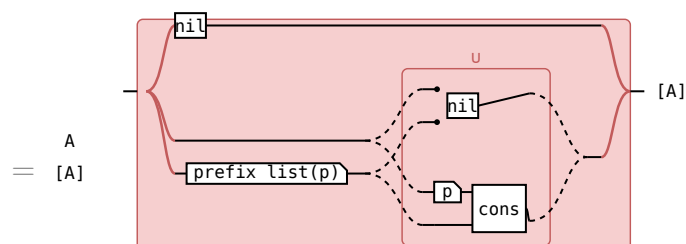
$F(\text{prefix}) [\text{nil}, \multimap \text{nil} \cup \text{cons}] \text{list}(p)$   
defining equation



$F(\text{prefix}) [\text{nil}, \multimap \text{nil} \cup (p \times \text{list}(p)) \text{cons}]$   
 $\text{list}(p)$  through cons

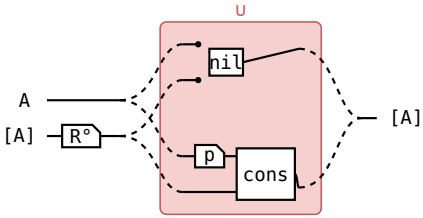
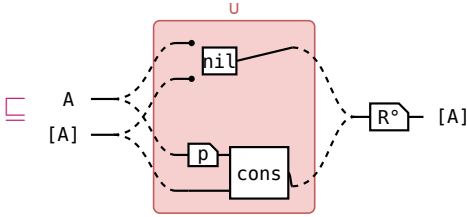


$[\text{nil}, \multimap \text{nil} \cup (p \times (\text{prefix list}(p))) \text{cons}]$   
relator, prefix entire

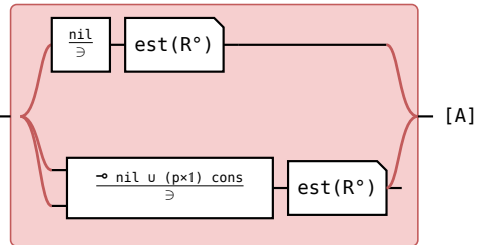
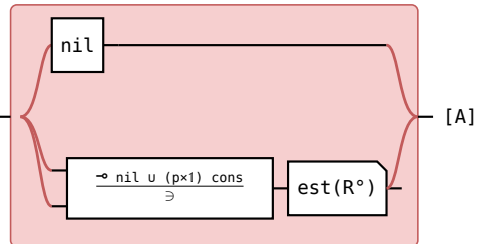
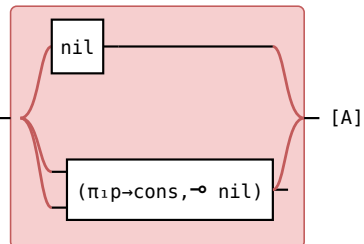


$F(\text{prefix list}(p))S$   
prefix list(p) entire

(11.6.1a) reads that off as  $\text{prefix list}(p) = \{S\}$ . (11.4.2a) cannot:  $\text{list}(p)$  is not entire,  $(1 \times \text{list}(p)) \multimap \text{nil} \sqsubseteq \multimap \text{nil}$ , and no algebra meets the side condition.

| $(1 \times R^\circ)(\neg \text{nil} \cup (p \times 1) \text{ cons}) \sqsubseteq (\neg \text{nil} \cup (p \times 1) \text{ cons}) R^\circ$ 3.3.3c)      |                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| formula — the cons branch of $F(R^\circ) \sqsubseteq SR^\circ$                                                                                         | reason                                                                                                                                                                                    |
| $(1 \times R^\circ) \neg \text{nil} \sqsubseteq \neg \text{nil} R^\circ$                                                                               | $\text{nil} R^\circ = \text{nil}$ (13.3.3a) — the right side is $\neg \text{nil}$ , and $(1 \times R^\circ) \neg \text{nil} \sqsubseteq \neg \text{nil} R^\circ$                          |
| $(1 \times R^\circ)(p \times 1) \text{ cons} \sqsubseteq (p \times 1) \text{ cons} R^\circ$                                                            | composition (13.2c) at $(1 \times R^\circ)$<br>$(p \times 1) = (p \times R^\circ) = (p \times 1)(1 \times R^\circ)$ and $(1 \times R^\circ) \text{ cons} \sqsubseteq \text{cons} R^\circ$ |
|  $(1 \times R^\circ)(\neg \text{nil} \cup (p \times 1) \text{ cons})$ |                                                                                                                                                                                           |
|  $(\neg \text{nil} \cup (p \times 1) \text{ cons}) R^\circ$           | union (13.2c) at $\varphi := \neg \text{nil}$ ,<br>$\psi := (p \times 1) \text{ cons}$                                                                                                    |

the **nil** branch is  $\text{nil} \sqsubseteq \text{nil} R^\circ$ .  $(1 \times R^\circ) \text{ cons} \sqsubseteq \text{cons} R^\circ$  is  $\text{cons length} = (1 \times \text{length}) \pi_2 \text{ succ}$  with **SUCC** monotone — a shorter tail makes a shorter list.

| $\frac{s}{\supseteq} \text{est}(R^\circ) = [\text{nil}, (\pi_1 p \rightarrow \text{cons}, \neg \text{nil})]$ 3.3.3d) |                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| formula                                                                                                              | reason                                                                                                                                                       |
| $F([A]) \xrightarrow{\frac{s}{\supseteq}} E([A]) \xrightarrow{\text{est}(R^\circ)} [A]$                              | $\frac{s}{\supseteq} \text{est}(R^\circ)$                                                                                                                    |
| $=$                               | $[\frac{\text{nil}}{\supseteq} \text{est}(R^\circ), \frac{\neg \text{nil} \cup (p \times 1) \text{ cons}}{\supseteq} \text{est}(R^\circ)]$ coproduct of maps |
| $=$                               | $[\text{nil}, \frac{\neg \text{nil} \cup (p \times 1) \text{ cons}}{\supseteq} \text{est}(R^\circ)]$ singleton, $R^\circ$ reflexive                          |
| $=$                               | $[\text{nil}, (\pi_1 p \rightarrow \text{cons}, \neg \text{nil})]$ $\text{nil} R = \top$                                                                     |

the set is  $\{\text{nil}\}$  where **p** fails on the head and  $\{\text{nil}, \text{cons}(a, xs)\}$  where it holds, and **nil** loses the second — (13.1a) at a two-element set.

$$\text{takewhile}(p) \triangleq \frac{\text{prefix list}(p)}{\supset} \quad \text{est}(R^\circ) = ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \rightarrow \text{nil})])$$

(13.3.3e)

Ex 7.39: the longest prefix all of whose elements pass  $p$ 

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                | Hinze–Marsden |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $[A] \vdash \frac{\text{prefix list}(p)}{\supset} E([A]) \vdash \text{est}(R^\circ) \vdash [A]$ <p><math>\frac{\text{prefix list}(p)}{\supset} \quad \text{est}(R^\circ)</math><br/>the specification — (13.1a)'s <math>\text{est}(R^\circ)</math></p>                                                                                                                                                                                 |               |
| $[A] \vdash \frac{(\frac{S}{\supset})}{\supset} E([A]) \vdash \text{est}(R^\circ) \vdash [A]$ <p><math>= \frac{(\frac{S}{\supset})}{\supset} \text{est}(R^\circ)</math><br/>(13.3.3b)</p>                                                                                                                                                                                                                                              |               |
| $[A] \vdash \frac{(\frac{S}{\supset} \text{est}(R^\circ))}{\supset} \vdash [A]$ <p><math>\sqsubseteq (\frac{S}{\supset} \text{est}(R^\circ))</math><br/>(13.3.2a) at <math>R^\circ</math>, with <math>F(R^\circ) S \subseteq SR^\circ</math> — (13.3.3c) — for its hypothesis: one longest <math>p</math>-prefix kept at each <math>\text{cons}</math>, instead of every <math>p</math>-prefix collected and one chosen at the end</p> |               |
| $[A] \vdash \frac{([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \rightarrow \text{nil})])}{\supset} \vdash [A]$ <p><math>= ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \rightarrow \text{nil})])</math><br/>(13.3.3d)</p>                                                                                                                                                                                                        |               |
| <p><math>\text{prefix}^\circ \text{ prefix} \cap R \cap R^\circ \subseteq 1</math><br/>(13.1b) at <math>\text{prefix}</math>: two prefixes of one list of equal length are equal, so <math>\frac{\text{prefix list}(p)}{\supset}</math><br/><math>\text{est}(R^\circ)</math> is simple — the longest, not a longest</p>                                                                                                                |               |
| <p><math>\frac{\text{prefix list}(p)}{\supset} \quad \text{est}(R^\circ)</math> entire<br/><math>\text{nil}</math> is always a <math>p</math>-prefix and <math>R</math> is connected on the prefixes of one list, so the longest exists</p>                                                                                                                                                                                            |               |
| <p><math>X \subseteq Y, X \text{ entire}, Y \text{ simple} \Rightarrow X=Y</math><br/><math>([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \rightarrow \text{nil})])</math> is a reduce of maps, hence entire — what turns the <math>\sqsubseteq</math> above into the heading's =</p>                                                                                                                                                |               |

### §13.3.4. $\text{mss} = ([\text{zero wrap}, ((1 \times \text{head}) \oplus, \pi_2) \text{ cons}]) \text{ est}(\geq)$

(13.3.4a)

$\text{FX} = 1 + A \times X$ ,  $A := \text{Int}$ ,  $\alpha \triangleq [\text{nil}, \text{cons}]$ ,  $\text{sum} = ([\text{zero}, \text{plus}])$  and  $\text{segment} = \text{suffix prefix}$  from (7a) and (12a).

$\text{head} \triangleq \text{cons}^\circ \pi_1$ ,  $\text{wrap} \triangleq (1, \rightarrow \text{nil}) \text{ cons}$  the head of a list and the one-element list, beside (12a)'s  $\text{tail} \triangleq \text{cons}^\circ \pi_2$

$\oplus \triangleq \frac{\rightarrow \text{zero} \cup \text{plus}}{\supset} \quad \text{est}(\geq)$  B&dM's  $\text{oplus} = \max(\wedge(\text{zerouplus}))$ ; the set at  $(a, b)$  is  $\{0, a+b\}$ , so  $\oplus$  is the larger of the two

| $\frac{\text{segment sum}}{\ni} \text{est}(\geq) = \frac{\text{suffix}}{\ni} E(\frac{\text{prefix sum}}{\ni} \text{est}(\geq)) \text{est}(\geq)$                                                                      |                                                                                                   |  |                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| formula — one wire from [A] to A, its type written along it                                                                                                                                                           |                                                                                                   |  | reason                                                                                                                                                                                                                                                                                                                 |
| $\cdot [A] \cdot \boxed{\frac{\text{segment sum}}{\ni}} \cdot EA \cdot \boxed{\text{est}(\geq)} \cdot A \cdot$                                                                                                        | $\frac{\text{segment sum}}{\ni} \text{est}(\geq)$                                                 |  |                                                                                                                                                                                                                                                                                                                        |
| $= \cdot [A] \cdot \boxed{\frac{\text{suffix (prefix sum)}}{\ni}} \cdot EA \cdot \boxed{\text{est}(\geq)} \cdot A \cdot$                                                                                              | $\frac{\text{suffix (prefix sum)}}{\ni} \text{est}(\geq)$                                         |  | segment=suffix<br>prefix<br>(12a), (13.3.4a)                                                                                                                                                                                                                                                                           |
| $= \cdot [A] \cdot \boxed{\frac{\text{suffix}}{\ni}} \cdot E([A]) \cdot \boxed{E(\text{prefix sum})} \cdot EA \cdot \boxed{\text{est}(\geq)} \cdot A \cdot$                                                           | $\frac{\text{suffix}}{\ni} E(\text{prefix sum}) \text{est}(\geq)$                                 |  | absorption<br>(10b) — $\text{frac}(S, \ni)$<br>$E(R) = \text{frac}(SR, \ni)$ at<br>$S := \text{suffix}, R := \text{prefix sum}$                                                                                                                                                                                        |
| $= \cdot [A] \cdot \boxed{\frac{\text{suffix}}{\ni}} \cdot E([A]) \cdot \boxed{E(\frac{\text{prefix sum}}{\ni})} \cdot E(EA) \cdot \boxed{\text{union}} \cdot EA \cdot \boxed{\text{est}(\geq)} \cdot A \cdot$        | $\frac{\text{suffix}}{\ni} E(\frac{\text{prefix sum}}{\ni}) \text{union est}(\geq)$               |  | $\frac{R}{\ni} \ni = R, \text{union} = E(\ni)$<br>(10b)'s $\text{frac}(R, \ni) \ni = R$ at<br>$R := \text{prefix sum}$ and<br>$E(R) \triangleq \text{frac}(\ni R, \ni)$ ;<br>(13.1b)'s<br>$\text{union} \triangleq \text{frac}(\ni \ni, \ni)$ ; the<br>middle equality is (11a)'s<br>$F(RS) = F(R)F(S)$ at<br>$F := E$ |
| $= \cdot [A] \cdot \boxed{\frac{\text{suffix}}{\ni}} \cdot E([A]) \cdot \boxed{E(\frac{\text{prefix sum}}{\ni})} \cdot E(EA) \cdot \boxed{E(\text{est}(\geq))} \cdot EA \cdot \boxed{\text{est}(\geq)} \cdot A \cdot$ | $\frac{\text{suffix}}{\ni} E(\frac{\text{prefix sum}}{\ni}) E(\text{est}(\geq)) \text{est}(\geq)$ |  | (13.1b), the sets non-empty                                                                                                                                                                                                                                                                                            |
| $= \cdot [A] \cdot \boxed{\frac{\text{suffix}}{\ni}} \cdot E([A]) \cdot \boxed{E(\frac{\text{prefix sum}}{\ni} \text{est}(\geq))} \cdot EA \cdot \boxed{\text{est}(\geq)} \cdot A \cdot$                              | $\frac{\text{suffix}}{\ni} E(\frac{\text{prefix sum}}{\ni} \text{est}(\geq)) \text{est}(\geq)$    |  | relator<br>(11a) —<br>$F(RS) = F(R)F(S)$ at<br>$F := E$                                                                                                                                                                                                                                                                |

B&dM's  $\max(P(\max(\wedge(\text{sum prefix})))) \wedge \text{suffix}$ , mirrored. The **union** step is (13.1b)'s  $P(\text{est}(R)) \text{est}(R) = P(\text{Dom}(\text{est}(R))) \text{union est}(R)$  — every suffix has the empty prefix, so **Dom** is **1** here — and  $P(f) = E(f)$  at the map it is applied to ((11.3d)).

(11.4.2a) at  $\alpha_B := [\text{nil}, \text{nil} \cup \text{cons}]$ ,  $S := \text{sum}$ : the side condition, so  $\text{prefix sum} = ([\text{zero}, \text{zero} \cup \text{plus}])$ .  $\text{prefix}$  is the reduce,  $\text{sum}$  the map fused into it — the intermediate list is gone.

| $(1 \times \geq)(\neg \text{zero} \cup \text{plus})E(\neg \text{zero} \cup \text{plus})\geq$<br>the <b>plus</b> branch of $F(\geq)SES\geq$ ; the <b>zero</b> branch is $\text{zero}E\text{zero}\geq$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| circuit — the head above, the running sum below; the tape is the $u$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Hinze–Marsden                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>A circuit diagram with a red box labeled <math>u</math> at the top. Inside the box, there are two components: a <b>zero</b> box and a <b>plus</b> box. The <b>zero</b> box has two inputs: a solid line from <math>A</math> and a dashed line from <math>\text{Int}</math>. The <b>plus</b> box has two inputs: a solid line from <math>\text{Int}</math> and a dashed line from <math>\text{Int}</math>. The output of the <b>zero</b> box is a dashed line to <math>\text{Int}</math>. The output of the <b>plus</b> box is a dashed line to <math>\text{Int}</math>. A small <math>\geq</math> box is on the left, with a solid line from <math>\text{Int}</math> and a dashed line to <math>\text{Int}</math>.</p> <p><math>(1 \times \geq)(\neg \text{zero} \cup \text{plus})</math></p>                                                                                                                                                                                                                                                                                                         | <p>A diagram showing a vertical line with a blue background on the left and a grey background on the right. A solid line labeled <math>A \times -</math> starts at the top left and goes down. A dashed line labeled <math>\text{Int}</math> starts at the top right and goes down. The solid line has a point labeled <math>\geq</math> and the dashed line has a point labeled <b>plus</b>. The solid line ends at the bottom left, and the dashed line ends at the bottom right, both labeled <math>\text{Int}</math>.</p> |
| <p>A circuit diagram with a red box labeled <math>u</math> at the top. Inside the box, there are two components: a <b>zero</b> box and a <b>plus</b> box. The <b>zero</b> box has two inputs: a solid line from <math>A</math> and a dashed line from <math>\text{Int}</math>. The <b>plus</b> box has two inputs: a solid line from <math>\text{Int}</math> and a dashed line from <math>\text{Int}</math>. The output of the <b>zero</b> box is a dashed line to <math>\text{Int}</math>. The output of the <b>plus</b> box is a dashed line to <math>\text{Int}</math>. A small <math>\geq</math> box is on the left, with a solid line from <math>\text{Int}</math> and a dashed line to <math>\text{Int}</math>.</p> <p><math>(1 \times \geq)\neg \text{zero} \cup (1 \times \geq) \text{plus}</math><br/> relator, composition over <math>U</math></p>                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p>A circuit diagram with a red box labeled <math>u</math> at the top. Inside the box, there are two components: a <b>zero</b> box and a <b>plus</b> box. The <b>zero</b> box has two inputs: a solid line from <math>A</math> and a dashed line from <math>\text{Int}</math>. The <b>plus</b> box has two inputs: a solid line from <math>\text{Int}</math> and a dashed line from <math>\text{Int}</math>. The output of the <b>zero</b> box is a dashed line to <math>\text{Int}</math>. The output of the <b>plus</b> box is a dashed line to <math>\text{Int}</math>. A small <math>\geq</math> box is on the left, with a solid line from <math>\text{Int}</math> and a dashed line to <math>\text{Int}</math>.</p> <p><math>\neg \text{zero} \cup \text{plus} \geq</math><br/> (5.1a), <math>(\geq \times \geq) \text{plus} E \text{plus} \geq</math>; <math>(\leq \times \leq) \text{plus} E \text{plus} \leq</math> is (13.3a), written <math>+</math> there, and <b>plus</b> is a map, so it is monotonic on an order and on its opposite together, which carries it to <math>\geq</math>.</p> | <p>A diagram showing a vertical line with a blue background on the left and a grey background on the right. A solid line labeled <math>A \times -</math> starts at the top left and goes down. A dashed line labeled <math>\text{Int}</math> starts at the top right and goes down. The solid line has a point labeled <b>plus</b> and the dashed line has a point labeled <math>\geq</math>. The solid line ends at the bottom left, and the dashed line ends at the bottom right, both labeled <math>\text{Int}</math>.</p> |
| <p>A circuit diagram with a red box labeled <math>u</math> at the top. Inside the box, there are two components: a <b>zero</b> box and a <b>plus</b> box. The <b>zero</b> box has two inputs: a solid line from <math>A</math> and a dashed line from <math>\text{Int}</math>. The <b>plus</b> box has two inputs: a solid line from <math>\text{Int}</math> and a dashed line from <math>\text{Int}</math>. The output of the <b>zero</b> box is a dashed line to <math>\text{Int}</math>. The output of the <b>plus</b> box is a dashed line to <math>\text{Int}</math>. A small <math>\geq</math> box is on the left, with a solid line from <math>\text{Int}</math> and a dashed line to <math>\text{Int}</math>.</p> <p><math>(\neg \text{zero} \cup \text{plus})\geq</math><br/> <math>\geq</math> reflexive</p>                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

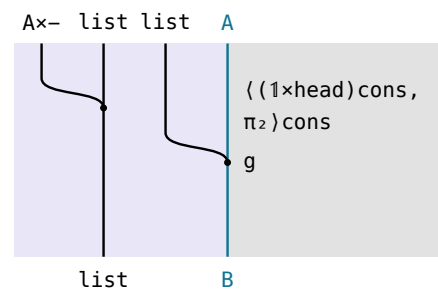
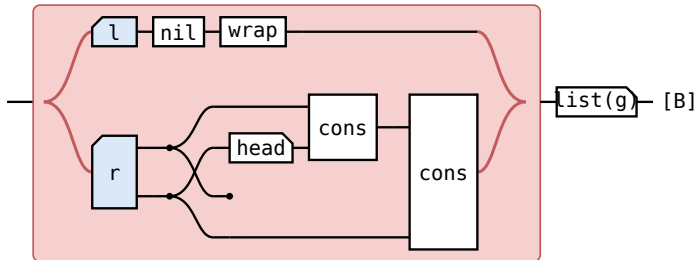
| $\frac{[zero, \multimap zero \cup plus]}{\supset} \quad est(\geq) = [zero, \oplus]$                                                                                                                                                                                                                                                                                       |                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <p><b>circuit</b> — the tape is the coproduct: zero's branch above, plus's below</p> <p><math display="block">\frac{[zero, \multimap zero \cup plus]}{\supset} \quad est(\geq)</math></p>                                                                                                                                                                                 | <p><b>Hinze–Marsden</b></p> |
| <p> <math display="block">1+A \times Int \rightarrow Int</math> <math display="block">[ \frac{zero}{\supset} est(\geq), \frac{\multimap zero \cup plus}{\supset} est(\geq) ]</math> <p>coproduct of maps — (11.2.1a) at <math>T := [zero, \multimap zero \cup plus]</math>, then<br/> <math>[U, V]Z = [UZ, VZ]</math> — (11.2a), composition over <math>U</math></p> </p> |                             |
| <p> <math display="block">1+A \times Int \rightarrow Int</math> <math display="block">[zero, \oplus]</math> <p>singleton, <math>\geq</math> reflexive — (13.1b)'s <math>\frac{1}{\supset} est(R) = 1 \cap R</math> at <math>R := \geq</math>, <b>zero</b> a map; the lower branch is <math>\oplus</math>'s definition, (13.3.4a), and no law</p> </p>                     |                             |

with (13.3.4d) the greedy theorem gives  $([zero, \oplus]) \sqsubseteq \frac{\text{prefix sum}}{\supset} est(\geq)$  — B&dM's own containment — and (13.3.4g) makes it an equality.

**[nil wrap, ((1×head) cons, π<sub>2</sub>) cons] list(g)=F(list(g))[c wrap, ((1×head) f, π<sub>2</sub>) cons]**

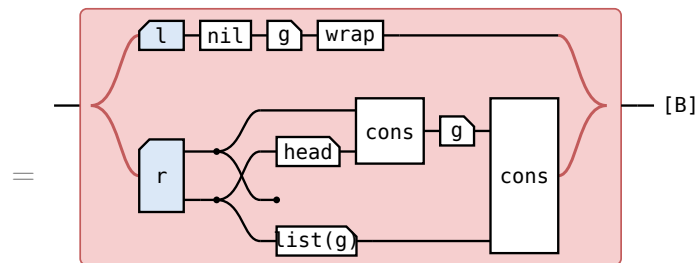
**circuit** — the 1 branch of 1+A×[A] above, the root and its list of tails below

**Hinze–Marsden**



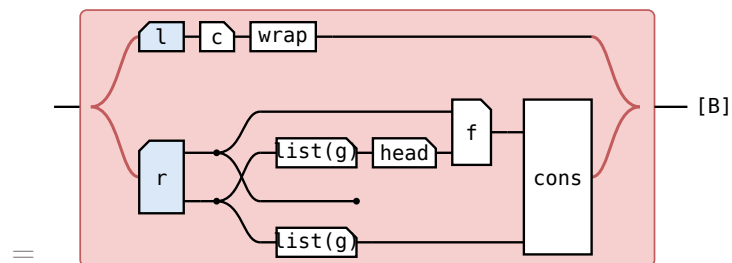
**[nil wrap, ((1×head) cons, π<sub>2</sub>) cons] list(g)**

$g \triangleq ([c, f])$  and  $\text{tails} = ([\text{nil wrap}, ((1 \times \text{head}) \text{ cons}, \pi_2) \text{ cons}])$ , since  $\text{tails}(\text{cons}(a, xs))$  is  $\text{cons}(\text{cons}(a, \text{head}(\text{tails } xs)), \text{tails } xs)$



**[nil g wrap, ((1×head) cons g, π<sub>2</sub> list(g)) cons]**

coproduct of maps, wrap and cons natural

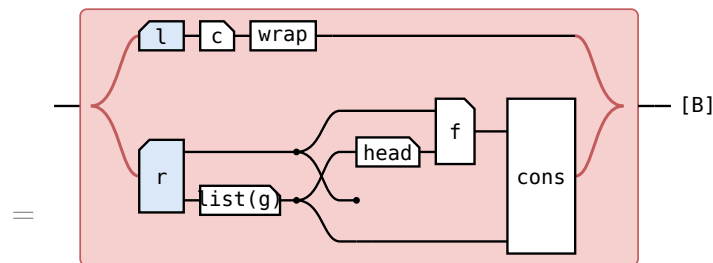


**[c wrap, ((1×(list(g) head)) f, (1×list(g)) π<sub>2</sub>) cons]**

$g$ 's defining equation,  $\text{list}(g) \text{ head} = \text{head } g$ ; the  $\pi_2$  slide

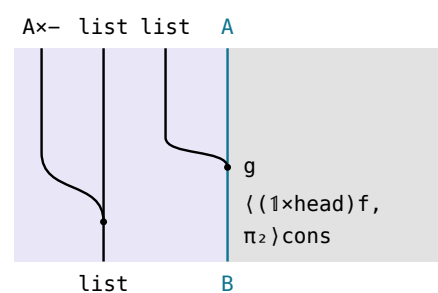
$\pi_2 \text{ list}(g) = (1 \times \text{list}(g)) \pi_2$  is 1 and 4 of (11.1.2b),  $\text{Dom}(\pi_1) = 1$  ((5c));  $\text{list}(g)$

$\text{head} = \text{head } g$  is the one step the note has no law for —  $\text{head}$  is undefined at  $\text{nil}$ , and the equality is a fact about  $\text{tails}$ , which never returns an empty list, not a naturality square



**F(list(g))[c wrap, ((1×head) f, π<sub>2</sub>) cons]**

fork, relator — the fork slide is 7 of (11.1.2b),  $1 \times \text{list}(g)$  a map there — and it is at  $c, f := \text{zero}, \oplus$ . (11.4.2a) then reads off  $\text{tails } \text{list}(g) = ([c \text{ wrap}, ((1 \times \text{head}) f, \pi_2) \text{ cons}])$ , the heading's fold.





| $\text{mss} = ([\text{zero wrap}, ((1 \times \text{head}) \oplus, \pi_2) \text{ cons}]) \text{ est}(\geq)$ $\text{Ex 7.40, } \oplus \stackrel{\text{zero u plus}}{\substack{\rightarrow \\ \supset}} \text{ est}(\geq) \text{ — (13.3.4a)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Hinze–Marsden |
| $[A] \text{ — } \boxed{\text{segment sum} \substack{\rightarrow \\ \supset}} \text{ — } \boxed{\text{est}(\geq)} \text{ — } A$ $\text{segment sum} \substack{\rightarrow \\ \supset} \text{ est}(\geq)$ <p>mss is the greatest of the segment sums — (13.3.4a)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |
| $[A] \text{ — } \boxed{\text{suffix} \substack{\rightarrow \\ \supset}} \text{ — } \boxed{E(\text{prefix sum} \substack{\rightarrow \\ \supset} \text{ est}(\geq))} \text{ — } \boxed{\text{est}(\geq)} \text{ — } A$ $= \text{suffix} \substack{\rightarrow \\ \supset} E(\text{prefix sum} \substack{\rightarrow \\ \supset} \text{ est}(\geq)) \text{ est}(\geq)$ <p>(13.3.4b)</p>                                                                                                                                                                                                                                                                                                                                                        |               |
| $[A] \text{ — } \boxed{\text{suffix} \substack{\rightarrow \\ \supset}} \text{ — } \boxed{E([\text{zero}, \oplus])} \text{ — } \boxed{\text{est}(\geq)} \text{ — } A$ $= \text{suffix} \substack{\rightarrow \\ \supset} E([\text{zero}, \oplus]) \text{ est}(\geq)$ <p>the greedy theorem (13.3.2a) at <math>R := \geq</math>, <math>S := [\text{zero}, \multimap \text{zero u plus}]</math> — (13.3.4d) is its condition and (13.3.4e) its <math>\substack{\rightarrow \\ \supset} \text{ est}(\geq)</math>; its <math>\sqsubseteq</math> is an <math>=</math> because <math>[\text{zero}, \oplus]</math> is entire and <math>\text{prefix sum} \substack{\rightarrow \\ \supset} \text{ est}(\geq)</math> simple (13.3.3e)'s last row</p> |               |
| $[A] \text{ — } \boxed{\text{tails}} \text{ — } \boxed{\text{list}([\text{zero}, \oplus])} \text{ — } \boxed{\text{est}(\geq)} \text{ — } A$ $= \text{tails list}([\text{zero}, \oplus]) \text{ est}(\geq)$ <p>tails implements <math>\text{suffix} \substack{\rightarrow \\ \supset}</math> and list(f) implements <math>E(f)</math> — (12a)</p>                                                                                                                                                                                                                                                                                                                                                                                            |               |
| $[A] \text{ — } \boxed{([\text{zero wrap}, ((1 \times \text{head}) \oplus, \pi_2) \text{ cons}])} \text{ — } \boxed{\text{est}(\geq)} \text{ — } A$ $= ([\text{zero wrap}, ((1 \times \text{head}) \oplus, \pi_2) \text{ cons}]) \text{ est}(\geq)$ <p>(11.4.2a) at (13.3.4f)'s side condition, <math>C, f := \text{zero}, \oplus</math></p>                                                                                                                                                                                                                                                                                                                                                                                                 |               |

one fold builds the  $n+1$  running maxima and the final  $\text{est}(\geq)$  reads them in one more pass, so  $\text{mss}$  is linear.

§13.3.5.  $\text{filter}(p) = ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)])$ 

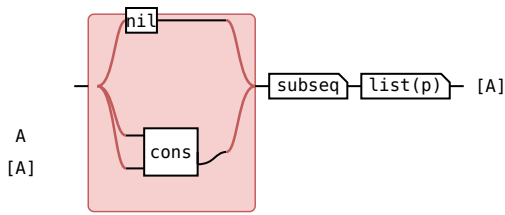
(13.3.5a)

| name                                           | definition                                                | type                           | example                                                                    | in words                                                                                                                                            |
|------------------------------------------------|-----------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| $F, \alpha, \rho, R$                           | as in (13.3.3a)                                           |                                |                                                                            |                                                                                                                                                     |
| $\pi_2$                                        | $\pi_2$ where (13.3.3a) has $\rightarrow$<br>$\text{nil}$ | $A \times [A] \rightarrow [A]$ | $\pi_2(3, [1, 2]) = [1, 2]$                                                | drop the head, keep the tail                                                                                                                        |
| subseq                                         | $([\text{nil}, \pi_2 \cup \text{cons}])$ (12a)            | $[A] \rightarrow [A]$          | $[3, 1, 2] \text{ subseq } [3, 2]$                                         | $xs \text{ subseq } ys \iff ys$ is $xs$ with elements dropped — at each $\text{cons}$ , drop the head or keep it                                    |
| $S$                                            | $[\text{nil}, \pi_2 \cup (p \times 1) \text{ cons}]$      | $F([A]) \rightarrow [A]$       | $(4, [2]) S [2]$ and<br>$(4, [2]) S [4, 2]$ , but<br>$(3, [2]) S [2]$ only | subseq's algebra with one extra $p$ — drop the head, or keep a head that passes $p$                                                                 |
| $1 \sqsubseteq \pi_2 R$<br>$\text{cons}^\circ$ |                                                           |                                |                                                                            | the tail is one shorter than the $\text{cons}$ , so $\pi_2$ loses the $\text{est}(R^\circ)$ at every step — where (13.3.3a)'s loser is $\text{nil}$ |

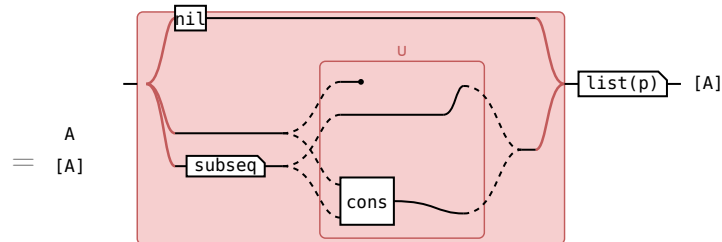
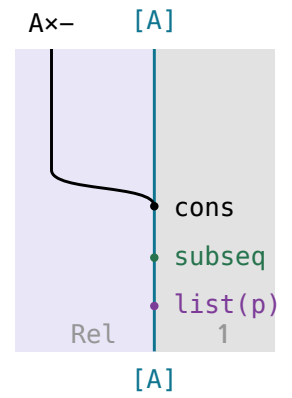
$$\alpha \text{ subseq list}(p) = F(\text{subseq list}(p))S$$

**circuit** — the fork is  $F([A]) = 1 + A \times [A]$ : nil above, the pair below

**Hinze–Marsden**

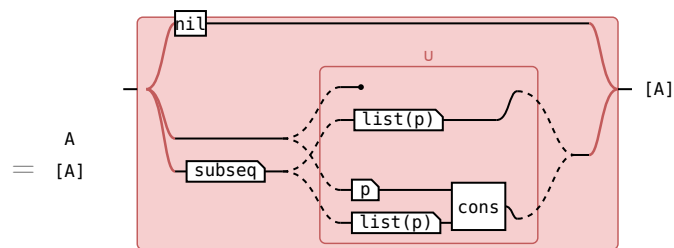
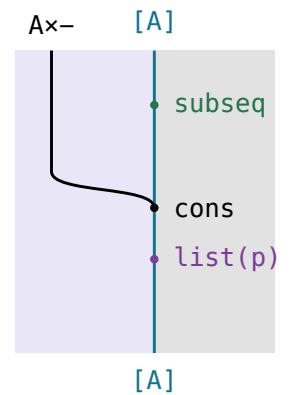


$\alpha \text{ subseq list}(p)$



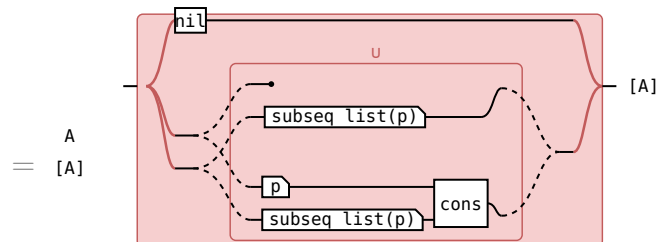
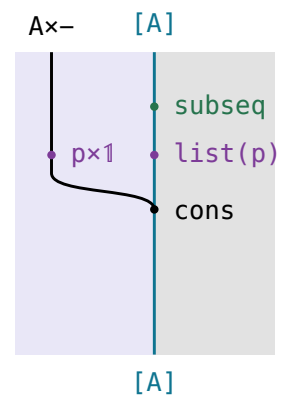
$F(\text{subseq}) [\text{nil}, \pi_2 \cup \text{cons}] \text{list}(p)$

defining equation



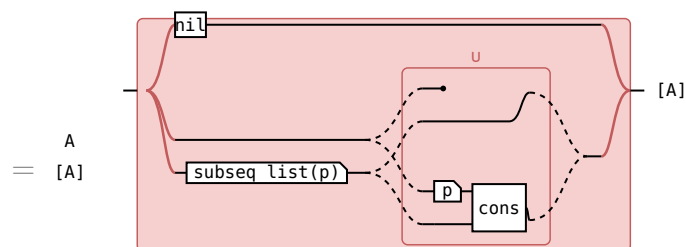
$F(\text{subseq}) [\text{nil}, (1 \times \text{list}(p)) \pi_2 \cup (p \times \text{list}(p)) \text{cons}]$

$\text{list}(p)$  through  $\text{cons}$ ,  $\pi_2$  natural



$[\text{nil}, (1 \times (\text{subseq list}(p))) \pi_2 \cup (p \times (\text{subseq list}(p))) \text{cons}]$

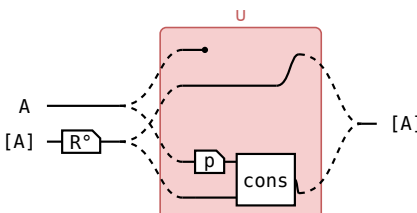
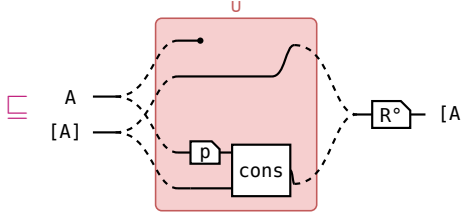
relator,  $\text{subseq}$  entire



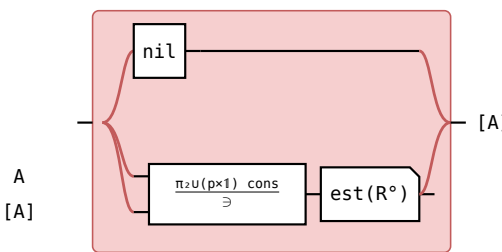
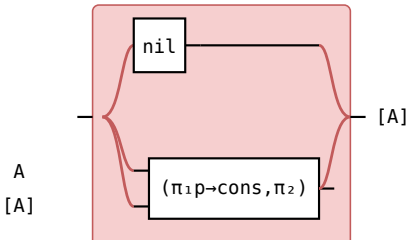
$F(\text{subseq list}(p))S$

$\text{subseq list}(p)$  entire

(11.6.1a) reads that off as  $\text{subseq list}(p) = (S)$  — (13.3.3b)'s chain with  $\pi_2$  for  $\rightarrow \text{nil}$ . Here (11.4.2a) is blocked for the same reason, and  $\pi_2$  is the branch that carries  $\text{list}(p)$  out by naturality ((12.1c) at  $\exists := \text{list}(p)$ ).

| $(1 \times R^\circ)(\pi_2 \cup (p \times 1) \text{ cons}) \sqsubseteq (\pi_2 \cup (p \times 1) \text{ cons}) R^\circ$ |                                                                                              |                                                                  | 3.3.5c) |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------|---------|
| formula                                                                                                               | — the cons branch of $F(R^\circ) S \sqsubseteq SR^\circ$                                     | reason                                                           |         |
|                                                                                                                       | $(1 \times R^\circ) \pi_2 = \pi_2 R^\circ$                                                   | $\pi_2$ natural (12.1c) — the right side is an EQUALITY here     |         |
|                                                                                                                       | $(1 \times R^\circ)(p \times 1) \text{ cons} \sqsubseteq (p \times 1) \text{ cons } R^\circ$ | (13.3.3c)'s cons branch, unchanged                               |         |
|                                      | $(1 \times R^\circ)(\pi_2 \cup (p \times 1) \text{ cons})$                                   |                                                                  |         |
|                                      | $(\pi_2 \cup (p \times 1) \text{ cons}) R^\circ$                                             | union (13.2c) at $X := \pi_2$ , $Y := (p \times 1) \text{ cons}$ |         |

$F(R^\circ) S \sqsubseteq SR^\circ$ , the nil branch again  $\text{nil} \sqsubseteq \text{nil } R^\circ$ . The first step is an equality: (13.3.3c) buys its  $\multimap \text{nil}$  branch with  $\text{nil } R^\circ = \text{nil}$ , and  $\pi_2$  needs only its naturality square.

| $\frac{S}{\sqsubseteq} \text{est}(R^\circ) = [\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)]$ |                                                                                          |                                            | 3.3.5d) |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------|---------|
| formula                                                                                              |                                                                                          | reason                                     |         |
| $F([A]) \sqsubseteq E([A]) \text{est}(R^\circ) [A]$                                                  | $\frac{S}{\sqsubseteq} \text{est}(R^\circ)$                                              |                                            |         |
| $=$<br>           | $[\text{nil}, \frac{\pi_2 \cup (p \times 1) \text{ cons}}{\supset} \text{est}(R^\circ)]$ | (13.3.3d)'s first two steps                |         |
| $=$<br>           | $[\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)]$                                 | $1 \sqsubseteq \pi_2 R \text{ cons}^\circ$ |         |

at  $(a, xs)$  the set is  $\{xs\}$  where  $p$  fails on  $a$  and  $\{xs, \text{cons}(a, xs)\}$  where it holds, and  $xs$  loses the second — (13.1a) at a two-element set. The head is dropped, not the whole tail: that is the one place  $\pi_2$  shows against (13.3.3d)'s  $\multimap \text{nil}$ .

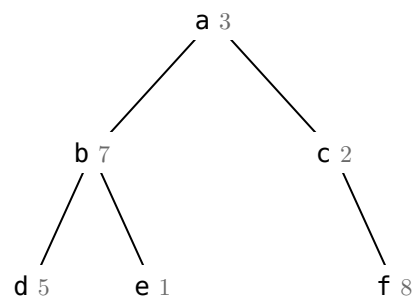
| $\text{filter}(p) = \llbracket \text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2) \rrbracket$ Ex 7.41, $R$ a preorder                                                                                                                                    |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit — one wire, its type written along it                                                                                                                                                                                                              | Hinze–Marsden |
| $[A] \rightarrow \boxed{\frac{\text{subseq list}(p)}{\supset}} E([A]) \rightarrow \boxed{\text{est}(R^\circ)} [A]$ $\text{filter}(p) \triangleq \frac{\text{subseq list}(p)}{\supset} \text{est}(R^\circ)$ (12a)                                           |               |
| $[A] \rightarrow \boxed{\frac{\llbracket S \rrbracket}{\supset}} E([A]) \rightarrow \boxed{\text{est}(R^\circ)} [A]$ $= \frac{\llbracket S \rrbracket}{\supset} \text{est}(R^\circ)$ $\text{subseq list}(p) = \llbracket S \rrbracket \text{ — (13.3.5b)}$ |               |
| $[A] \rightarrow \boxed{\llbracket \frac{S}{\supset} \text{est}(R^\circ) \rrbracket} [A]$ $\llbracket \frac{S}{\supset} \text{est}(R^\circ) \rrbracket$ (13.3.2a) at $R^\circ$ , whose hypothesis $F(R^\circ) S \subseteq SR^\circ$ is (13.3.5c)           |               |
| $[A] \rightarrow \boxed{\llbracket \text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2) \rrbracket} [A]$ $= \llbracket \text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2) \rrbracket$ (13.3.5d)                                                          |               |
| $\llbracket \text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2) \rrbracket$ entire and $\text{filter}(p)$ simple $=$ an entire relation below a simple one equals it; the simplicity is (12a)'s Ex 7.41, not a law of this note                           |               |

$(\text{subseq list}(p))^\circ (\text{subseq list}(p)) \cap R \cap R^\circ \subseteq 1$  fails — two  $p$ -subsequences of one list can be of equal length and different — so §13.3.3's uniqueness argument does not transfer. What survives  $\text{est}(R^\circ)$  is the one keeping **exactly** the passing elements: a subsequence that drops a passing element is beaten by the one that keeps it.

## §13.4. Planning a company party

| definition                                                                               | type                                              | note                                                                                           | (13.4a) |
|------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------|---------|
| $\text{tree } A ::= \text{node } (A, [\text{tree } A])$                                  | $A \rightarrow A$                                 | The company hierarchy: an employee, and the list of subtrees under them.                       |         |
| $F(A, B) = A \times [B]$                                                                 | $A \times A \rightarrow A$                        | The base functor <b>tree</b> folds: an employee beside the recursive position, one layer deep. |         |
| rating                                                                                   | $A \rightarrow \text{Real}$                       | What one employee is worth as a guest.                                                         |         |
| $\text{cost} \triangleq \text{list}(\text{rating}) \text{ sum}$                          | $[A] \rightarrow \text{Real}$                     | What a guest list is worth.                                                                    |         |
| $R \triangleq \text{cost} \leq \text{cost}^\circ$                                        | $[A] \rightarrow [A]$                             | The preorder the guest list is maximised over.                                                 |         |
| $\text{choose} \triangleq \pi_1 \cup \pi_2$                                              | $[A] \times [A] \rightarrow [A]$                  | Takes one of the two parties a subtree returns.                                                |         |
| $\text{include} \triangleq (1 \times (\text{list}(\pi_2) \text{ concat})) \text{ cons}$  | $F(A, [A] \times [A]) \rightarrow [A]$            | The party that invites the root, which puts every immediate subtree's root out. A map.         |         |
| $\text{exclude} \triangleq (1 \times (\text{list}(\text{choose}) \text{ concat})) \pi_2$ | $F(A, [A] \times [A]) \rightarrow [A]$            | The party that leaves the root out, so each subtree is free to choose. Not a map.              |         |
| $S \triangleq (\text{include}, \text{exclude})$                                          | $F(A, [A] \times [A]) \rightarrow [A] \times [A]$ | The algebra: one step returns both parties of a subtree at once.                               |         |
| $\text{party} \triangleq (S) \text{ choose}$                                             | $\text{tree } A \rightarrow [A]$                  | Every guest list the president's ruling allows.                                                |         |
| the specification<br>$\frac{\text{party}}{\exists} \text{ est}(R^\circ)$                 | $\text{tree } A \rightarrow [A]$                  | A guest list of greatest total conviviality.                                                   |         |

## §13.4.1. include and exclude



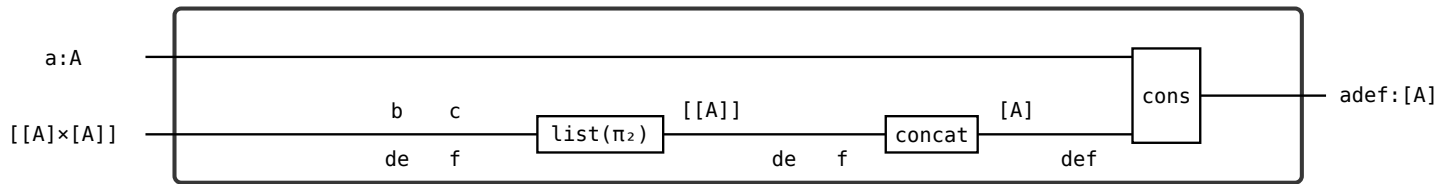
the small grey number is the employee's rating

| node    | the algebra's input<br>$A \times [[A] \times [A]]$ | include                                         | exclude                                | (13.4.1b) |
|---------|----------------------------------------------------|-------------------------------------------------|----------------------------------------|-----------|
| d, e, f | $(d, [])$                                          | $[d]$                                           | $[]$                                   |           |
| b       | $(b, ([d], []), ([e], []))$                        | $[b]$                                           | $[d, e], [d], [e], []$                 |           |
| c       | $(c, ([f], []))$                                   | $[c]$                                           | $[f], []$                              |           |
| a       | $(a, ([b], \dots), ([c], \dots))$                  | $[a] \# \text{exclude}(b) \# \text{exclude}(c)$ | $\text{choose}(b) \# \text{choose}(c)$ |           |

with those ratings,  $\text{est}(R^\circ)$  keeps  $([b]=7, [d, e]=6)$  at **b**, where **include** wins, and  $([c]=2, [f]=8)$  at **c**, where **exclude** wins, so at the root **include**  $= [a, d, e, f] = 17$  beats **exclude**  $= [b, f] = 15$ , and **choose** takes 17.

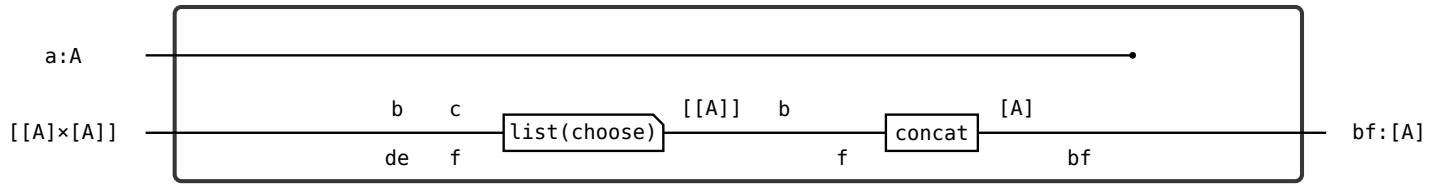
include=

(13.4.1c)



exclude=

(13.4.1d)



$\text{list}(\text{choose}) \quad [[d], []], ([e], [])$   
 1st item  $([d], [])$  choose  $\rightarrow [d]$  or  $[]$   
 2nd item  $([e], [])$  choose  $\rightarrow [e]$  or  $[]$   
 2x2=4 combinations:  $[[d], [e]] \quad [[d], []] \quad [[]], [e]] \quad [[]], []]$   
 concat flattens:  $[d, e] \quad [d] \quad [e] \quad []$

(13.4.1e)

### §13.4.2. $\text{list}((R \times R)^\circ)$

$[ ([b], [d, e]) , ([c], [f]) ]$   
 $\downarrow$   
 $[ ([b], [d]) , ([c], []) ]$

$\text{list}((R \times R)^\circ)$ , elementwise

(13.4.2a)

$([b], [d, e])$   
 $\begin{cases} \text{exclude: the best party in b's subtree when b does not come} \\ \text{include: the best party in b's subtree when b comes} \end{cases}$

| element              | 1st component<br>include                      | 2nd component<br>exclude                                                                                |
|----------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 1st, $([b], [d, e])$ | $[b] R^\circ [b]$<br>$7 \geq 7$ , reflexivity | $[d, e] R^\circ [d]$<br>$6 \geq 5$ , and $([b], [d, e])$ is the pair $\text{est}(R^\circ)$ keeps at $b$ |
| 2nd, $([c], [f])$    | $[c] R^\circ [c]$<br>$2 \geq 2$ , reflexivity | $[f] R^\circ []$<br>$8 \geq 0$ , and $([c], [f])$ is the pair $\text{est}(R^\circ)$ keeps at $c$        |

(13.4.2b)

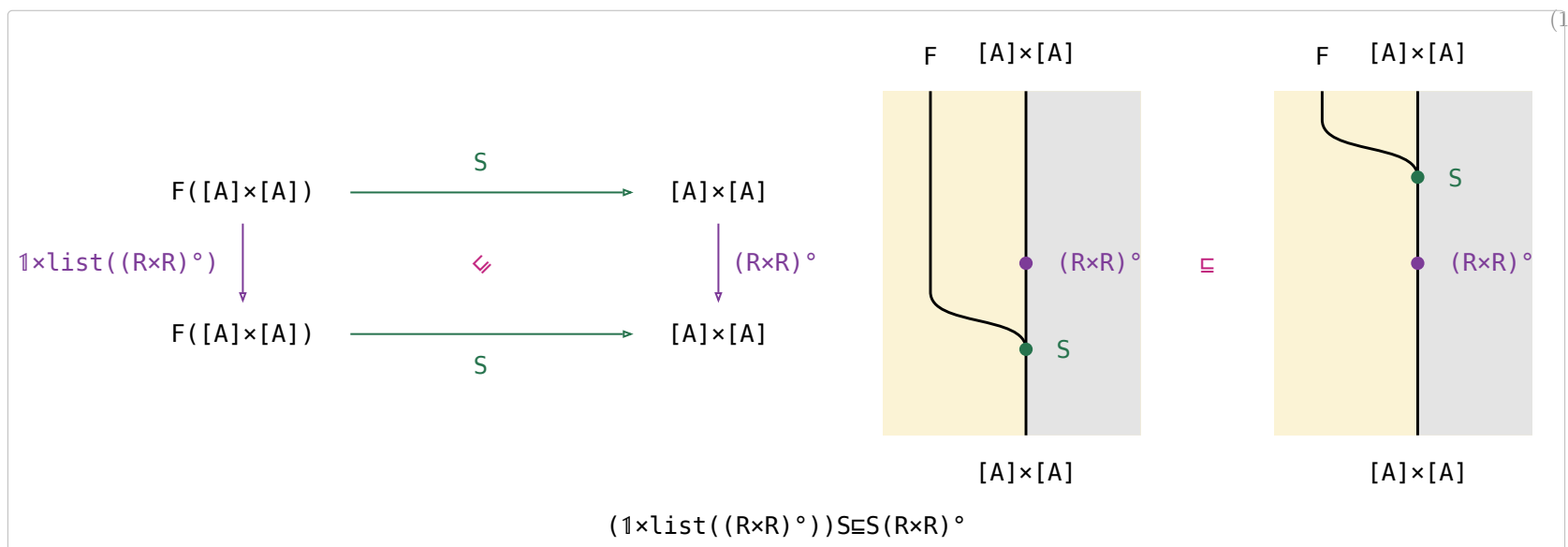
the first component of  $1 \times \text{list}((R \times R)^\circ)$  is the root employee.

$R \triangleq \text{cost} \leq \text{cost}^\circ$  is a preorder, so the components that do not move are related by reflexivity —  $(R \times R)^\circ$  demands a relation in **both** components, it cannot skip one.

$R$  compares cost only, not contents:  $[f] R^\circ []$  is legal though  $[f]$  has an element  $[]$  does not. That is why the three leaves of §13.4.3's proof all reduce to " $\text{cost}$  is a sum".

### §13.4.3. $(1 \times \text{list}((R \times R)^\circ)) S \subseteq S(R \times R)^\circ$ — $S : F([A] \times [A]) \rightarrow [A] \times [A]$ monotonic on $(R \times R)^\circ$

(13.4.3a)



$$\begin{array}{c}
 \frac{\langle S \rangle \text{ choose}}{\ni} \\
 \text{tree } A \xrightarrow{\quad} E[A] \quad = \quad \text{tree } A \xrightarrow{\frac{\langle S \rangle}{\ni}} E([A] \times [A]) \xrightarrow{E(\text{choose})} E[A] \\
 \frac{\langle S \rangle \text{ choose}}{\ni} = \frac{\langle S \rangle}{\ni} E(\text{choose}) \quad (10b), \text{ absorption}
 \end{array}$$

| $  \begin{array}{l}  S \triangleq (\text{include}, \text{exclude}) \\  (1 \times \text{list}((R \times R)^\circ)) S \sqsubseteq S(R \times R)^\circ \\  (1 \times \text{list}((R \times R)^\circ)) \text{include} \sqsubseteq \text{include } R^\circ \\  (1 \times \text{list}((R \times R)^\circ)) \text{exclude} \sqsubseteq \text{exclude } R^\circ  \end{array}  $ |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                                                                 | Hinze–Marsden |
|                                                                                                                                                                                                                                                                                                                                                                         |               |
| $\sqsubseteq$                                                                                                                                                                                                                                                                                                                                                           |               |
| $\sqsubseteq$                                                                                                                                                                                                                                                                                                                                                           |               |
| $\sqsubseteq$                                                                                                                                                                                                                                                                                                                                                           |               |

- $g$   $(R \times R)^\circ g \in g R^\circ$   
 $g := \pi_2$  is  $(R \times R)^\circ \pi_2 = (\text{Dom}(\pi_1 R^\circ)) \pi_2 R^\circ \subseteq \pi_2 R^\circ$ ,  $g := \pi_1$  its mirror  $(\text{Dom}(\pi_2 R^\circ)) \pi_1 R^\circ \subseteq \pi_1 R^\circ$  — 1 and 4 of (11.1.2b), then  $\text{Dom} \subseteq 1$ ;  $g := \text{choose} \triangleq \pi_1 \cup \pi_2$  is the union of the two (13.2c). **list** monotonic, (11a)  
 $\text{concat}$   $\text{list}(R^\circ) \text{concat} \subseteq \text{concat } R^\circ$   
 a LEAF: no law above it. **cost** is a sum, so  $\text{cost}(\text{concat } xss) = \text{sum}(\text{list}(\text{cost}) xss)$  and a cheaper part makes a cheaper whole. B&dM's exercise.  
 $h$   $(1 \times R^\circ) h \subseteq h R^\circ$   
 a LEAF:  $\text{cost}(\text{cons}(a, xs)) = \text{rating}(a) + \text{cost}(xs)$ , so a cheaper tail makes a cheaper list. B&dM's exercise. For  $h := \pi_2$  it is an EQUALITY  $(1 \times R^\circ) \pi_2 = (\text{Dom}(\pi_1)) \pi_2 R^\circ = \pi_2 R^\circ$ :  $\pi_1$  is a map, hence entire, so  $\text{Dom}(\pi_1) = 1$  — (5c)

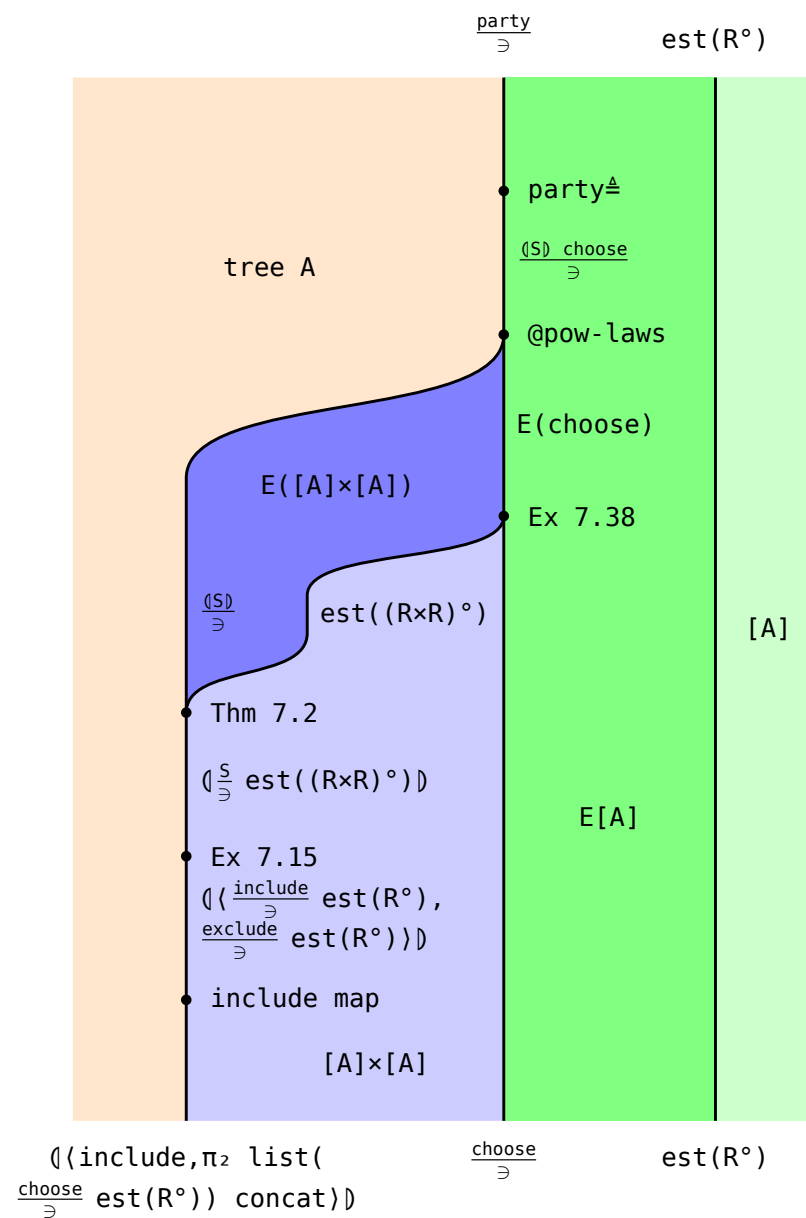
|                                                |               |             |
|------------------------------------------------|---------------|-------------|
| $(1 \times (\text{list}(g) \text{ concat})) h$ | $g$           | $h$         |
| <b>include</b>                                 | $\pi_2$       | <b>cons</b> |
| <b>exclude</b>                                 | <b>choose</b> | $\pi_2$     |



|                                                                           |                                                                                                                                   |           |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------|
| $R \triangleq \{(\emptyset, \emptyset)\} : \{0, 1\} \rightarrow \{0, 1\}$ | not entire — undefined at 1                                                                                                       | (13.4.3d) |
| $(R \times R) \text{ choose at } (\emptyset, 1)$                          | nothing — $R \times R$ needs BOTH components related, and $R$ is undefined at 1                                                   |           |
| choose $R$ at $(\emptyset, 1)$                                            | $\emptyset$ — choose picks the first component $\emptyset$ , and $R \emptyset \emptyset$ holds                                    |           |
| $((\emptyset, 1), \emptyset)$                                             | in choose $R$ , not in $(R \times R) \text{ choose}$                                                                              |           |
| $(R \times R) \text{ choose} = \text{choose } R$                          | iff $R$ is entire — $\text{Dom} \sqsubseteq 1$ is the g row's only inequality step, and it is an equality iff $\text{Dom}(R) = 1$ |           |

#### §13.4.4. The derivation

| $\frac{\text{party}}{\exists} \text{est}(R^\circ) \ni ((\text{include}, \pi_2 \text{ list}(\frac{\text{choose}}{\exists} \text{est}(R^\circ)) \text{ concat})) \frac{\text{choose}}{\exists} \text{est}(R^\circ)$ | $\frac{\text{choose}}{\exists} \text{est}(R^\circ)$                                          | 3.4.4a)                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>formula</b> — one wire, tree A to [A]; the pair [A]×[A] is where it runs as two                                                                                                                                | <b>Hinze–Marsden</b><br>— outside the ( ) above, inside it below; a fork drawn at one branch | <b>reason</b>                                         |
|                                                                                                                                                                                                                   |                                                                                              |                                                       |
| $\frac{\text{party}}{\exists} \text{est}(R^\circ)$<br>                                                                                                                                                            |                                                                                              | def. party                                            |
|                                                                                                                                                                                                                   |                                                                                              | (10b), absorption                                     |
|                                                                                                                                                                                                                   |                                                                                              | Ex 7.38 at choose, $R^\circ$<br>(13.4.3c)'s g row     |
|                                                                                                                                                                                                                   |                                                                                              | Theorem 7.2 at $(R \times R)^\circ$ (13.4.3a)         |
|                                                                                                                                                                                                                   |                                                                                              | Ex 7.15, the fork splits                              |
|                                                                                                                                                                                                                   |                                                                                              | include a map, $\text{est}(R^\circ)$ into each branch |



include map  $\sqsubseteq$  include a map,  $\text{est}(R^\circ)$  into each branch

Ex 7.15  $\sqsubseteq$  the fork splits

Thm 7.2  $\sqsubseteq (1 \times \text{list}((R \times R)^\circ)) \text{S} \sqsubseteq \text{S}(R \times R)^\circ$

Ex 7.38  $\sqsubseteq (R \times R)^\circ \text{choose} \sqsubseteq \text{choose } R^\circ$

@pow-laws = absorption

party<sup>A</sup> = definition of party

## §13.5. Shortest paths on a cylinder

| definition                                                                                                  | type                                                     | note                                                                                         | (13.5a) |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------|---------|
| $F(A, X) = A + A \times X$                                                                                  | $\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$ | The base functor: one square of the new column, alone or in front of the path so far.        |         |
| $L = \text{list}^+$                                                                                         | $\mathcal{A} \rightarrow \mathcal{A}$                    | A path is a non-empty list of squares, one per column crossed.                               |         |
| $\alpha$ the initial algebra                                                                                | $F(A, L\mathcal{A}) \rightarrow L\mathcal{A}$            | $\alpha = [\text{wrap}, \text{cons}]$ : a path is started by one square, or extended by one. |         |
| $N$<br>the n-tuple relator                                                                                  | $\mathcal{A} \rightarrow \mathcal{A}$                    | One component per row of a column, so the fold carries $n$ answers at once.                  |         |
| $R \triangleq \text{sum} \leq \text{sum}^\circ$                                                             | $L \text{ Nat} \rightarrow L \text{ Nat}$                | The cost of a path, which the cheapest minimises.                                            |         |
| setify                                                                                                      | $NA \rightarrow EA$                                      | Forgets which row a component came from, so the $n$ answers become one set.                  |         |
| moves                                                                                                       | $NA \rightarrow E(NA)$                                   | The three columns a path may step to: rotated up, unrotated, rotated down.                   |         |
| trans                                                                                                       | $E(NA) \rightarrow N(EA)$                                | Transposes a set of tuples, so every row collects its own paths.                             |         |
| zip                                                                                                         | $F(NA, NB) \rightarrow NF(A, B)$                         | Commutates $N$ with the base functor, pairing each new square with its own row's paths.      |         |
| $\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$                                                        | $F(A, EB) \rightarrow E(F(A, B))$                        | Turns a square paired with a set of paths into the set of extensions of those paths.         |         |
| $\text{generate} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp } P(\alpha))$ | $F(NA, N(E(LA))) \rightarrow N(E(LA))$                   | One fold step: extends every path of every row by the new column.                            |         |
| $\text{paths} \triangleq (\text{generate}) \text{ setify union}$                                            | $L \text{ N Nat} \rightarrow E(L \text{ Nat})$           | Every path across the cylinder.                                                              |         |
| the specification<br>$\text{paths est}(R)$                                                                  | $L \text{ N Nat} \rightarrow L \text{ Nat}$              | A cheapest path from the entry side to the exit side.                                        |         |

§13.5.1.  $N(E(L \ A))$ 

$$\begin{array}{rcl} 1 & 5 & \\ 2 & 6 & \\ 3 & 7 & [(1,2,3,4), (5,6,7,8)] : L \text{ N Nat} \\ 4 & 8 & \end{array} \quad (13.5.1a)$$

$$(\text{generate})[(5,6,7,8)] = (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}) \quad (13.5.1b)$$

$$\begin{aligned} &\text{generate}((1,2,3,4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = \\ &(\{[1,5], [1,6], [1,8]\}, \\ &\{[2,5], [2,6], [2,7]\}, \\ &\{[3,6], [3,7], [3,8]\}, \\ &\{[4,5], [4,7], [4,8]\}) \end{aligned}$$

|     |                                                        |           |
|-----|--------------------------------------------------------|-----------|
| $N$ | the 4-tuple, one component per row of the entry column | (13.5.1c) |
| $E$ | the set of paths that can start in that row            |           |
| $L$ | a path is the non-empty list of the squares it crosses |           |

row 1's set has no  $[1,7]$  — from row 1 only rows 4, 1 and 2 are reachable — and it does have  $[1,8]$ , because the cylinder glues the bottom row to the top. That is **moves trans  $N(\text{union})$**  of (13.5a): component  $k$  collects the paths of rows  $k-1$ ,  $k$ ,  $k+1$ .

## §13.5.2. The derivation

| <b>paths est(R) <math>\exists(Q)</math> setify est(R)</b><br>B&dM §7.4, p. 182; $Q$ is (13.5.2c)'s algebra                                                                                                                                                                                                      |               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                         | Hinze–Marsden |
| $L \ N \ Nat \rightarrow \boxed{\text{paths}} \rightarrow \boxed{\text{est}(R)} \rightarrow L \ Nat$<br>$\text{paths est}(R)$                                                                                                                                                                                   |               |
| $L \ N \ Nat \rightarrow \boxed{(\text{generate})} \rightarrow \boxed{\text{setify}} \rightarrow \boxed{\text{union}} \rightarrow \boxed{\text{est}(R)} \rightarrow L \ Nat$<br>$= (\text{generate}) \text{ setify union est}(R)$<br>(13.5a) at $\text{paths}$                                                  |               |
| $L \ N \ Nat \rightarrow \boxed{(\text{generate})} \rightarrow \boxed{\text{setify}} \rightarrow \boxed{P(\text{est}(R))} \rightarrow \boxed{\text{est}(R)} \rightarrow L \ Nat$<br>$\sqsupset (\text{generate}) \text{ setify } P(\text{est}(R)) \text{ est}(R)$<br>(13.1b) at $\text{union}$ , $R$ a preorder |               |
| $L \ N \ Nat \rightarrow \boxed{(\text{generate})} \rightarrow \boxed{N(\text{est}(R))} \rightarrow \boxed{\text{setify}} \rightarrow \boxed{\text{est}(R)} \rightarrow L \ Nat$<br>$\sqsupset (\text{generate})N(\text{est}(R)) \text{ setify est}(R)$<br>$\text{setify lax natural}$                          |               |
| $L \ N \ Nat \rightarrow \boxed{(Q)} \rightarrow \boxed{\text{setify}} \rightarrow \boxed{\text{est}(R)} \rightarrow L \ Nat$<br>$\sqsupset (Q) \text{ setify est}(R)$<br>(11.4.2a) at (13.5.2b)                                                                                                                |               |

(13.5.2b)

| <b>generate <math>N(\text{est}(R)) \sqsupseteq F(1, N(\text{est}(R)))Q</math></b><br>the fusion condition of (13.5.2a)'s last step                                          |               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                     | Hinze–Marsden |
| $F(NA, N(E(LA))) \rightarrow \text{generate} \rightarrow N(\text{est}(R)) \rightarrow N(LA)$<br>generate $N(\text{est}(R))$                                                 |               |
| $F(NA, N(E(LA))) \rightarrow F(1, N(\text{est}(R))) \rightarrow Q \rightarrow N(LA)$<br>$\sqsupseteq F(1, N(\text{est}(R)))Q$<br>(7.13), then zip, trans, moves lax natural |               |

(7.13) is  $F(1, \text{est}(R)) \alpha \sqsubseteq_{\text{cp}} P(\alpha) \text{ est}(R)$ , (13.3.1c) at the map  $\alpha$  with  $\frac{F(1, \exists)\alpha}{\exists} = \text{cp } P(\alpha)$ : extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.

(13.5.2c)

| <b><math>Q = [N(\text{wrap}), (1 \times \text{moves trans } N(\text{est}(R))) \text{ zip}' N(\text{cons})]</math></b><br>the algebra (13.5.2a)'s last step folds                                                                                           |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit — the fork is $F(NA, N(LA)) = NA + NA \times N(LA)$                                                                                                                                                                                                | Hinze–Marsden |
| $F(NA, N(LA)) \rightarrow Q \rightarrow N(LA)$<br>$Q$                                                                                                                                                                                                      |               |
| $F(NA, N(LA)) \rightarrow F(1, \text{moves trans } N(\text{est}(R))) \rightarrow \text{zip} \rightarrow N(\alpha) \rightarrow N(LA)$<br>$= F(1, \text{moves trans } N(\text{est}(R))) \text{ zip}' N(\alpha)$<br>the fusion condition read as a definition |               |
| $NA$<br>$=$<br>$NA$<br>$N(LA)$<br>$[N(\text{wrap}), (1 \times \text{moves trans } N(\text{est}(R))) \text{ zip}' N(\text{cons})]$<br>$\text{zip} = 1 + \text{zip}', \alpha = [\text{wrap}, \text{cons}]$                                                   |               |

## §13.6. The security van problem

| definition                                                                                                                   | type                                                                            | note                                                                                            | (13.6a) |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------|
| $R \triangleq \text{length} \leq \text{length}^\circ$                                                                        | $[[\text{Int}]] \rightarrow [[\text{Int}]]$                                     | The order the schedule is minimised over: fewer van visits is better.                           |         |
| $\text{ceiling} \triangleq \frac{\text{prefix sum}}{\ni} \text{est}(\geq)$                                                   | $[\text{Int}] \rightarrow \text{Int}$                                           | The highest the bank's balance reaches over a stretch of transactions.                          |         |
| $\text{floor} \triangleq \frac{\text{prefix sum}}{\ni} \text{est}(\leq)$                                                     | $[\text{Int}] \rightarrow \text{Int}$                                           | The lowest it reaches, so $\text{ceiling} - \text{floor}$ is the cash the stretch has to carry. |         |
| <b>secure</b><br>the coreflexive on $x$ with<br>$\text{bmax}(\text{ceiling } x, \text{ceiling } x - \text{floor } x) \leq N$ | $[\text{Int}] \rightarrow [\text{Int}]$                                         | The stretches one van visit can serve: some starting reserve keeps the cash between 0 and N.    |         |
| <b>ok</b><br>the coreflexive on $(a, xs)$ with $xs$ non-empty and $[a] \# \text{head } xs$ secure                            | $\text{Int} \times [[\text{Int}]] \rightarrow \text{Int} \times [[\text{Int}]]$ | The test the final program runs, in place of <b>old</b> 's <b>secure</b> .                      |         |
| $\text{new} \triangleq (\text{wrap} \times 1) \text{ cons}$                                                                  | $\text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$                   | Calls the van: the transaction opens a segment of its own.                                      |         |
| $\text{glue} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } (\text{cons} \times 1) \text{ cons}$                    | $\text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$                   | Puts the transaction on the front of the first segment.                                         |         |
| $\text{old} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } ((\text{cons } \text{secure}) \times 1) \text{ cons}$    | $\text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$                   | <b>glue</b> restricted to a first segment that stays secure.                                    |         |
| $\text{partition} = ([\text{nil}, \text{newuglue}])$                                                                         | $[\text{Int}] \rightarrow [[\text{Int}]]$                                       | Every splitting of the transactions into consecutive segments.                                  |         |
| $S \triangleq [\text{nil}, \text{newuold}]$                                                                                  | $1 + \text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$               | The algebra whose fold is every splitting of the transactions into secure segments.             |         |
| $\text{partition list}(\text{secure}) = (S)$                                                                                 | $[\text{Int}] \rightarrow [[\text{Int}]]$                                       | Fusion: keeping only secure segments is what turns <b>glue</b> into <b>old</b> .                |         |
| $H \triangleq (\text{head prefix}^\circ \text{ head}^\circ) \cup (\text{nil}^\circ \text{ nil})$                             | $[[\text{Int}]] \rightarrow [[\text{Int}]]$                                     | One schedule's first segment is a prefix of the other's, or both are empty.                     |         |
| $R; H \triangleq R \cap (R^\circ \Rightarrow H)$                                                                             | $[[\text{Int}]] \rightarrow [[\text{Int}]]$                                     | The refinement of $R$ that makes both halves of $S$ monotonic.                                  |         |
| $ R  \triangleq R \cap R^\circ$                                                                                              | $[[\text{Int}]] \rightarrow [[\text{Int}]]$                                     | The strict part $R$ splits into: $R; H =  R  \cup (R \cap H)$ .                                 |         |
| the specification<br>$\frac{\text{partition list}(\text{secure})}{\ni} \text{est}(R)$                                        | $[\text{Int}] \rightarrow [[\text{Int}]]$                                       | A schedule with the fewest secure segments.                                                     |         |

$$\frac{\text{partition list}(\text{secure})}{\ni} \text{est}(R) \quad (11.4.2a), \text{ secure prefix} \ni \text{prefix secure} \quad \frac{(S)}{\ni} \text{est}(R) \quad (7.14) \text{ holds, } (7.15) \text{ FALSE; } R; H \in R \quad \frac{(S)}{\ni} \text{est}(R; H) \quad (13.6b)$$

$$\text{Theorem 7.2, } S \text{ monotonic on } R; H \text{ by (7.16) and (7.17)} \quad \frac{(S)}{\ni} \text{est}(R; H) \quad \text{old} \ni \text{new} \quad (R; H)^\circ \quad ([\text{nil}, (\text{ok} \rightarrow \text{glue}, \text{new})])$$

$$\frac{(1 \times (R; H)) \text{ old}}{R; H = |R| \cup (R \cap H), \text{ } \cup \text{ distributes}} \quad \frac{(1 \times |R|) \text{ old} \cup (1 \times (R \cap H)) \text{ old}}{(7.19) \text{ and } (7.20) \text{ on } |R|, (7.21) \text{ on } R \cap H} \quad \text{new } (R \cap H) \cup \text{old } (R \cap H)$$

$$X \cap Y \subseteq X; Y, \text{ converses} \quad \frac{}{(newuold)(R; H)}$$

(7.15)  $(1 \times R) \circ \text{old} \sqsubseteq (\text{new} \circ \text{old})R$  is FALSE — the shorter partition need not stay secure — and that is what refining  $R$  to  $R;H$  buys; the second chain is (7.17), and (7.16) rests the same way on (7.18)  $(1 \times T) \text{ new} \sqsubseteq \text{new } H$ .



## §14. Thinning Algorithms

### §14.1. Thinning

For  $Q : A \rightarrow A$ ,  $\text{thin } Q \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : EA \rightarrow EA$  (8.1).

$ys \text{ (thin } Q) \text{ } xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \text{ } Q \text{ } a)$

(14.1a)

| the law                                                                                                                                                                   | what it says                                                                                                              | (14.1b) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------|
| $X \subseteq \frac{S}{\exists} \text{ thin } Q \iff X \subseteq S \text{ and } S^\circ X \subseteq Q^\circ \epsilon$                                                      | the universal property: everything kept is an $S$ -value, and every $S$ -value has a $Q$ -lower bound among the kept ones |         |
| $Q \subseteq R \implies \text{thin } Q \subseteq \text{thin } R$                                                                                                          | the fewer pairs $Q$ relates, the fewer subsets count as thinnings                                                         |         |
| $1 \subseteq \text{thin } Q$ , and $\text{thin } Q$ is a preorder if $Q$ is                                                                                               | keeping everything is always a legal thinning                                                                             |         |
| $\text{est}(R) = \text{thin } Q \text{ est}(R) \quad Q \subseteq R, \text{ both preorders}$                                                                               | <b>thin-introduction</b> : thinning first cannot lose an $R$ -minimum                                                     |         |
| $\text{thin } Q \supseteq \text{est}(Q) \quad \frac{1}{\exists} \quad (8.2)$                                                                                              | <b>thin-elimination</b> : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on     |         |
| $\frac{S}{\exists} \text{ thin } Q \supseteq \frac{S}{\exists} \text{ est}(R) \quad \frac{1}{\exists}$<br>(8.3), $R \cap (S^\circ S) \subseteq Q$                         | the usable variant: $R$ need only refine $Q$ between values $S$ gives one argument                                        |         |
| $\text{union thin } Q \supseteq P(\text{thin } Q) \text{ union} \quad (8.4)$                                                                                              | thinning each member set is a thinning of the union                                                                       |         |
| $Q \left( \frac{F(\exists, 1) S}{\exists} \text{ thin } Q \right) \subseteq \frac{Q(S)}{\exists} \text{ thin } Q$<br>Theorem 8.1, $S$ monotonic on $Q$                    | the <b>thinning theorem</b> : thinning at every step beats thinning only at the end                                       |         |
| $Q \left( \frac{F(\exists, 1) S}{\exists} \text{ thin } Q \right) \text{ est}(R) \subseteq \frac{Q(S)}{\exists} \text{ est}(R)$<br>Corollary 8.1, $Q \subseteq R$ as well | the same against the optimisation problem itself, by thin-introduction                                                    |         |

### §14.2. Paths in a layered network

$F(A, X) = A + A \times X$ ,  $L = \text{list}^+$  with initial algebra  $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$ .

$\text{wrapz} \triangleq (\text{wrap}, \text{zero})$ ,  $\text{consw } (a, (xs, n)) = (\text{cons } (a, xs), \text{wt } (a, \text{head } xs) + n)$ .

$\text{cost} \triangleq ([\text{wrapz}, \text{consw}]) \pi_2$ ,  $Q([\text{wrapz}, \text{consw}]) = (1, \text{cost})$ ,  $R \triangleq \text{cost} \leq \text{cost}^\circ$ .

$Q \triangleq R \cap (\text{head } \text{head}^\circ)$ ,  $S \triangleq F(1, \exists) \alpha$ ,  $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$ ,  $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$ ,  $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{ est}(R)$ .

(14.2a)

| the law                                                                                                                                                                                                                                                   | what it says                                                                                                             | (14.2b) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{Q(F(\exists, 1) \alpha)}{\exists} \text{ est}(R) = \frac{L(\exists)}{\exists} \text{ est}(R)$                                                                                                                                                      | the specification: a least-cost path across the network                                                                  |         |
| $F(\exists, Q) \alpha \subseteq F(\exists, 1) \alpha Q$                                                                                                                                                                                                   | $S$ is monotonic on $Q$ ; on $R$ it is not, since the next edge can cost arbitrarily much                                |         |
| $Q \left( \frac{F(\exists, \exists) \alpha}{\exists} \text{ thin } Q \right) \text{ est}(R) \subseteq \frac{Q(F(\exists, 1) \alpha)}{\exists} \text{ est}(R)$<br>Corollary 8.1                                                                            | thinning applies: one partial path per starting vertex is enough                                                         |         |
| $S \text{ head} \subseteq [1, \pi_2]$<br>$S \text{ head}$ simple, so $R \cap (S^\circ S) \subseteq Q$                                                                                                                                                     | the side condition of (8.3): between two paths $S$ builds from one argument, equal cost and equal head already means $Q$ |         |
| $\frac{F(\exists, \exists) \alpha}{\exists} \text{ thin } Q$<br>$\supseteq \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists)}{\exists} P(\alpha) \text{ est}(R) \right)$<br>(8.4), (8.3), $P \left( \frac{1}{\exists} \right) \text{ union} = 1$ | thin eliminated: split the algebra at $F(\exists, 1) F(1, \exists)$ , thin each part, one minimum per part               |         |
| $\frac{F(1, \exists)}{\exists} P(\alpha) \text{ est}(R) = [\text{wrap}, \text{step}]$                                                                                                                                                                     | that inner part read off the two branches of $\alpha$                                                                    |         |
| $\frac{Q(F(\exists, 1) \alpha)}{\exists} \text{ est}(R) \supseteq ([P(\text{wrap}), \text{cpl } P(\text{step})]) \text{ est}(R)$                                                                                                                          | the program: one fold, a best path per vertex of the current layer                                                       |         |

### §14.3. Implementing thin

(14.3a)

setify :  $[A] \rightarrow EA$ ,  $\text{cup} : EA \times EA \rightarrow EA$ ,  $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$ ,  $\text{listcp}(F) : F([A]) \rightarrow [FA]$ ,  $\text{sort } P \triangleq \text{setify}^\circ \text{ ordered } P$  for  $P$  a connected preorder.

$\text{thinlist } Q$  is any  $\text{thinlist } Q \subseteq \text{subseq}$  with  $\text{thinlist } Q \text{ setify} \subseteq \text{setify thin } Q$ ; one is  $\langle [\text{nil}, \text{bump } Q] \rangle$ ,  $\text{bump } Q (a, []) = [a]$ ,  $\text{bump } Q (a, [b] \# xs) = (b \ Q \ a \rightarrow [a] \# xs, a \ Q \ b \rightarrow [b] \# xs, [a] \# [b] \# xs)$ .

**Binary thinning** data:  $S = (f_1 p_1) \cup (f_2 p_2)$  with  $p_1, p_2$  coreflexive;  $Q$  a preorder with  $Q \subseteq R$  and both  $f_1 p_1, f_2 p_2$  monotonic on  $Q$ ;  $P$  a connected preorder with both  $f_1, f_2$  monotonic on  $P$ ;  $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$ .

| the law                                                                                                                                                                                   | what it says                                                                                                     | (14.3b) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------|
| $\text{thinlist } Q \text{ xs} = [\text{minlist } Q \text{ xs}]$<br>(8.5), $Q$ connected, $XS$ non-empty                                                                                  | what thinning should come to when it can: one element                                                            |         |
| $\text{sort } P \text{ thinlist } Q \subseteq \text{thin } Q \text{ sort } P$ (8.6)                                                                                                       | thinning a sorted list is a thinning of the set — this is what $\text{thinlist } Q \subseteq \text{subseq}$ buys |         |
| $\text{sort } P \text{ minlist } Q \subseteq \text{est}(Q)$ (8.7)                                                                                                                         | a minimum of the sorted list is a minimum of the set                                                             |         |
| $\text{sort}(f P f^\circ) \text{ list}(f) \subseteq P(f) \text{ sort } P$ (8.8)                                                                                                           | shunt a function through a sort                                                                                  |         |
| $\text{sort } P \text{ filter}(p) \subseteq E(p) \text{ sort } P$ (8.9), $p$ coreflexive                                                                                                  | filtering a sorted list sorts the restricted set                                                                 |         |
| $(\text{sort } P \times \text{sort } P) \text{ merge } P \subseteq \text{cup sort } P$ (8.10)                                                                                             | merging two sorted lists sorts their union                                                                       |         |
| $F(\text{sort } P) \text{ listcp}(F) \subseteq \text{cp}(F) \text{ sort}(FP)$<br>(8.11), $F$ linear                                                                                       | $\text{listcp}(F)$ is the list implementation of the cartesian product $\text{cp}(F)$                            |         |
| $F(\text{sort } P) \text{ listcp}(F) \text{ list}(f) \text{ filter}(p) \subseteq \frac{F(\exists) f p}{\exists} \text{ sort } P$<br>Lemma 8.1, $f$ monotonic on $P$ , $p$ coreflexive     | one sorted list built from sorted arguments, instead of a set built and then sorted                              |         |
| $\langle \text{listcp}(F) \langle g_1, g_2 \rangle \text{ merge } P \text{ thinlist } Q \rangle \text{ minlist } R$<br>$\subseteq \frac{Q \cup R}{\exists} \text{ est}(R)$<br>Theorem 8.2 | the <b>binary thinning theorem</b> : a fold on sorted lists of partial solutions, thinned at every step          |         |

### §14.4. The knapsack problem

(14.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}$ ,  $\text{value} \triangleq \text{list}(\text{vol}) \text{ sum}$ ,  $\text{weight} \triangleq \text{list}(\text{wt}) \text{ sum}$ .

$\text{subseq} = \langle [\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2] \rangle$ , within  $w$  the coreflexive on  $xs$  with  $\text{weight } xs \leq w$ .

$R \triangleq \text{value} \geq \text{value}^\circ$ ,  $Q \triangleq R \cap (\text{weight} \leq \text{weight}^\circ)$ ,  $P \triangleq R$ .

$FA = 1 + \text{Item} \times A$ ,  $\text{listcp}(F) = \text{wrap} + \text{cpr}$ ,  $g_1 \triangleq \text{list}([\text{nil}, \text{cons}]) \text{ filter}(\text{within } w) = [\text{list}(\text{nil}), h_1]$ ,  
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2]$ .

$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w)$ ,  $h_2 \triangleq \text{list}(\pi_2)$ .

| the law                                                                                                                                                             | what it says                                                                                              | (14.4b) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{subseq (within } w)}{\exists} \text{ est}(R)$                                                                                                          | the specification: a selection of greatest total value that stays within the capacity $w$                 |         |
| $\text{subseq (within } w) = \langle ([\text{nil}, \text{cons}] (\text{within } w)) \cup [\text{nil}, \pi_2] \rangle$<br>fusion, weights non-negative               | the selections that fit are themselves a fold, and the fold's algebra already has binary thinning's shape |         |
| $(1 \times R) (\text{cons (within } w)) \subseteq \text{cons (within } w) R$<br>FALSE                                                                               | not monotonic on $R$ : a selection of greater value need not still fit once one more item goes in         |         |
| $(1 \times Q) (\text{cons (within } w)) \subseteq \text{cons (within } w) Q$<br>$(1 \times Q) \pi_2 \subseteq \pi_2 Q$                                              | both halves are monotonic on $Q$ once ties in value are broken by weight                                  |         |
| $\langle \text{listcp}(F) \langle g_1, g_2 \rangle \text{ merge } R \text{ thinlist } Q \rangle \text{ minlist } R$<br>Theorem 8.2 at $P \triangleq R$ , $F$ linear | binary thinning, sorting in descending order of value                                                     |         |
| $\text{knapsack } w = \langle [\text{nil}, \text{cpr } \langle h_1, h_2 \rangle \text{ merge } R \text{ thinlist } Q] \rangle \text{ minlist } R$                   | the program; $\text{minlist } R$ becomes <b>head</b> , since packings come out in descending value        |         |

## §14.5. The paragraph problem

(14.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp(F)=wrap+cpr.

new (a,xs)=[a]+xs, glue (a,xs)=[a]+head xs+tail xs, partition≐([wrap wrap,newuglue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), fits w the coreflexive on a line x with width x≤w, ok w the coreflexive
on [x]+xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T.

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

| the law                                                                                                                                                                                          | what it says                                                                                                             | (14.5b) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{partition list}^+(\text{fits } w)}{\ni} \text{ est}(R)$                                                                                                                             | the specification: least waste among the paragraphs whose every line fits                                                |         |
| $\text{partition list}^+(\text{fits } w) = ([\text{wrap wrap}, \text{newu}(\text{glue } (ok\ w))])$<br>fusion, every word fits on a line by itself                                               | the fitting paragraphs are themselves a fold                                                                             |         |
| $[\text{wrap wrap}, \text{newu}(\text{glue } (ok\ w))]$<br>$= [\text{wrap wrap}, \text{new}] \cup ([\text{wrap wrap}, \text{glue}] (ok\ w))$                                                     | rewritten into binary thinning's shape $(f_1 p_1) \cup (f_2 p_2)$                                                        |         |
| $(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$                                                                                                                        | glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this |         |
| $(1 \times Q) \text{ new} \sqsubseteq \text{new } Q$<br>$(1 \times Q) (\text{glue } (ok\ w)) \sqsubseteq \text{glue } (ok\ w) Q$<br>cons monotonic on $\text{collect} \leq \text{collect}^\circ$ | both halves are monotonic on Q once ties in waste are broken by the first line                                           |         |
| merge $T = \text{cat}$ ; $P \triangleq \text{head prefix head}^\circ$ also serves                                                                                                                | T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input                        |         |
| $(\text{listcp}(F) \langle g_1, g_2 \rangle \text{ cat thinlist } Q) \text{ minlist } R$<br>Theorem 8.2 at $P \triangleq T$                                                                      | binary thinning, one candidate kept per first line                                                                       |         |
| paragraph $w = ([\text{start}, \text{cpr } \langle h_1, h_2 \rangle \text{ cat thinlist } Q]) \text{ minlist } R$                                                                                | the program, $g_1$ and $g_2$ split along the coproduct                                                                   |         |

## §14.6. Bitonic tours

(14.6a)

```

FA=(City×City)+(City×A), the base functor of cons-lists of length at least two; listcp(F)=wrap+cpr; tc :
City×City→Real, neither positive nor symmetric.

start (a,b)=[a,b],[a,b], dropl (a,([b]+xs,ys))=[a]+xs,[a]+ys, dropr (a,(xs,[b]+ys))=[a]+xs,
[a]+ys, tour≐([start,dropludropr]).

cost (xs,ys)=outcost xs+incost ys, outcost [a0,...,an]=tc (a0,a1)+...+tc (an-1,an), incost [a0,...,an]=tc
(a1,a0)+...+tc (an,an-1).

next≐tail head, next2≐next×next, R≐cost≤cost°, Q≐Rn(next2 next2°), P≐T, g1≐list([start,dropl]),
g2≐list([start,dropr]).

```

| the law                                                                                                                                                                      | what it says                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| $\frac{\text{tour}}{\ni} \text{ est}(R)$                                                                                                                                     | the specification: a least-cost bitonic tour, outward journey and return kept as a pair of lists                              |
| $(1 \times R) \text{ dropl} \sqsubseteq \text{dropl } R \quad \text{FALSE}$<br>$(1 \times R) \text{ dropr} \sqsubseteq \text{dropr } R \quad \text{FALSE}$                   | neither drop is monotonic on <b>R</b> : the two edges it adds and removes depend on <b>head</b> and <b>next</b> of both lists |
| $(1 \times Q) \text{ dropl} \sqsubseteq \text{dropl } Q$<br>$(1 \times Q) \text{ dropr} \sqsubseteq \text{dropr } Q$                                                         | both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input            |
| $\langle \text{listcp}(F) \langle g_1, g_2 \rangle \text{ cat thinlist } Q \rangle \text{ minlist } R$<br>Theorem 8.2 at $P \triangleleft T$ , merge $\tau = \text{cat}$     | binary thinning, one candidate per pair of second cities                                                                      |
| $\text{mintour} = \langle [\text{start wrap, cpr } \langle \text{list}(\text{dropl}), \text{list}(\text{dropr}) \rangle \text{ cat thinlist } Q] \rangle \text{ minlist } R$ | the program: quadratic, because each step adds just two tours to the list kept                                                |

(14.6b)

## §15. Dynamic Programming

### §15.1. Theory

$h : FB \rightarrow B$  a map,  $T : FA \rightarrow A$  an  $F$ -algebra,  $R : B \rightarrow B$ .

(15.1a)

$H \triangleq (T)^\circ \circ (h) : A \rightarrow B$ ,  $M \triangleq \frac{H}{\supseteq} \text{est}(R)$  the problem to be solved,  $(\mu X : G(X))$  as in (11.6.3a).

| the law                                                                                                                                                                                                                                                                   | what it says                                                                                       | (15.1b) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|---------|
| $(\mu X : \frac{T^\circ}{\supseteq} P(F(X)h) \text{ est}(R)) \sqsubseteq M$<br>Theorem 9.1, $h$ monotonic on $R$                                                                                                                                                          | <b>dynamic programming:</b> decompose in every way, solve each part, keep one optimum per part     |         |
| $\frac{T^\circ}{\supseteq} P(F(M)h) \text{ est}(R) \sqsubseteq M \quad (9.1)$                                                                                                                                                                                             | all Knaster–Tarski leaves to prove                                                                 |         |
| $\frac{T^\circ}{\supseteq} P(F(M)h) \text{ est}(R) \sqsubseteq H \quad (9.2)$<br>$H^\circ \frac{T^\circ}{\supseteq} P(F(M)h) \text{ est}(R) \sqsubseteq R \quad (9.3)$                                                                                                    | (9.1) split by $M = H \cap (H^\circ \setminus R^\circ)$                                            |         |
| $P(X) \text{ est}(R) \sqsubseteq (\exists X) \cap (\epsilon \setminus (XR^\circ))$<br>(9.4) = (7.10), (13.1.8a)                                                                                                                                                           | the only fact about <b>est</b> either proof uses                                                   |         |
| $(\mu X : \frac{T^\circ}{\supseteq} \text{thin } Q P(F(X)h) \text{ est}(R)) \sqsubseteq M$<br>Theorem 9.2, $Q$ a preorder with $QF(H)h \sqsubseteq F(H)hR$ ; <b>thin</b> $Q$ as in (14.1b)                                                                                | the same with a thinning step: decompositions that can never win are dropped                       |         |
| the fixed point is unique, and entire<br>Theorem 6.3, $T^\circ$ followed by $F$ 's membership relation inductive; $\frac{T^\circ}{\supseteq}$ finite and non-empty, $R$ connected                                                                                         | when the recursion can be refined to a recursive function                                          |         |
| $\frac{[V_1, V_2]^\circ}{\supseteq} \text{thin}(Q_1 + Q_2) P([U_1, U_2]) \text{ est}(R)$<br>$= (\text{Ran } V_1 \rightarrow W_1, W_2)$<br>$W_i \triangleq \frac{V_i^\circ}{\supseteq} \text{thin}(Q_i) P(U_i) \text{ est}(R)$<br>Proposition 9.1, $V_2 V_1^\circ = \perp$ | $FA$ is usually a coproduct, and disjoint ranges split the fixed point into one branch per summand |         |
| $F(R)h \sqsubseteq hR$<br>$QF(H)h \sqsubseteq F(H)hR$                                                                                                                                                                                                                     | the two conditions to be checked, in the order the three propositions below discharge them         |         |
| $F(R)h \sqsubseteq hR$<br>Proposition 9.2, $R \triangleq \text{cost} \leq \text{cost}^\circ$ , $h \text{ cost} = F(\text{cost})k$ , $F(\leq)k \sqsubseteq k \leq$                                                                                                         | monotonicity when the cost is itself a fold with a step $k$ monotonic on $\leq$                    |         |
| $F(R \cap (H^\circ H))h \sqsubseteq hR$<br>Proposition 9.3, $R \triangleq \text{cost} \leq \text{cost}^\circ$ , $h \text{ cost} = F(\langle \text{cost}, H^\circ \rangle)k$ , $F(\leq \times \mathbb{1})k \sqsubseteq k \leq$ , $H^\circ$ simple                          | monotonicity <b>in context</b> : $k$ may also read the input the part was built from               |         |
| $Q \triangleq F(U, V)$<br>Proposition 9.4, $U, V$ preorders, $F(U, R)h \sqsubseteq hR$ , $VH \sqsubseteq HR$                                                                                                                                                              | both conditions of Theorem 9.2 at once, split along the two arguments of a bifunctor               |         |

### §15.2. The string edit problem

$Op ::= \text{cpy Char} \mid \text{del Char} \mid \text{ins Char}$ ,  $F(A, B) = 1 + (A \times B)$ ,  $\alpha \triangleq [\text{nil}, \text{cons}]$ .

(15.2a)

$\text{edit} \triangleq ([\text{base}, \text{step}]) : [Op] \rightarrow [\text{Char}] \times [\text{Char}]$ , **base** returning  $([], [])$ , **step**  $(\text{cpy } a, (xs, ys)) = ([a] \# xs, [a] \# ys)$ , **step**  $(\text{del } a, (xs, ys)) = ([a] \# xs, ys)$ , **step**  $(\text{ins } a, (xs, ys)) = (xs, [a] \# ys)$ .

$\text{length} \triangleq ([\text{zero}, \pi_2 \text{ succ}])$ ,  $R \triangleq \text{length} \leq \text{length}^\circ$ ,  $U \triangleq T$ ,  $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$ ,  $Q \triangleq 1 + (U \times V)$ , **empty** the coreflexive on  $(xs, ys)$  with both lists empty.

**unstep** implements  $\frac{\text{step}^\circ}{\supseteq} \text{thin}(U \times V)$ : **unstep**  $([a] \# xs, []) = [(\text{del } a, (xs, []))]$ , **unstep**  $([], [b] \# ys) = [(\text{ins } b, ([], ys))]$ , **unstep**  $([a] \# xs, [b] \# ys) = (a \rightarrow [(\text{cpy } a, (xs, ys))], [(\text{del } a, (xs, [b] \# ys)), (\text{ins } b, ([a] \# xs, ys))])$ .

| the law                                                                                                                                                                                                                                                                                                                               | what it says                                                                                                                                          | (15.2b) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{edit}^\circ}{\supset} \text{est}(R)$                                                                                                                                                                                                                                                                                     | the specification: a shortest edit sequence from which both strings can be reconstituted                                                              |         |
| $F(R) \alpha \sqsubseteq \alpha R$<br>Proposition 9.2 at <code>length</code> , <code>succ</code> monotonic on $\leq$                                                                                                                                                                                                                  | <code>cons</code> is monotonic on $R$ , so Theorem 9.1 already applies                                                                                |         |
| $QF(1, \text{edit}^\circ) \alpha \sqsubseteq F(1, \text{edit}^\circ) \alpha R$                                                                                                                                                                                                                                                        | the thinning condition sought, over $F(0p, [\text{Char}] \times [\text{Char}])$                                                                       |         |
| $F(T, R) \alpha \sqsubseteq \alpha R$ left as an exercise                                                                                                                                                                                                                                                                             | $U \triangleq T$ costs nothing: any two operations may be compared                                                                                    |         |
| $\text{edit}(1 \times \text{suffix}) \sqsubseteq R^\circ \text{edit}$<br>$\text{edit}(\text{suffix} \times 1) \sqsubseteq R^\circ \text{edit}$                                                                                                                                                                                        | the other half of Proposition 9.4 for $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$                                                   |         |
| $\text{edit}(1 \times \text{tail}) \sqsubseteq R^\circ \text{edit}$<br>$\text{edit}(\text{tail} \times 1) \sqsubseteq R^\circ \text{edit}$<br>$\text{suffix} = \text{tail}^*$ , and $BA \sqsubseteq CB \implies BA^* \sqsubseteq C^*B$                                                                                                | one step is enough: drop the operation that produced the head, or weaken its <code>cpy</code> to a <code>del</code> — never lengthening the sequence  |         |
| $(\mu X : \frac{[\text{base}, \text{step}]^\circ}{\supset} \text{thin } Q P([\text{nil}, (1 \times X) \text{ cons}]) \text{ est}(R)) \sqsubseteq \frac{\text{edit}^\circ}{\supset} \text{est}(R)$<br>Theorem 9.2                                                                                                                      | a copy, when available, beats a delete or an insert                                                                                                   |         |
| $X = (\text{empty} \rightarrow \text{nil}, \frac{\text{step}^\circ}{\supset} \text{thin}(U \times V) P((1 \times X) \text{ cons}) \text{ est}(R))$<br>Proposition 9.1                                                                                                                                                                 | <code>base</code> and <code>step</code> have disjoint ranges                                                                                          |         |
| $\text{mle} = (\text{empty} \rightarrow \text{nil}, \text{unstep list}((1 \times \text{mle}) \text{ cons}) \text{ minlist } R)$                                                                                                                                                                                                       | the program: at most two decompositions survive, but the same subproblem is solved many times, so the running time is exponential                     |         |
| $\text{column } xs \text{ } ys = [\text{mle } (u, ys) \mid u \leftarrow \text{tails } xs]$<br>$\text{column } xs = ([\text{fstcol } xs, \text{nextcol } xs]), \text{fstcol} = \text{list}(\text{del})$<br>$\text{tails}$                                                                                                              | the tabulation: $\text{mle } (xs, ys)$ needs $\text{mle } (u, v)$ for every tail $u$ of $xs$ and $v$ of $ys$ , so the columns are built right to left |         |
| $\text{column } xs \text{ } ([b] \# ys) = \text{nextcol } xs \text{ } (b, \text{column } xs \text{ } ys)$<br>$\text{nextcol } xs \text{ } (b, us)$<br>$= ([\text{base } (b, \text{last } us), \text{step } b]) \# us$<br>$us = \text{zip } (xs, \text{zip } (\text{init } us, \text{tail } us))$                                      | each column is a fold built bottom to top, over $xs$ zipped with the adjacent pairs of the column to its right                                        |         |
| $\text{base } (b, u) = [[\text{ins } b] \# u]$<br>$\text{step } b \text{ } ((a, (u, v)), ws) =$<br>$(a = b \rightarrow [[\text{cpy } a] \# v] \# ws,$<br>$[b \text{ min}(R) ([\text{del } a] \# w, [\text{ins } b] \# u)] \# ws)$<br>$w = \text{head } ws$ ; these <code>base</code> , <code>step</code> are not <code>edit</code> 's | an entry depends on the one below it (a delete), the one to its right (an insert), and the one below that (a copy) — quadratic in the two lengths     |         |

### §15.3. Optimal bracketing

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <p><code>tree A ::= tip A   bin (tree A, tree A)</code>, <math>FX = A + X^2</math>, so <math>F(R) = 1 + R^2</math>; <math>h \triangleq [\text{tip}, \text{bin}]</math>, <math>\text{flatten} \triangleq ([\text{wrap}, \text{cat}])</math> : <code>tree A</code> <math>\rightarrow</math> <code>list</code> <math>+</math> <code>A</code>, <math>H = \text{flatten}^\circ</math>.</p> <p><math>(\text{cost}, \text{size}) \triangleq ([\text{opt}, \text{opb}])</math>, <math>\text{opt} \triangleq (\text{zero}, \text{st})</math>, <math>\text{opb } ((cx, sx), (cy, sy)) = (\text{cb } (sx, sy) + cx + cy, \text{sb } (sx, sy))</math>.</p> <p><code>sb</code> associative, so <math>\text{size} = \text{flatten } sz</math> for a map <math>sz</math>; <math>R \triangleq \text{cost} \leq \text{cost}^\circ</math>, <math>g \triangleq [\text{zero}, (1 \times sz)^2 \text{ opb } \pi_1]</math>, single the coreflexive on singleton lists.</p> <p><math>\text{splits} \triangleq (\text{inits}^+, \text{tails}^+)</math> <code>zip</code>, an implementation of <math>\frac{\text{cat}^\circ}{\supset}</math>; <math>\text{array} \triangleq \text{inits list}(\text{row})</math>, <math>\text{row} \triangleq \text{tails list}(\text{mct})</math>, <math>\text{col} \triangleq \text{inits list}(\text{mct})</math>.</p> <p><math>\text{mix} \triangleq \text{zip list}(\text{bin}) \text{ minlist } R</math>, <math>\text{next} \triangleq (\pi_1, \text{mix}) \text{ snoc}</math>, <math>\text{process} \triangleq ((\text{tip wrap}) \times 1) \text{ loop}(\text{next})</math>.</p> | (15.3a) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|

| the law                                                                                                                                                                                                                                                                                                                                                                                             | what it says                                                                                                                           | (15.3b) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{flatten}^\circ}{\ni} \text{est}(R)$                                                                                                                                                                                                                                                                                                                                                    | the specification: a least-cost bracketing of $a_1 \oplus \dots \oplus a_n$ , the tree whose flattening is the given list              |         |
| $[\text{tip}, \text{bin}] \text{ cost} = (1 + (\text{cost}, \text{flatten})^2) \text{ g} \quad (9.5)$                                                                                                                                                                                                                                                                                               | the cost of a node reads only the cost and the flattening of its two subtrees                                                          |         |
| $(1 + (\leq \times 1)^2) \text{ g} \in \text{g} \leq \quad (9.6)$                                                                                                                                                                                                                                                                                                                                   | $\text{g}$ is monotonic on $\leq$ in its two cost arguments                                                                            |         |
| $F(Rn(\text{flatten } \text{flatten}^\circ))h \in hR$<br>Proposition 9.3, $H^\circ = \text{flatten}$ a map                                                                                                                                                                                                                                                                                          | monotonicity in context: only trees with the same flattening are compared                                                              |         |
| $(\mu X : \frac{[\text{wrap}, \text{cat}]^\circ}{\ni} P([\text{tip}, (X \times X) \text{ bin}] \text{ est}(R)) \in \frac{\text{flatten}^\circ}{\ni} \text{est}(R) \quad \text{Theorem 9.1}$                                                                                                                                                                                                         | split the list in every way, bracket both halves, join                                                                                 |         |
| $X = (\text{single} \rightarrow \text{wrap}^\circ \text{ tip}, \frac{\text{cat}^\circ}{\ni} P((X \times X) \text{ bin}) \text{ est}(R))$<br>Proposition 9.1                                                                                                                                                                                                                                         | <code>wrap</code> and <code>cat</code> have disjoint ranges                                                                            |         |
| $\text{mct} = (\text{single} \rightarrow \text{head } \text{tip}, \text{splits } \text{list}((\text{mct} \times \text{mct}) \text{ bin}) \text{ minlist } R)$                                                                                                                                                                                                                                       | the program; exponential, since the segments of one list overlap                                                                       |         |
| $\text{mct} = (\text{single} \rightarrow \text{head } \text{tip}, (\text{init } \text{col}, \text{tail } \text{row}) \text{ mix}) \quad (9.7)$                                                                                                                                                                                                                                                      | the tabulation: <code>mct xs</code> is needed for every non-empty segment <code>xs</code> , so the values are held as an array of rows |         |
| $\text{col} = (\text{single} \rightarrow \text{head } \text{tip } \text{wrap}, (\text{init } \text{col}, \text{tail } \text{row}) \text{ next}) \quad (9.8)$<br>$\text{cons } \text{col} = (1 \times \text{array}) \text{ process} \quad (9.9)$<br>$\text{row} = (\text{single} \rightarrow \text{head } \text{tip } \text{wrap}, (\text{mct}, \text{tail } \text{row}) \text{ cons}) \quad (9.10)$ | a column extends the column to its left, a row the row below it; (9.9) is (9.8) rewritten as a loop                                    |         |
| $\text{array} = ([\text{fstcol}, \text{addcol}]), \text{fstcol} \hat{=} \text{tip } \text{wrap } \text{wrap}$<br>$\text{addcol} \hat{=} (\pi_1 \text{ tip } \text{wrap}, \text{step}) \text{ cons}$<br>$\text{step} \hat{=} (\text{process } \text{tail}, \pi_2) \text{ zip } \text{list}(\text{cons})$                                                                                             | the program: one fold building the array column by column, cubic in the length of the input                                            |         |

## §15.4. Data compression

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <pre> [A] ::= nil   snoc ([A], A), list+ A ::= wrap A   snoc (list+ A, A), String = [Char], Code ::= sym Char   ptr (String, String+), F(Code, String) = 1 + (String × Code), α̂ = [nil, snoc].  decodê = ([nil, extend]) : [Code] → String, extend (xs, sym a) = xs # [a], extend (xs, ptr (ys, zs)) = xs # zs when ys # zs is a proper prefix of xs # zs; H = decode°.  sizê = ([zero, distr [1 × c, 1 × p] plus]) with c, p the constant costs of a symbol and a pointer; R̂ = size ≤ size°, Q̂ = F(T + T, prefix°) = 1 + (prefix° × (T + T)), the two T on symbols and on pointers.  lrt ws = est(prefix° × (T + T)) {(xs, (ys, zs))   xs # zs = ws, ys # zs a proper prefix of ws}, the longest repeated tail; reduce (ws # [a]) = (zs ≠ [] → [(ws, sym a), (xs, ptr (ys, zs))], [(ws, sym a)]) with (xs, (ys, zs)) = lrt (ws # [a]). </pre> | (15.4a) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|

| the law                                                                                                                                                                                                                                                     | what it says                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{\text{decode}^\circ}{\ni} \text{est}(R)$                                                                                                                                                                                                             | the specification: a smallest code sequence decoding to the given string                                                             |
| $F(R) \alpha \sqsubseteq \alpha R \quad (\tau + \tau)[c, p] = [c, p]$                                                                                                                                                                                       | monotonicity is routine, the two costs being constants                                                                               |
| $F(\tau + \tau, R) \alpha \sqsubseteq \alpha R$<br>decode prefix $\in R^\circ$ decode<br>Proposition 9.4 at $Q \triangleq F(\tau + \tau, \text{prefix}^\circ)$                                                                                              | the thinning condition split in two: pointers are ordered only by how much input they consume, symbols and pointers not at all       |
| decode init $\in R^\circ$ decode                                                                                                                                                                                                                            | enough for the second: drop the last character of the output by shortening or removing the last code element, never raising the cost |
| $(\mu X : \frac{[\text{nil}, \text{extend}]^\circ}{\ni} \text{thin } Q P([\text{nil}, (X \times 1) \text{ snoc}]) \text{est}(R)) \sqsubseteq \frac{\text{decode}^\circ}{\ni} \text{est}(R)$<br>Theorem 9.2                                                  | between a symbol and a pointer nothing can be decided in advance; between two pointers the longer match wins                         |
| $X = (\text{null} \rightarrow \text{nil}, \frac{\text{extend}^\circ}{\ni} \text{thin}(\text{prefix}^\circ \times (\tau + \tau)) P((X \times 1) \text{ snoc}) \text{est}(R))$<br>Proposition 9.1                                                             | <code>nil</code> and <code>extend</code> have disjoint ranges                                                                        |
| $\frac{\text{extend}^\circ}{\ni} (\text{ws} \# [a]) = \{(\text{ws}, \text{sym } a)\} \cup \{(x\text{s}, \text{ptr } (y\text{s}, z\text{s})) \mid x\text{s} \# z\text{s} = \text{ws} \# [a], y\text{s} \# z\text{s} \text{ a proper prefix of } \text{ws}\}$ | the decompositions of one string: take the last character as a symbol, or end with a pointer                                         |
| reduce implements $\frac{\text{extend}^\circ}{\ni} \text{thin}(\text{prefix}^\circ \times (\tau + \tau))$                                                                                                                                                   | thinning leaves at most two: the symbol, and the pointer of <code>lrt</code>                                                         |
| encode = ( <code>null</code> $\rightarrow$ <code>nil</code> , reduce list(( <code>encode</code> $\times$ 1) <code>snoc</code> ) <code>minlist</code> R)                                                                                                     | the program, again exponential; the book gives no tabulation for it                                                                  |



## §16. Greedy Algorithms

### §16.1. Theory

$h, T, R, H, M$  as in (15.1a); additionally  $Q$  a **connected** preorder on the sets  $\frac{T^\circ}{\supset}$  returns, so that  $\frac{T^\circ}{\supset} \text{est}(Q)$  is entire. (16.1a)

| the law                                                                                                                                                                                                                          | what it says                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $(\mu X : \frac{T^\circ}{\supset} \text{est}(Q) F(X)h) \sqsubseteq M$<br>Theorem 10.1, the hypotheses of Theorem 9.2                                                                                                             | <b>greedy</b> : one decomposition is kept at every step, so $P$ and $\frac{T^\circ}{\supset}$ disappear from the recursion (16.1b)                                                                                                              |
| $\frac{[V_1, V_2]^\circ}{\supset} \text{est}(Q_1 + Q_2) [U_1, U_2]$<br>$= (\text{Ran } V_1 \rightarrow W_1, W_2)$<br>$W_i \triangleq \frac{V_i^\circ}{\supset} \text{est}(Q_i) U_i$<br>Proposition 10.1, $V_2 V_1^\circ = \perp$ | Proposition 9.1 with <b>est</b> for <b>thin</b>                                                                                                                                                                                                 |
| $Q \triangleq F(U, V)$ Proposition 9.4                                                                                                                                                                                           | still the way to get both conditions, but $\frac{T^\circ}{\supset} \text{est}(Q)$ must now be entire as well                                                                                                                                    |
| Theorem 7.2                                                                                                                                                                                                                      | that greedy theorem chooses among the <b>results</b> of one relational step of a reduce; Theorem 10.1 chooses among the <b>decompositions</b> of the input, for an arbitrary $T$ rather than an initial algebra, at the cost of $h$ being a map |

### §16.2. The detab-entab problem

$\text{detab} \triangleq ([\text{nil}, \text{expand}]) : \text{String} \rightarrow \text{String}$  over snoc-lists,  $\alpha \triangleq [\text{nil}, \text{snoc}]$ ,  $H = \text{detab}^\circ$ ;  $\text{expand } (xs, a) = (a = \text{TB} \rightarrow \text{fill } xs, xs \# [a])$ ,  $\text{fill } xs = xs \# \text{blanks } (n - (\text{col } xs) \bmod n)$ . (16.2a)

$\text{col} \triangleq ([\text{zero}, \text{count}])$ ,  $\text{count } (c, a) = (a = \text{NL} \rightarrow 0, c + 1)$ ; TB the tab, BL the blank, NL the newline, tab stops every  $n$  columns.

$R \triangleq \text{length} \leq \text{length}^\circ$ ,  $U$  the preorder with  $a U b \iff a = \text{TB} v a = b$ ,  $V \triangleq \text{prefix}^\circ n (\text{fill } \text{fill}^\circ)$ ,  $Q \triangleq 1 + (V \times U)$ .

$\text{unfill } xs$  the shortest prefix of  $xs$  with  $\text{fill } (\text{unfill } xs) = \text{fill } xs$ ;  $\text{tbc}$  the trailing blank count,  $\text{triple} \triangleq (\text{unfill } \text{entab}, \text{tbc}, \text{col})$ .

| the law                                                                                                                                                                                                                                                                                                  | what it says                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{\text{detab}^\circ}{\supset} \text{est}(R)$                                                                                                                                                                                                                                                       | the specification: a shortest input <b>detab</b> expands to the given output — <b>detab entab=1</b> and nothing shorter does                   |
| $F(T,R) \alpha \sqsubseteq \alpha R$ left as an exercise                                                                                                                                                                                                                                                 | $U$ may be any preorder on characters; $a \sqcup b \iff a=TBva=b$ puts <b>TB</b> below every character, so <b>est</b> prefers a tab to a blank |
| $\text{detab prefix} \in R^\circ \text{ detab}$ FALSE                                                                                                                                                                                                                                                    | at $n=8$ , <b>detab [a,b,c,d,e,TB]=[a,b,c,d,e,BL,BL,BL]</b> , whose prefix <b>[a,b,c,d,e,BL,BL]</b> is longer than any input giving it         |
| $\text{detab } V^\circ \in R^\circ \text{ detab } V \triangleq \text{prefix}^\circ n(\text{fill fill}^\circ)$                                                                                                                                                                                            | true once only the prefixes that do not cross a tab stop are allowed                                                                           |
| $\text{nil } V^\circ = \text{nil fill } V^\circ = \text{fill}$<br>$\text{snoc } V^\circ \sqsubseteq \text{snoc } U(\pi_1 V^\circ)$ left as exercises                                                                                                                                                     | the three properties of $V$ the derivation rests on; the second is what <b>fill fill</b> <sup>°</sup> was added for                            |
| $\text{expand } V^\circ \sqsubseteq \text{expand } U(\pi_1 V^\circ)$                                                                                                                                                                                                                                     | the claim they add up to: shortening the output either leaves the last step alone or discards it                                               |
| $\text{detab } V^\circ \sqsubseteq \text{detab } U(\text{init detab } V^\circ)$<br><i>init</i> inductive, so the greatest solution is the unique one                                                                                                                                                     | hence $\text{detab } V^\circ \sqsubseteq \text{prefix detab}$ , and $\text{prefix} \in R^\circ$ finishes Proposition 9.4                       |
| $(\mu X : \frac{[\text{nil}, \text{expand}]^\circ}{\supset} \text{est}(Q) (1+(X \times 1)) [\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{detab}^\circ}{\supset} \text{est}(R)$<br>Theorem 10.1                                                                                                      | greedy: one character of input is decided at each step                                                                                         |
| $X = (\text{null} \rightarrow \text{nil}, \frac{\text{expand}^\circ}{\supset} \text{est}(V \times U) (X \times 1) \text{snoc})$ Proposition 10.1                                                                                                                                                         | <b>nil</b> and <b>expand</b> have disjoint ranges                                                                                              |
| $\frac{\text{expand}^\circ}{\supset} (xs \# [a]) = \{(ys, TB) \mid \text{fill } ys = xs \# [a]\} \cup \{(xs, a)\}$<br>the first set is non-empty iff $a=BL$ and $\text{col } (xs \# [a]) \bmod n = 0$                                                                                                    | the last output character came from a tab only if it is a blank landing exactly on a tab stop                                                  |
| $\frac{\text{expand}^\circ}{\supset} \text{est}(V \times U) (xs \# [a]) =$<br>$(a=BL \wedge \text{col } (xs \# [a]) \bmod n = 0 \rightarrow (\text{unfill } xs, TB), (xs, a))$                                                                                                                           | the greedy step: emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop                                    |
| $\text{entab } xs = \text{entab } (\text{unfill } xs) \# \text{blanks } (tbc \ xs)$ (10.1)                                                                                                                                                                                                               | what makes <b>triple</b> a <b>snoc-list</b> reduce: the output splits at the last tab stop                                                     |
| $\text{triple} = ([\text{base}, \text{op}])$ and $\text{entab} = \text{triple assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$                                                                                                                                                                       | the program: one pass carrying the column and the count of pending blanks                                                                      |
| <b>base</b> returns $([], (0, 0))$<br><b>op</b> $((xs, (t, c)), a) =$<br>$(a=BL \wedge (c+1) \bmod n \neq 0 \rightarrow (xs, (t+1, c+1)),$<br>$a=BL \rightarrow (xs \# [TB], (0, c+1)),$<br>$a=NL \rightarrow (xs \# \text{blanks } t \# [NL], (0, 0)),$<br>$(xs \# \text{blanks } t \# [a], (0, c+1)))$ | hold a blank back, cash the held blanks in for a tab at a tab stop, flush them at a newline, flush them before anything else                   |

### §16.3. The minimum tardiness problem

$FX = 1 + (X \times \text{Job})$ ,  $\alpha \triangleq [\text{nil}, \text{snoc}]$  on schedules,  $\beta \triangleq [\text{nil}, \text{snag}]$  on bags, **snag** putting a job into a bag;  $\text{bagify} \triangleq ([\beta]) : [\text{Job}] \rightarrow \text{Bag Job}$ ,  $H = \text{bagify}^\circ$ .

$\text{ct}, \text{dt}, \text{wt} : \text{Job} \rightarrow \text{Real}$  the completion, due and weighting quantities of a job;  $\text{penalty } (xs, j) = (\text{sum } (\text{list}(\text{ct}) \ xs) + \text{ct } j - \text{dt } j) \times \text{wt } j$ .

$\text{cost} \triangleq \frac{\text{prefix}}{\supset} P(\alpha^\circ [\text{zero}, \text{penalty}]) \text{est}(\geq)$ ,  $\text{cost } [] = 0$ ,  $\text{cost } (xs \# [j]) = \text{bmax } (\text{cost } xs, \text{penalty } (xs, j))$ ,  $R \triangleq \text{cost} \leq \text{cost}^\circ$ .

$\text{perm} \triangleq \text{bagify bagify}^\circ = ([\text{nil}, \text{add}])$ ,  $\text{add } (xs, j) = ys \# [j] \# zs$  for some  $xs = ys \# zs$ .

$k \triangleq [\text{zero}, \text{assocr } (1 \times ((\text{bagify}^\circ \times 1) \text{penalty})) \text{bmax}]$ ,  $f \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{penalty}]$ ,  $Q \triangleq f \leq f^\circ$ .

| the law                                                                                                                                                                                                                    | what it says                                                                                                                                                                                      | (16.3b) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{bagify}^\circ}{\supset} \text{est}(R)$                                                                                                                                                                        | the specification: an ordering of the given bag of jobs with least maximum penalty                                                                                                                |         |
| $F(Rn(\text{bagify } \text{bagify}^\circ))\alpha \sqsubseteq \alpha R \quad (10.2)$                                                                                                                                        | monotonicity in context — only schedules of one bag are compared                                                                                                                                  |         |
| $(Qn(\beta\beta^\circ))F(\text{bagify}^\circ)\alpha \sqsubseteq F(\text{bagify}^\circ)\alpha R \quad (10.3)$                                                                                                               | the greedy condition, also in context                                                                                                                                                             |         |
| $\text{cost } (xs\#[j]) = \text{bmax } (\text{cost } xs, \text{penalty } (\text{perm } xs, j))$<br>$\alpha \text{ cost} = F((\text{cost}, \text{bagify}^\circ))k, F(\leq \times 1)k \sqsubseteq k \leq$<br>Proposition 9.3 | (10.2): <b>penalty</b> sums the completion times of $xs$ , and a sum does not see the order                                                                                                       |         |
| $\alpha \text{ cost} = (g, h) \text{ bmax} \quad (10.4)$<br>$g \triangleq [\text{zero}, \text{penalty}] \quad (10.5)$<br>$h \triangleq [\text{zero}, \pi_1 \text{ cost}] \quad (10.6)$                                     | the cost of a schedule split into the penalty of its last job and the cost of the rest                                                                                                            |         |
| $\text{add} \sqsubseteq \pi_1 R \quad (10.7)$<br>$\beta \text{ bagify}^\circ = F(\text{bagify}^\circ) [\text{nil}, \text{add}] \quad (10.8)$                                                                               | adding a job never lowers the cost; and <b>bagify</b> <sup>°</sup> is itself a reduce on bags. Together: $\beta \text{ bagify}^\circ \sqsubseteq F(\text{bagify}^\circ) h \leq \text{cost}^\circ$ |         |
| $F(\text{bagify})Q^\circ F(\text{bagify}^\circ) = g \geq g^\circ$ , met by<br>$Q \triangleq f \leq f^\circ, f \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{ penalty}]$                                    | the specification $Q$ has to satisfy, and the choice that meets it: a $Q$ -minimum is a job of least penalty                                                                                      |         |
| no greedy <b>reduce</b> exists                                                                                                                                                                                             | a greedy snoc-list reduce would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole                                           |         |
| $X = (\text{null} \rightarrow \text{nil}, \frac{\text{snag}^\circ}{\supset} \text{est}(Q) (X \times 1) \text{ snoc}) \quad \text{Theorem 10.1, Proposition 10.1}$                                                          | $\text{nil}$ and $\text{snag}$ have disjoint ranges                                                                                                                                               |         |
| $\text{schedule} = (\text{null} \rightarrow \text{nil}, \text{pick } (\text{schedule} \times 1) \text{ snoc})$<br>$\text{pick} \sqsubseteq \frac{\text{snag}^\circ}{\supset} \text{est}(Q)$ , a partial function           | the program: repeatedly remove a job of least penalty and put it last; quadratic in the number of jobs                                                                                            |         |

## §16.4. The TeX problem

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $\text{intern} \triangleq \text{val } \text{round} : \text{Decimal} \rightarrow [0, 2^{16}), \text{val} \triangleq ([\text{zero}, \text{shift}]), \text{shift } (d, r) = (d+r)/10, \text{round } r \text{ rounds } 2^{16}r \text{ to the nearest integer: } \text{round } r = n \iff 2n-1 < 2^{17}r < 2n+1.$<br>$\text{interval } n = ((2n-1)/2^{17}, (2n+1)/2^{17}), r \text{ inrange } (a, b) \iff a < r < b, \text{round}^\circ = \text{interval inrange}, R \triangleq \text{length} \leq \text{length}^\circ.$<br>Interval the pairs $(a, b)$ with $0 < b < 1$ and $a < b$ (10.9) $[\text{arb}, \text{step}] : 1 + (\text{Digit} \times \text{Interval}) \rightarrow \text{Interval}, \text{step } (d, (a, b)) = ((d+a)/10, (d+b)/10).$<br>$FX = 1 + (\text{Digit} \times X), \alpha \triangleq [\text{nil}, \text{cons}], h \triangleq ([\text{arb}, \text{step}]) = H^\circ, ! : \text{Digit} \times \text{Interval} \rightarrow 1, Q \triangleq (l^\circ !^\circ r) \cup 1, w \triangleq 2^{17}.$ | (16.4a) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|

| the law                                                                                                                                                                                        | what it says                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{\text{intern}^\circ}{\ni} \text{ est}(R) = \text{interval } \frac{\text{inrange val}^\circ}{\ni} \text{ est}(R)$                                                                        | the specification: a shortest decimal whose internal representation is the given multiple of $2^{-16}$ . $\text{round}^\circ$ is not a map, but $\text{interval}$ is, so it comes out of the $\ni$ |
| $\text{val inrange}^\circ = ([\text{arb}, \text{step}])$<br>fusion; $\text{zero inrange}^\circ = \text{arb}$ , $\text{shift inrange}^\circ = (1 \times \text{inrange}^\circ)$<br>$\text{step}$ | the converse of $\text{val}$ , cut down to intervals, is a reduce on cons-lists                                                                                                                    |
| $F(R) \alpha \sqsubseteq \alpha R$                                                                                                                                                             | $\text{cons}$ is monotonic on $R$ — routine                                                                                                                                                        |
| $\alpha^\circ F(h) Q^\circ \sqsubseteq R^\circ \alpha^\circ F(h)$                                                                                                                              | the greedy condition of Theorem 10.1, converted                                                                                                                                                    |
| $\frac{\text{arb}^\circ}{\ni} (a, b) = (a < 0 \rightarrow \{*\}, \{\})$<br>$\frac{\text{step}^\circ}{\ni} (a, b) = \{(d, (10a-d, 10b-d)) \mid 0 < 10b-d < 1\}$                                 | the decompositions of an interval: stop, or take one more digit                                                                                                                                    |
| $0 < 10b-d_1 < 1$ and $0 < 10b-d_2 < 1$ imply $d_1 = d_2$                                                                                                                                      | $\text{step}^\circ$ is in fact a map, $d$ the digit with $0 < 10b-d < 1$ , so $\frac{[\text{arb}, \text{step}]^\circ}{\ni}$ returns at most two elements and $Q$ need only choose between them     |
| $Q \triangleq (l^\circ !^\circ r) \cup 1$ , and $! \text{ nil} \sqsubseteq \text{cons } R^\circ$<br>from $! \text{ nil length} \sqsubseteq \text{cons length} \geq$                            | stop whenever stopping is legal: the empty decimal is shorter than any other                                                                                                                       |
| $(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{\ni} \text{ est}(Q) F(X) \alpha) \sqsubseteq \frac{\text{intern}^\circ}{\ni} \text{ est}(R)$ Theorem 10.1                                     | greedy: emit the one digit the interval allows, until the interval contains zero                                                                                                                   |
| $\text{extern} = \text{interval } f$<br>$f(a, b) = (a < 0 \rightarrow [], [d] \# f(10a-d, 10b-d))$                                                                                             | the program, with $d$ the digit above                                                                                                                                                              |
| $\text{extern } n = f(2n-1, 2n+1)$<br>$f(p, q) = (p \leq 0 \rightarrow [], [d] \# f(10p-w d, 10q-w d))$<br>$d = (10q) \text{ div } w, w = 2^{17}$                                              | the same in integer arithmetic only, as chapter 3 required of $\text{intern}$ : every interval reached is $(p/w, q/w)$                                                                             |

## §17. Appendix

### §17.1. $P(S) \text{ est}(R) = (\exists S) \cap (\epsilon \setminus (SR^\circ))$

$(p, q)$  tabulates  $W \triangleq (\exists S) \cap (\epsilon \setminus (SR^\circ))$ , and  $y \triangleq \frac{(p \exists S) \cap (qR)}{\exists}$ , a map.

(17.1a)

(17.1b)

$$\begin{array}{c} \boxed{\begin{array}{c} W \\ P(S) \text{ est}(R) \end{array}} \xleftarrow{p^\circ q = W, 1 \sqsubseteq yy^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \\ \boxed{\begin{array}{c} y^\circ q \\ \text{est}(R) \end{array}} \end{array}$$

(17.1c)

$$\begin{array}{c} \boxed{\begin{array}{c} y^\circ q \\ \exists \end{array}} \xleftarrow{f^\circ \cdot \neg f \cdot, \cdot \exists \neg \neg} \boxed{\begin{array}{c} q \\ (p \exists S) \cap (qR) \end{array}} \xleftarrow{\Delta \neg \cap, f^\circ \cdot \neg f \cdot} \begin{array}{c} \boxed{\begin{array}{c} p^\circ q \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} 1 \\ R \end{array}} \end{array}$$

R reflexive

(17.1d)

$$\begin{array}{c} \boxed{\begin{array}{c} \epsilon y^\circ q \\ R^\circ \end{array}} \xleftarrow{\cdot \exists \neg \neg, \circ, \text{meet}} \boxed{\begin{array}{c} R^\circ q^\circ q \\ R^\circ \end{array}} \quad \text{q simple} \\ \\ \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \xleftarrow{\Delta \neg \cap, \cdot T \neg / T, \circ, f^\circ \cdot \neg f \cdot} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \end{array} \\ \\ \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \xleftarrow{\cdot \exists \neg \neg, \text{meet}} \boxed{\begin{array}{c} p^\circ p \exists S \\ \exists S \end{array}} \quad \text{p simple} \\ \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} p \exists \\ (p \exists) \cap (qRS^\circ) \end{array}} \xleftarrow{1 \sqsubseteq qq^\circ} \boxed{\begin{array}{c} q^\circ p \exists \\ RS^\circ \end{array}} \xleftarrow{\circ, T \cdot \neg T \setminus} \boxed{\begin{array}{c} W \\ \epsilon \setminus (SR^\circ) \end{array}} \quad \text{W's right half} \end{array}$$

### §17.2. $P(\text{est}(R)) \text{ est}(R) = P(\text{Dom}(\text{est}(R))) \text{ union est}(R)$

(17.2a)

$$\boxed{P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{Dom}(\text{est}(R)) \text{ est}(R) = \text{est}(R)} \boxed{P(\text{Dom}(\text{est}(R)) \text{ est}(R)) \text{ est}(R)}$$

$$\xrightarrow{P \text{ a relator}} \boxed{P(\text{Dom}(\text{est}(R))) P(\text{est}(R)) \text{ est}(R)} \xrightarrow{(13.1.9a)} \boxed{P(\text{Dom}(\text{est}(R))) \text{ union est}(R)}$$

(17.2b)

$$\boxed{\begin{array}{c} P(\text{Dom}(\text{est}(R))) \text{ union est}(R) \\ P(\text{est}(R)) \text{ est}(R) \end{array}} \xleftarrow{(17.1b) \text{ at } S := \text{est}(R), \Delta \neg \cap, T \cdot \neg T \setminus} \boxed{\begin{array}{c} P(\text{Dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \quad \boxed{\begin{array}{c} \epsilon P(\text{Dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}}$$

$$\boxed{\begin{array}{c} P(\text{Dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \xleftarrow{P(\text{Dom}(\text{est}(R))) \sqsubseteq 1} \boxed{\begin{array}{c} \text{union est}(R) \\ \exists \text{est}(R) \end{array}}$$

$$\xleftarrow{T(\text{Un}V) \sqsubseteq T \text{Un}TV, \cdot \exists \neg \neg, (13.1.4a)} \boxed{\begin{array}{c} (\exists \exists) \cap (\epsilon \setminus (\epsilon \setminus R^\circ)) \\ \exists \text{est}(R) \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} \exists (\exists \cap (\epsilon \setminus (\epsilon \setminus R^\circ))) \\ \exists (\exists \cap (\epsilon \setminus R^\circ)) \end{array}} \quad \text{count of } T \cdot \neg T \setminus$$

$$\boxed{\begin{array}{c} \epsilon P(\text{Dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\epsilon \text{ lax natural}} \boxed{\begin{array}{c} \text{Dom}(\text{est}(R)) \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}}$$

